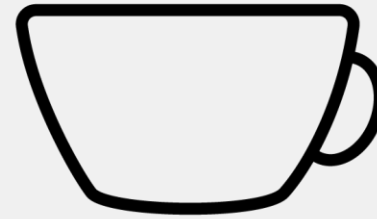
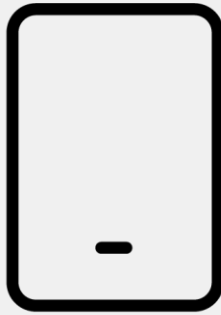


Network Systems Administration



Housekeeping



How to complete this module

As part of your Live event, you will need to complete the activity book and Labs.

- Open your **Learner Activity Document** for this module and use it to document your answers.
- Add your **full name and date** in the **header** of your **Learner Activity Document**.
- You may need to ask / do some research to complete some of the activities.
- Once your **Learner Activity Document** is complete, upload it to **Bud** along with your **Certificate of completion**.
- Most importantly, if you need **help, message or email your tutor**.



Learning outcomes

You will:

- Explain server hardware and operating system (OS) roles, features, and functions.
- Compare server types, storage options, Redundant Array of Independent Disks (RAID) levels, and disaster recovery (DR) sites.
- Evaluate system configurations, storage solutions, and network maintenance strategies.
- Identify tools and steps for server installation, configuration, and post-installation.
- Apply virtualisation, secure installation, and RAID setup in case studies and labs.



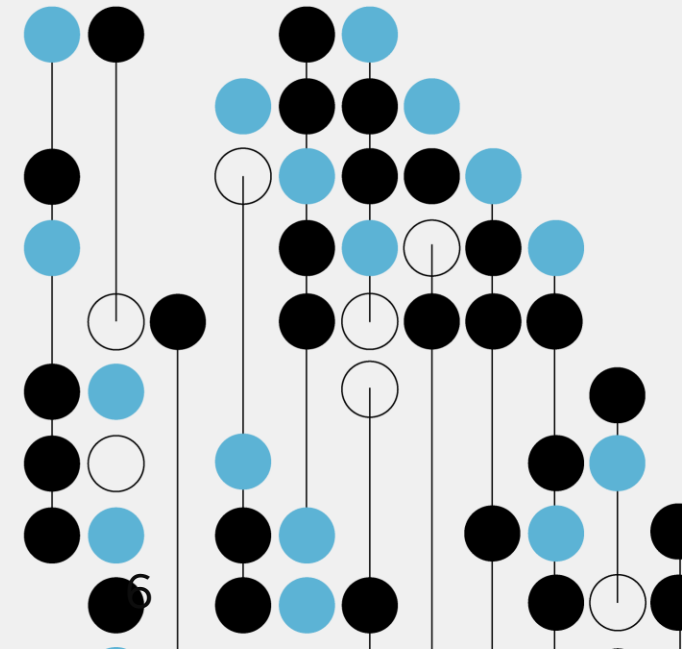
Learning outcomes

You will:

- Discuss automation, Simple Network Management Protocol (SNMP), and network automation risks / benefits.
- Describe OS deployment, monitoring, and performance optimisation for Linux and Windows.
- Construct and optimise file systems, partitions, memory, and troubleshoot server OS, Active Directory, print servers, and performance tools.
- Recommend best practices for Service Level Agreements (SLAs), change management (CM), and documentation.



Day 1: The architecture, physical and virtual, of typical IT systems



Objectives: Day 1

- Explain how server hardware, OS, redundancy, virtualisation, and continuity strategies support reliability and uptime.
- Evaluate server types, configurations, and storage options based on performance, cost, and business needs.
- Analyse rack / blade design considerations (ventilation, security, chassis) and compare storage types by features, durability, and use case.
- Recommend and justify storage and RAID solutions using case studies.
- Outline, set up, and troubleshoot Redundant Array of Independent Disks (RAID) configurations.



Understand server features: Discussion and Q&A

- What are the main features of a server, and what is the difference between a server and a standard desktop computer?
- How do Windows and Linux server operating systems differ in features and functionality?
- What measures can be put in place to ensure hardware redundancy and why is this important in servers?
- What is the impact of virtualisation to server performance?
- How can businesses ensure continuity when they deploy server to meet business needs?



Server usage and deployment

- Used by multiple people at once.
- Used to provide public services (File transfer, Web services).
- Provide the means for businesses to restrict access to internal users only.
- Used to provide network services.
- **Additional deployment requirements:**
 - Provides high availability / fault tolerant.
 - Used for load balancing.
 - Can scale very well to meet demand.



Types of server hardware: Research

Research the following types of server hardware which will be followed by Q&A and group discussions:

- Blade server (midplane / backplane)
- Rack
- Tower
- Server cluster

You have **20 minutes**.



Types of server hardware: Discussion and Q&A

- Can anyone tell us the key differences between blade, rack, tower, and cluster servers?
- What consideration should be in place before choosing the right server hardware (rack, blade, tower)?
- How does blade optimise redundancy and power usage?
- What benefits does server cluster provide in terms of performance and resilience?
- Can anyone provide a scenarios where each type of server (blade, rack, tower, or cluster) would be deployed?

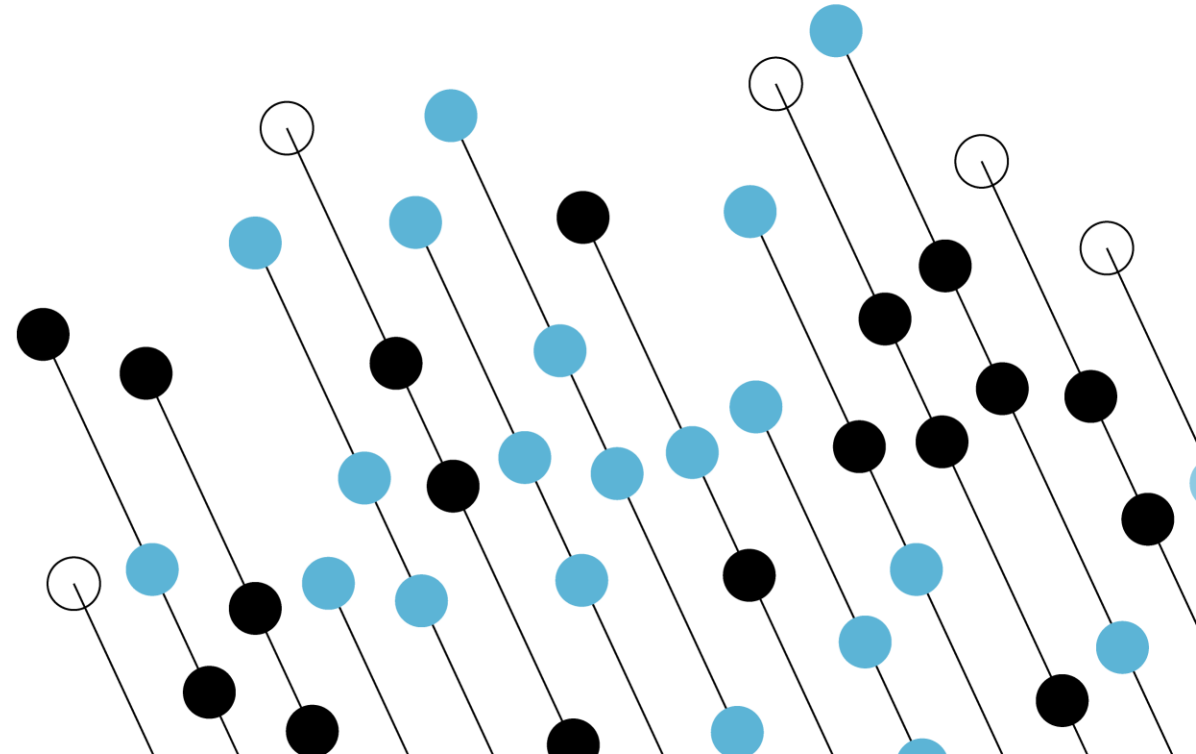


Server deployment consideration

- Physical space
- Services provided
- Space requirement
- Power consumptions (120V, 208V, 240V)
- Data centre deployment
- Environmental factors
- Expansion requirements



Server hardware types



Tower server



Tower server characteristics

- Servers are stacked in a standardised rack.
- Normally used in data centres and large enterprise.
- Rectangular, but shape can vary.
- Requires no mounting hardware.
- Can be place on top of a desk.
- Parts can be replaced by removing the side panel.
- Heat sink.



Tower server advantages and disadvantages

Advantages

- Racks are more affordable, making them a good choice for small businesses.
- Easier to cool as they are usually standalone device.
- Can be quite as they do not generate noise like blade and rack mount.
- Doesn't require much space if deployed as a single tower.

Disadvantages

- Can be bulky and heavy to move around.
- Difficult to stack when a couple of towers are needed.
- Not efficient for large organisation deployment.
- Has limitations for expansion.

Tower server use case

- In a small business where basic networking services are required.
- Can be used a dedicated backup system.
- Used where space is an issue.
- Used for development and testing.
- Commonly used in a warehouse printing / scanning terminal.



Rack mount server



Rack mount server characteristics

- Allows stacking of multiple servers mounted in rack enclosures.
- Servers can easily be pulled out and configured.
- Includes rail and cable management arms.
- Normal rack width size is 19inches.
- Depth range is from 31.5 to 41.25inches.
- Height varies from 1U(1.75") to 42U inches.



Rack enclosures

- Ventilation is from front to back.
- Hot air is sucked from the back.
- The rack has to be secured to the floor.
- Rack door has lock to prevent unauthorised access.
- Efficient colling as rack has integrated cooling system.



Rack server chassis components

System board includes:

- Central processing unit (CPU).
- Random access memory (RAM).
- Host bus adapter (HBA).
- Input / Output ports.
- Drive slots (HDD / SSD).



Rack mount advantages and disadvantages

Advantages

- Improved server density.
- Provides cable management.
- Improved floor space usage.
- Lockable enclosures to protect cabling from emanation.
- They are very scalable.

Disadvantages

- Might have heat dissipation issue if not correctly placed.
- A specialised server mounting procedures must be followed.
- They are pricier than tower servers.
- Setup can be complex as rack mounting and power distribution is required.

Rack servers use case

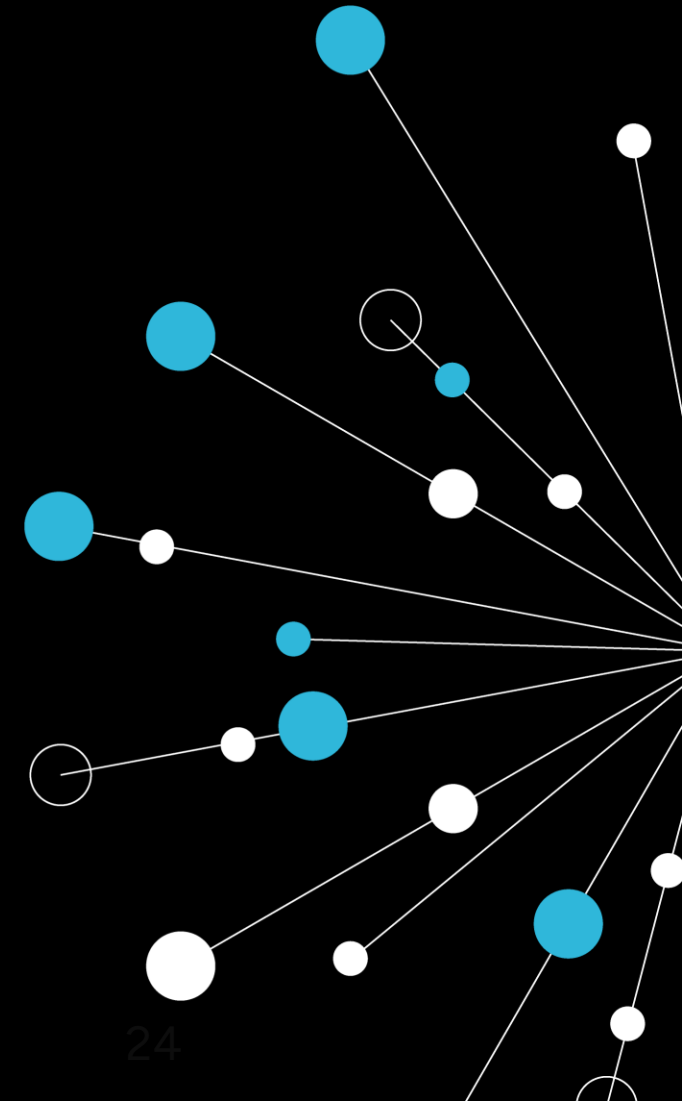
- Medium to large enterprise businesses
- Consolidation of different servers
- When floor space consideration is necessary
- Where easy cable management is required
- For easy scalability
- Widely used in data centres



Activity 1: Choosing tower or rack mount server

Find the activity in your **Learner Activity Document** and read the instructions.

Read the case study, analyse the business needs, and recommend a tower or rack mount server.



Blade server technology



Blade server technology

- High density servers
- Requires a lot of CPU / memory
- Fantastic for virtualisation
- High availability
- Blade enclosure can provide:
 - Integrated power and cooling.
 - High speed network switching.
 - Storage blades.
 - Secure management access.



Blade server advantages and disadvantages

Advantages

- Highly redundant enclosure for blade servers.
- Supply power and connectivity to all blades.
- Greatly reduces number of cables in a full rack.
- Most components are hot-pluggable.
- Provide a highly redundant enclosures

Disadvantages

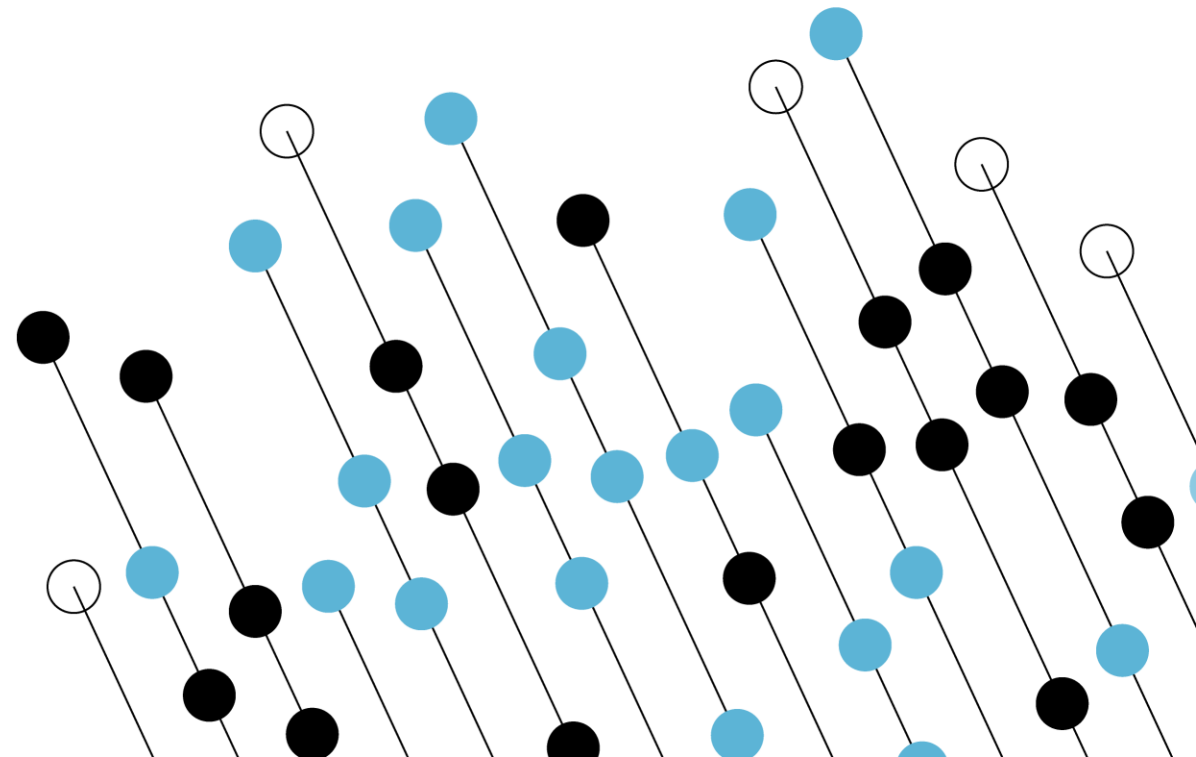
- Can be very heavy.
- Not usually portable.
- Requires advanced cooling system to maintain performance.
- Cooling fans can be loud.

Blade servers use case

- Used in large enterprises
- Data centres
- Virtualisation
- Where a high scalability is required
- Deploy enterprise multi-purpose applications
- Where high processing power is required
- Centralised monitoring and performance
- Where there is limited space



Server room cooling



Server room cooling system

Servers produce a lot of heat, which must be controlled to prevent overheating, using an **airflow control system** (front to back).

- Internally done using heatsink (conduction)
- Externally done using air-conditioning (convection)
- Fans
- Liquid cooling
- Clear server ventilation point

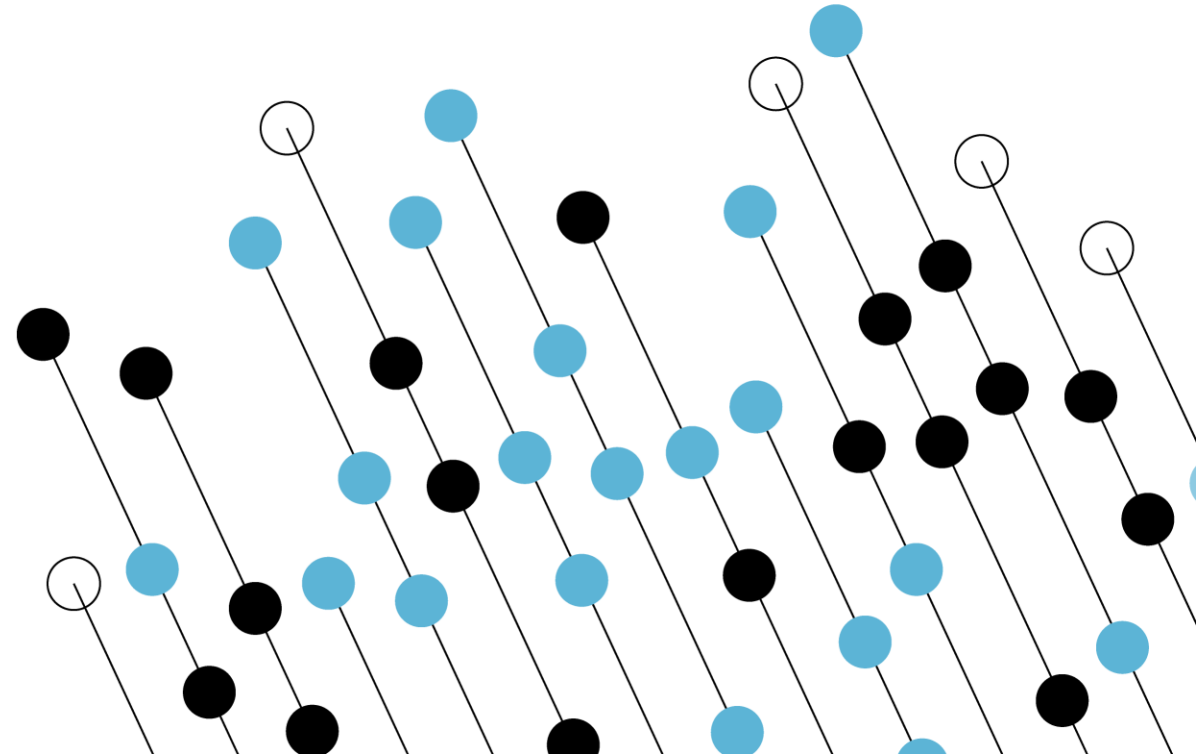


Server location and environment cooling consideration

- Maintaining continuous availability is a critical factor when deploying servers.
- The first step is to understand the server form factor power requirements.
- Server placement and heat dissipation.
- Identify source of heat generation.
- Understand power distribution between the server units.
- Deploy appropriate cooling systems.



Storage



Server storage: Research

Research the following types of storage solutions and provide some examples of their usage:

- Direct Attached Storage (**DAS**)
- Network Attached Storage (**NAS**)
- Storage Area Network (**SAN**)
- Redundant Array of Independent Disks (**RAID**)
- Characteristics of a Solid-State Drive (**SSD**)
- Characteristics of a Hard Disk Drive (**HDD**)

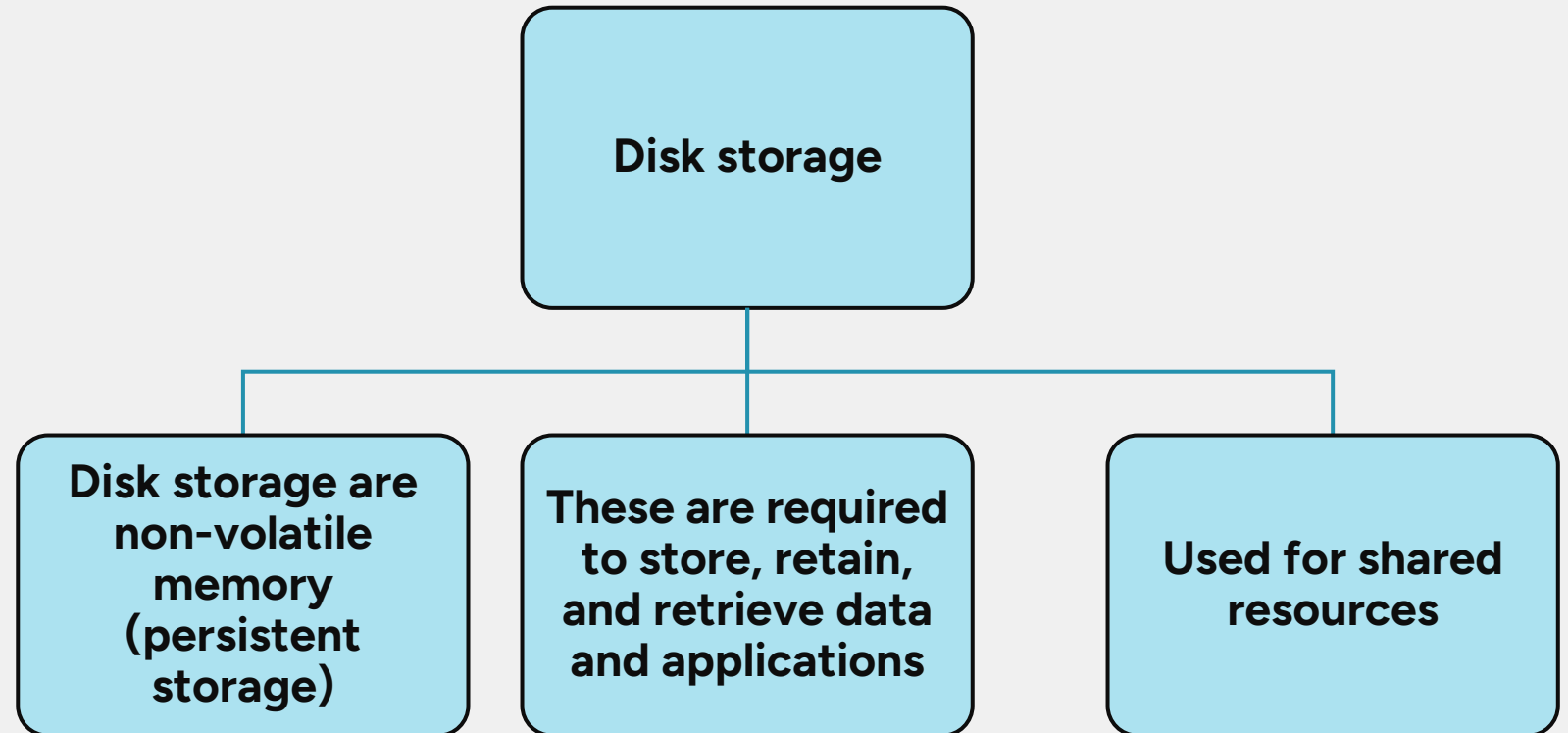
You have **20 minutes**.



Server storage options

Factors to look for when selecting disk drive for servers:

- Storage format
- Performance
- Storage capacity
- Interface



Storage form factor

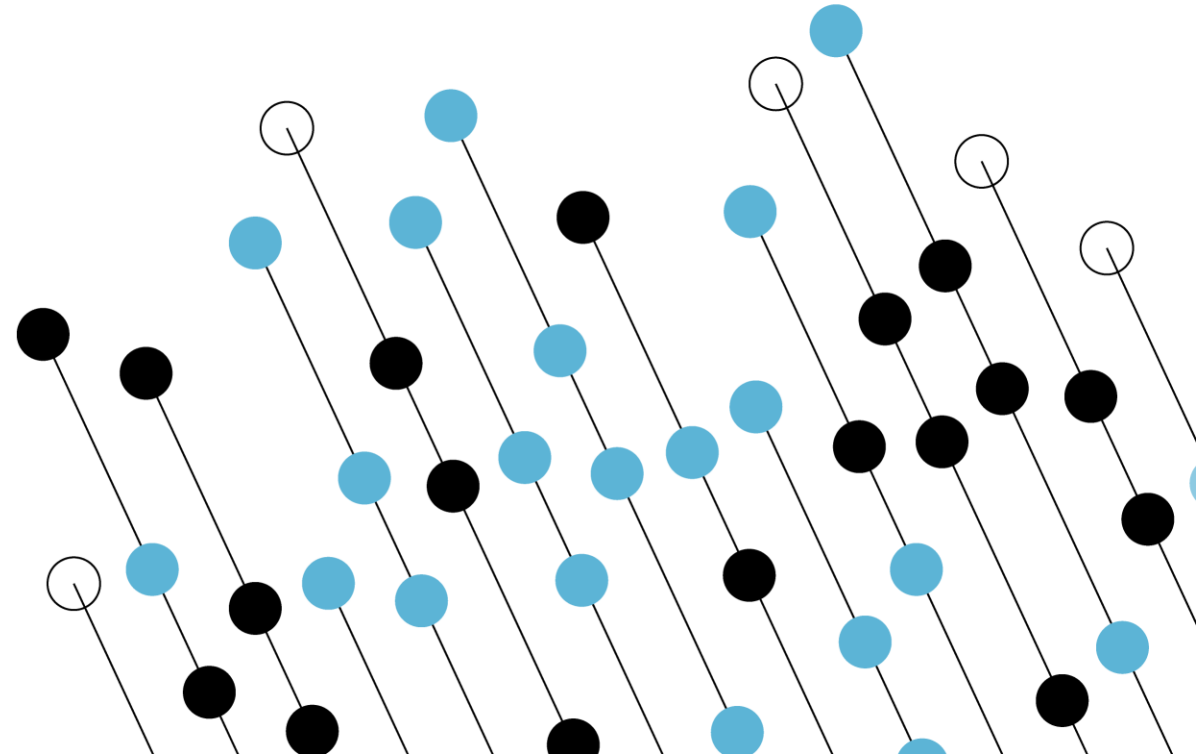
- HDD
- SSD
- Drive form factors:
 - 3.5"
 - 2.5"

Connectivity Interface

- IDE / SCSI / SAS / SATA



Hard Disk Drive (HDD)



Hard Disk Drive (HDD): Discussion and Q&A

- How does HDD fragmentation impact performance?
- How does the mechanical nature of HDDs affect their performance and reliability compared to SSDs?
- Why are HDDs still widely used despite the reduction in cost of SSD?
- What factors can impact the performance of HDD?
- In what situation would the deployment of HDD be a better choice than SSD?

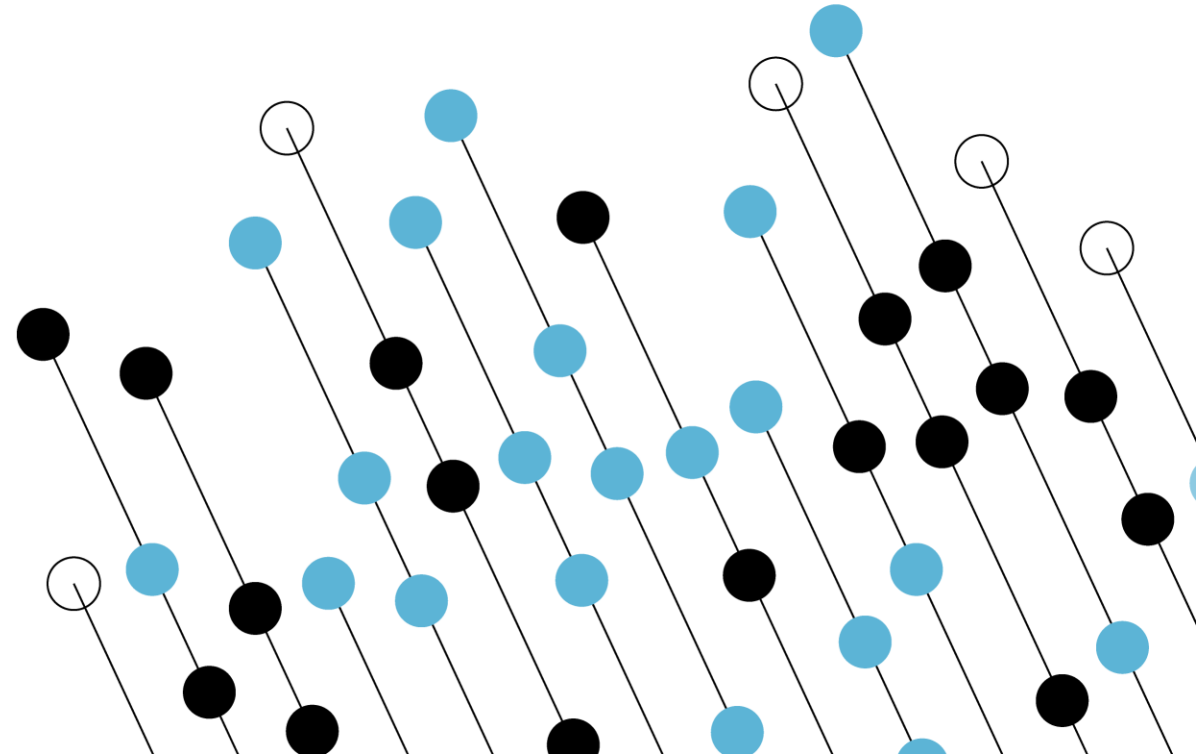


Hard Disk Drive (HDD)

- Spinning disks read by actuator arm
- Noisy and spinning platters can cause vibration
- Higher capacity
- Usually come with SATA / PATA or SCSI interface
- May require defragmenting



Solid-State Drive (SSD)



Solid-State Drive (SSD): Discussion and Q&A

- How do SSDs differ from HDDs in terms of performance and durability?
- What are the key differences between NAND and DRAM-based SSDs, and how do they impact performance?
- What are the cost implications of using SSDs instead of HDDs for data storage?
- What are the main factors that affect the lifespan of an SSD, and how can users maximise it?
- How does wear levelling and TRIM help extend the lifespan of SSDs?



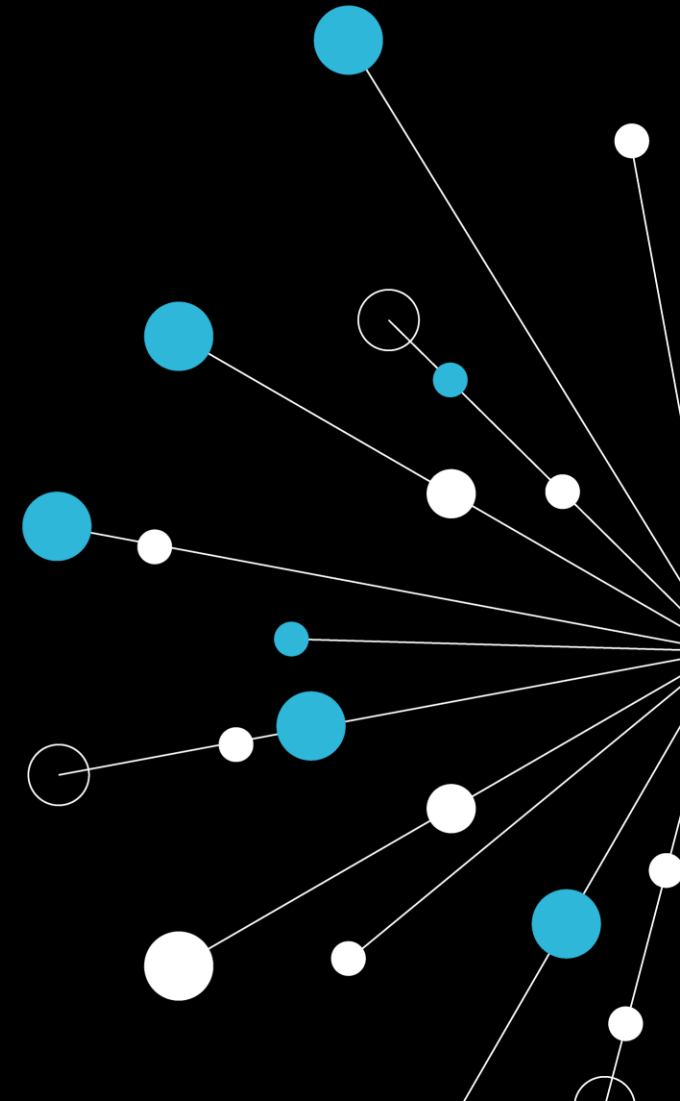
Solid State Drive (SSD)

- NAND based flash memory
- Similar to USB drive technology
- No sound or vibration produced
- Faster read / write speed
- More expensive
- Usually come with SATA / eSATA or PCIe interface
- Wear levelling and Trim

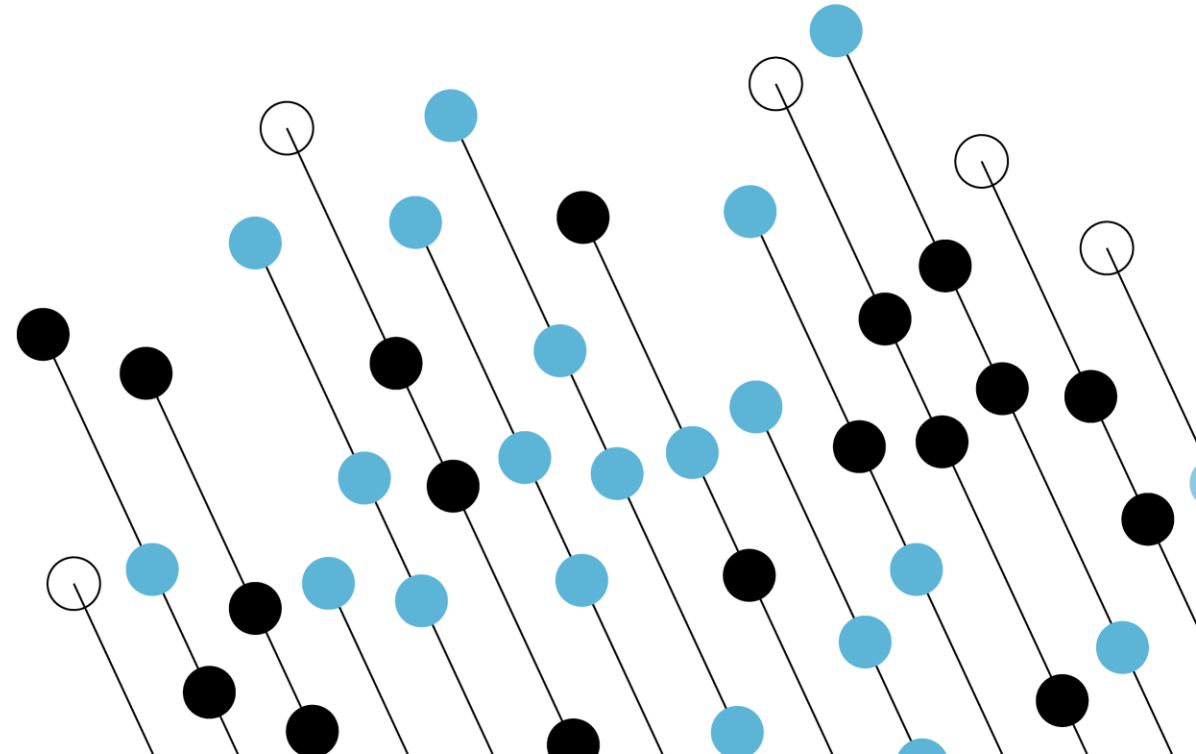


Activity 2: Recommend HDD vs SSD

Find the activity in your **Learner Activity Document** and read the instructions.



Direct Attached Storage (DAS)



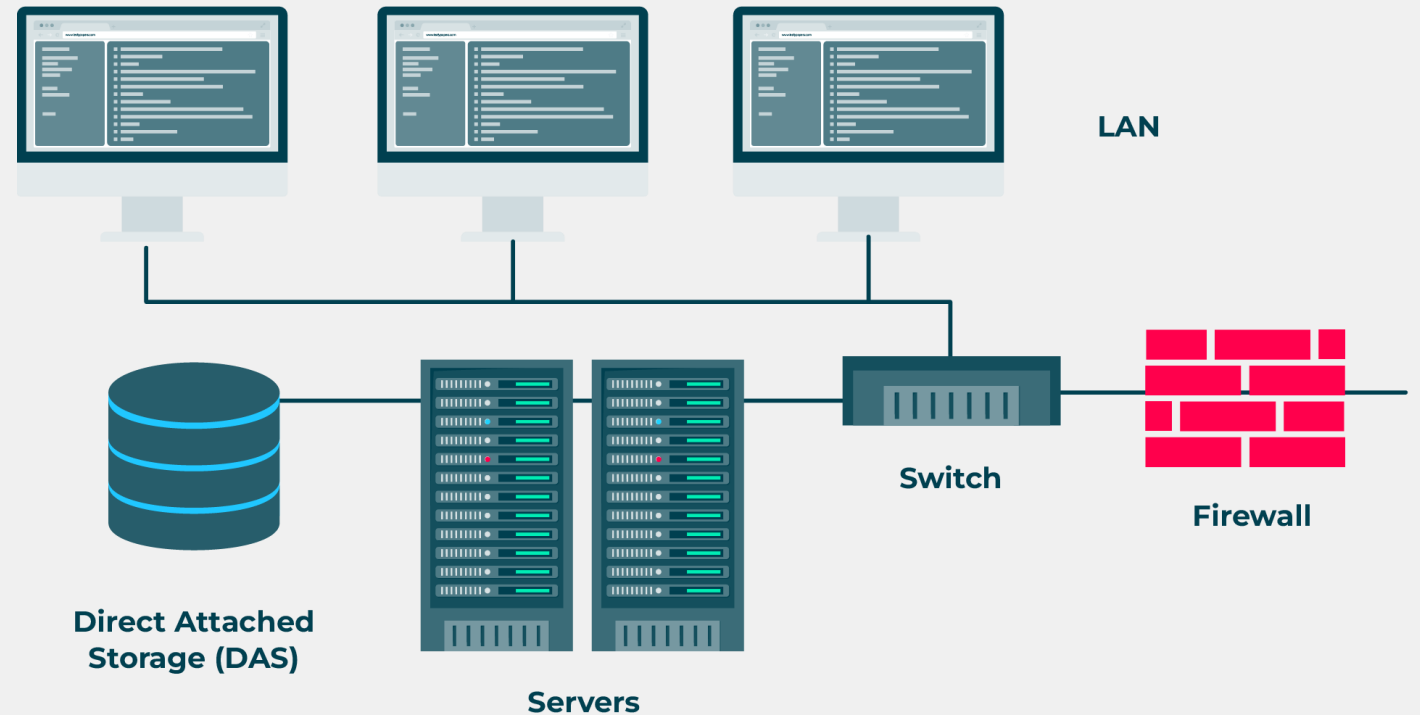
Direct-Attached Storage (DAS): Discussion and Q&A

- What are the primary use cases for DAS compared to NAS or SAN?
- What are the advantages and disadvantages of using DAS?
- What are the limitations of DAS in terms of performance and scalability?
- Can anyone recommend some use cases for DAS?
- What are the security implications of using DAS in a network environment?

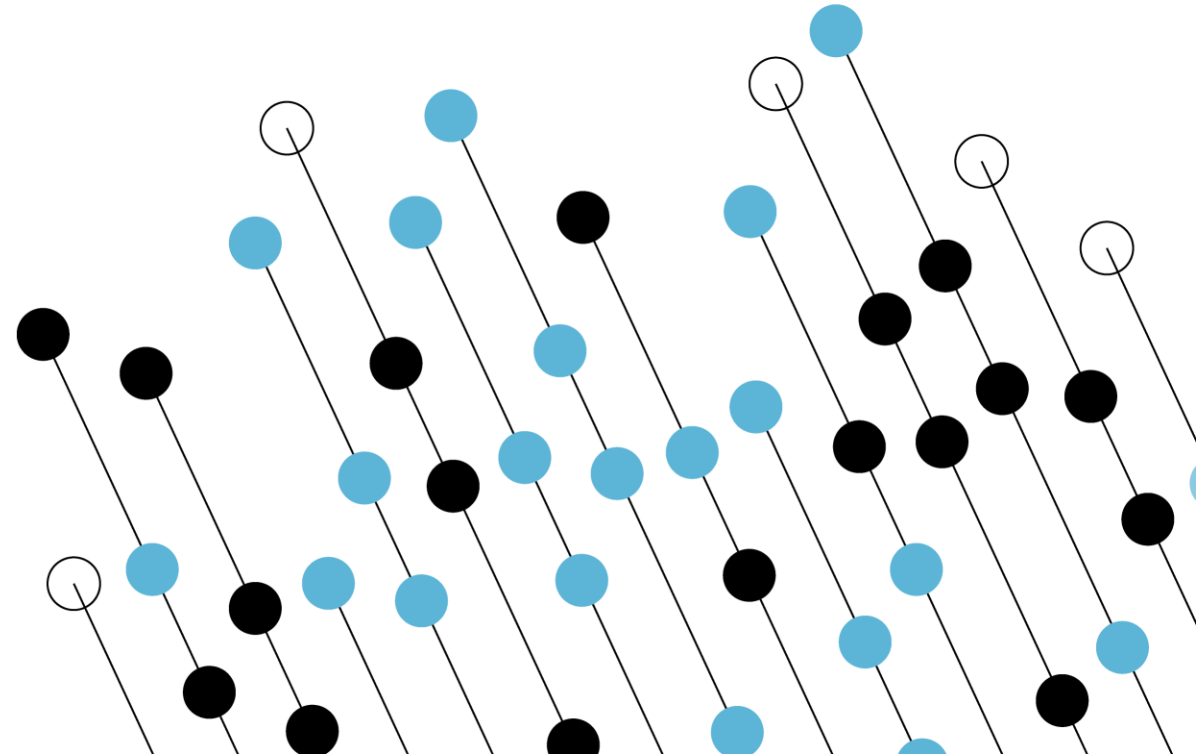


Direct Attached Storage

- Direct Attached Storage (DAS) uses standard drives.
- This can be setup as an internal or external drive for sharing over the network.
- Connectors used in DAS implementation include:
 - IDE.
 - SATA.
 - USB.
 - SCSI.



Network Attached Storage (NAS)



Network-Attached Storage (NAS): Discussion and Q&A

- How does NAS improve data accessibility and collaboration within a networked environment compared to DAS?
- What are the key performance issues when using NAS compared to DAS?
- In what types of business network would NAS be the best storage solution?
- What security risks are associated with NAS, and what measures can be taken to address the security risk?
- What can be configured to ensure data redundancy in a NAS configured storage?



Network Attached Storage (NAS)

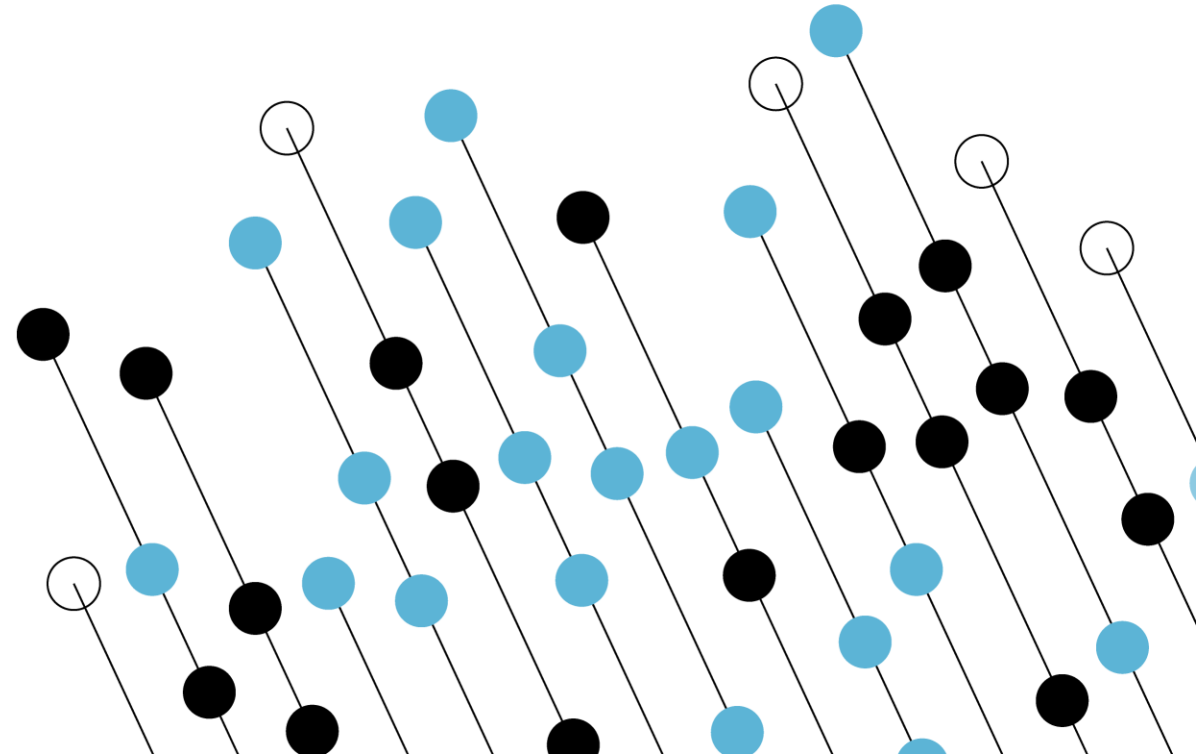
- File level storage
- Deployed on an IP network
- Bunch of aggregated disks managed centrally on a network
- Often deployed using RAID configurations

Uses the following network protocols to share data:

- SMB / CIFS
- NFS
- AFP



Storage Area Network (SAN)



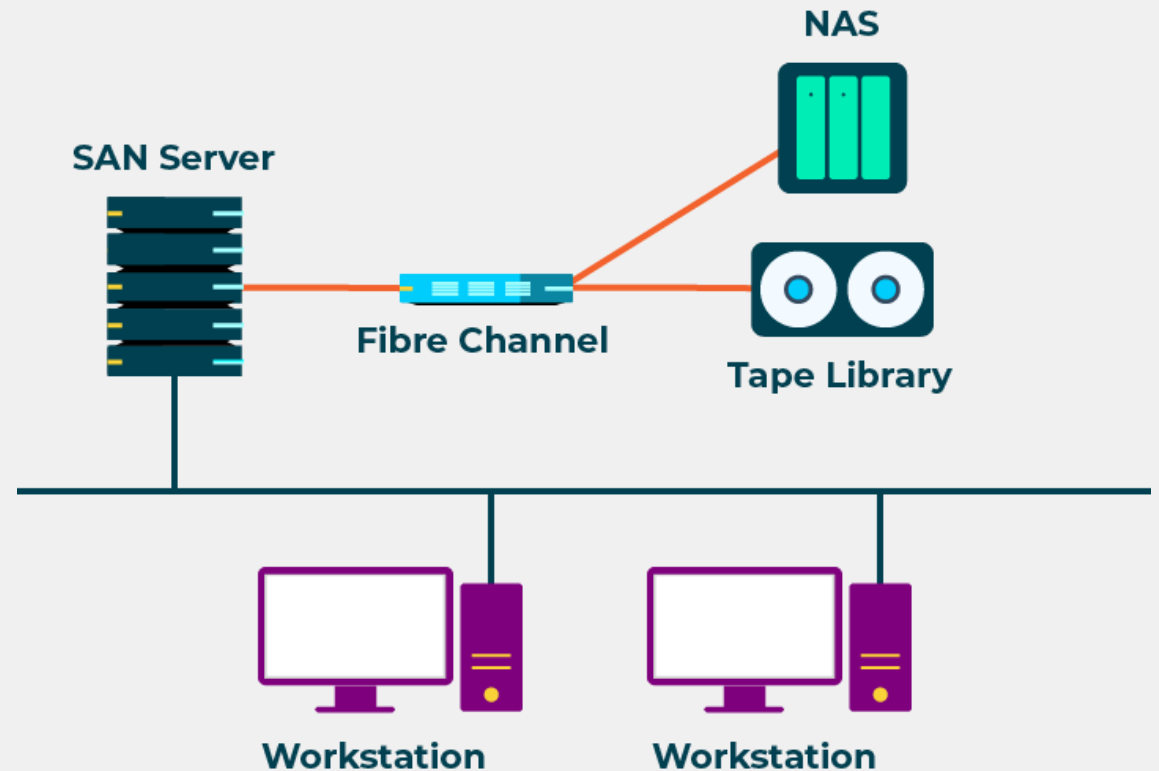
Storage Area Network (SAN): Discussion and Q&A

- What are the main differences between SAN and NAS in terms of deployment and architecture?
- What are the key factors for SAN deployment compared to NAS and DAS?
- How does SAN technology ensure scalability and high availability?
- What are the cost implications associated with SAN?
- How does Fibre Channel (FC) and iSCSI impact the speed and efficiency of SAN?
- In what scenario is SAN preferred enterprise solution?



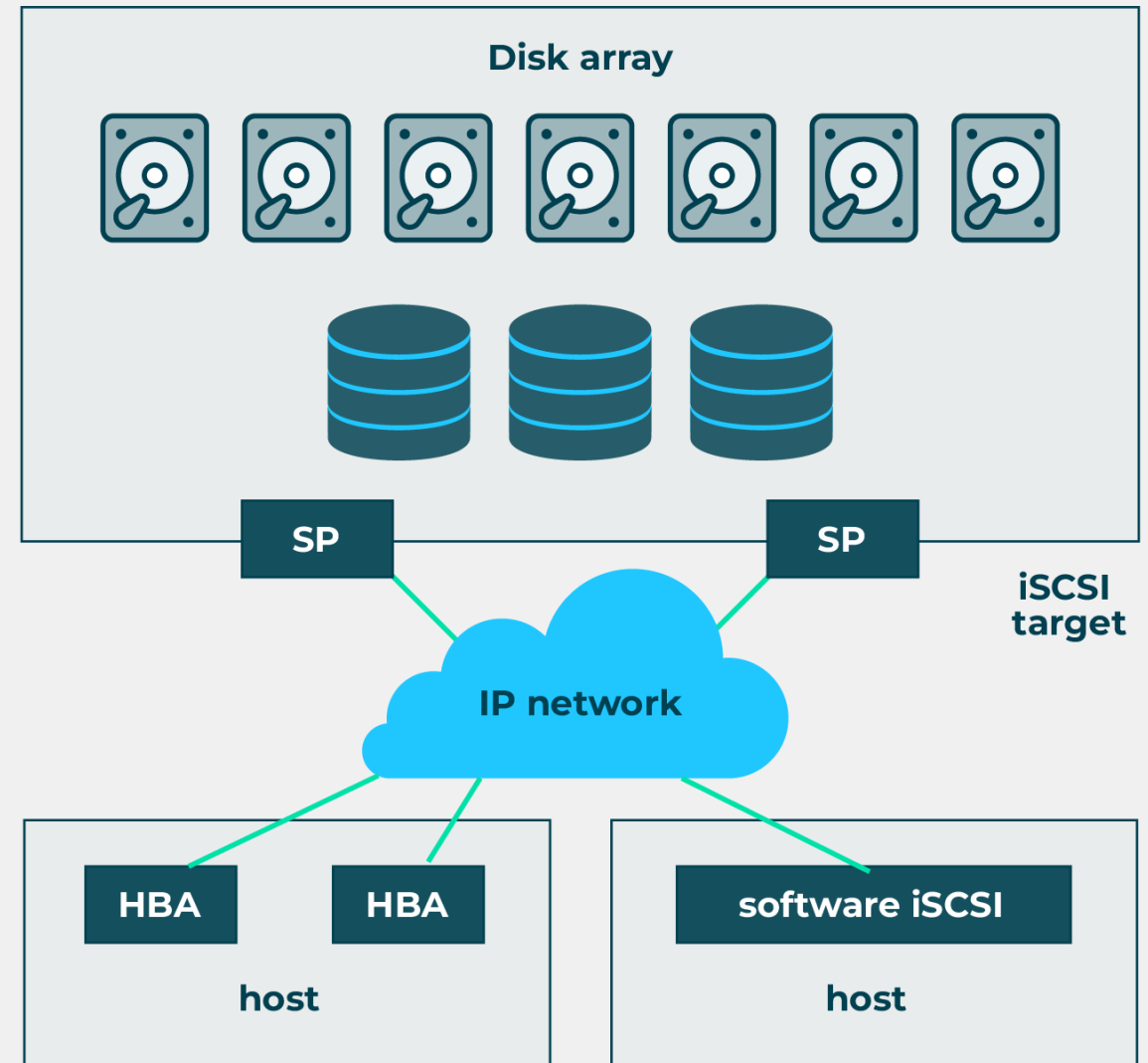
Introduction to Storage Area Network (SAN)

- Block level storage
- High-capacity storage connected through a high-speed network
- Dedicated network of storage devices
 - Tape libraries
 - Disk arrays
- Transfers data over network:
 - iSCSI
 - Fibre Channel Protocol (FCP)
- SAN Server to manage and share storage
- SAN components:
 - Initiator
 - Target
 - FC Switch
 - LUN – Logical Unit Number



SAN: iSCSI components

- SAN protocol for SCSI over TCP / IP
- Components
- Initiators
- Clients sending SCSI commands
- Host Bus Adapter (HBA)
- Targets
- Storage device executes commands
 - Uses conventional Ethernet technology
 - High administrative and network overhead



SAN: Fibre channel

SAN Protocol for high-speed data transfer

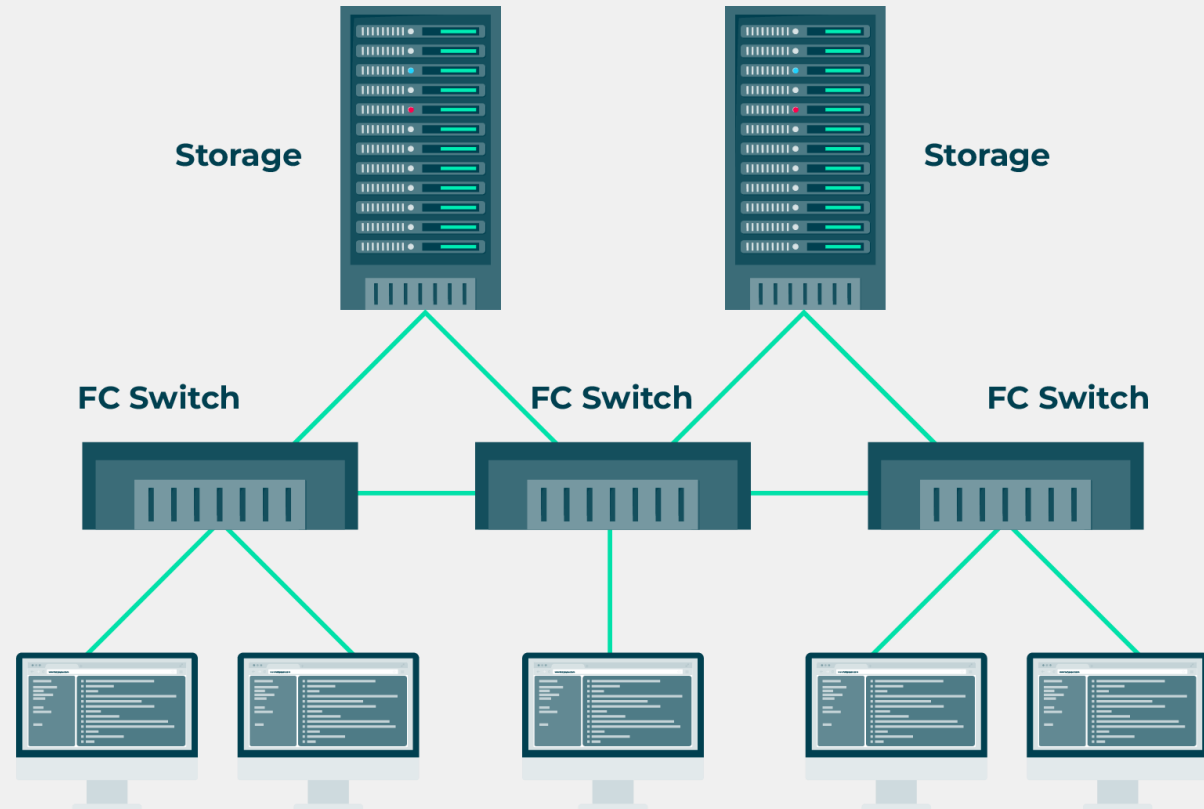
- Lossless transport
- Flexible network topology

Components

- Initiators
- Targets
- FC-capable switch

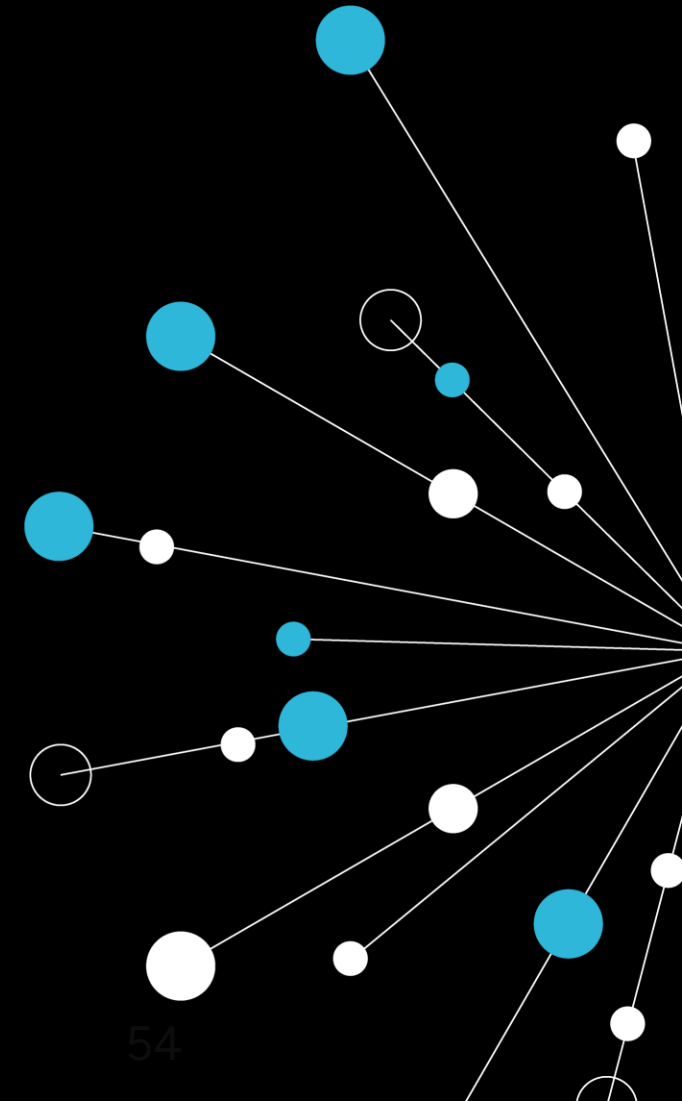
FCoE

- Fibre Channel over Ethernet

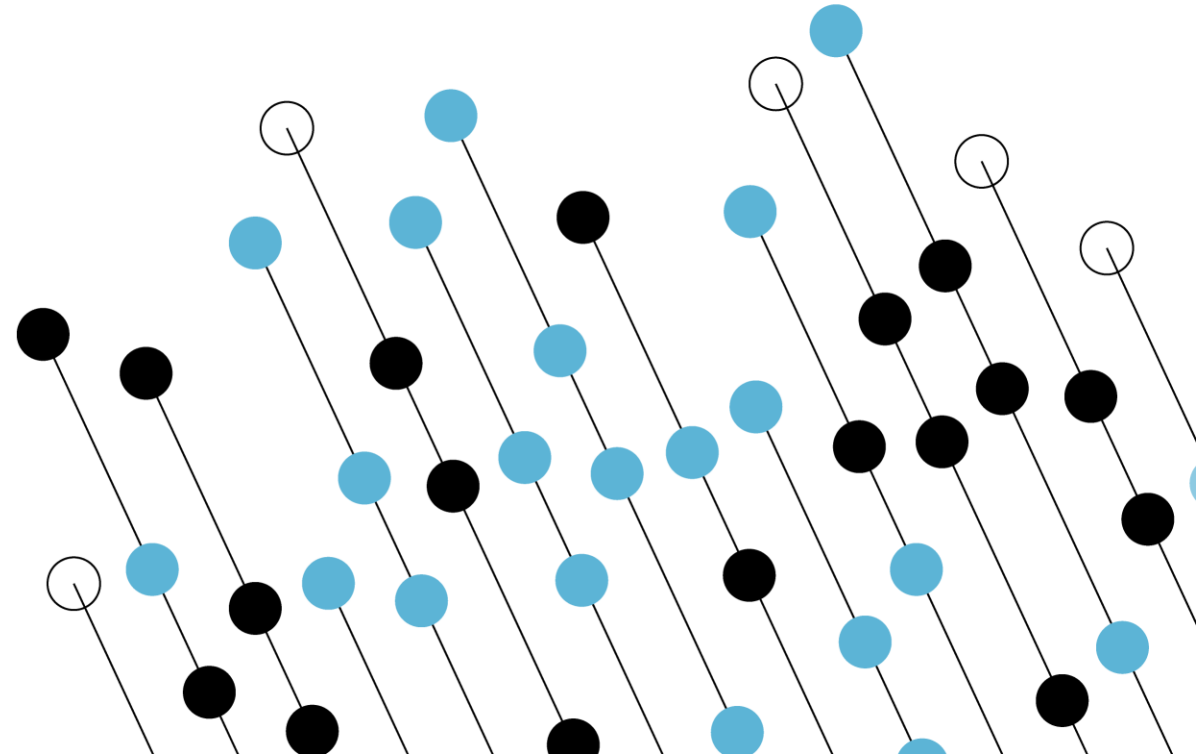


Activity 3: Recommend DAS-NAS-SAN storage

Find the activity in your **Learner Activity Document** and read the instructions.



Redundant Array of Independent Disks (RAID)



Redundant Array of Independent Disks (RAID): Discussion and Q&A

- What are the different RAID levels and how are they implemented for performance and redundancy?
- What factors need to be considered when selecting RAID type?
- How does striping differs from mirroring in terms of RAID deployment?
- In what scenario would hardware RAID be preferred over software RAID?
- Are there any downsides to RAID usage?

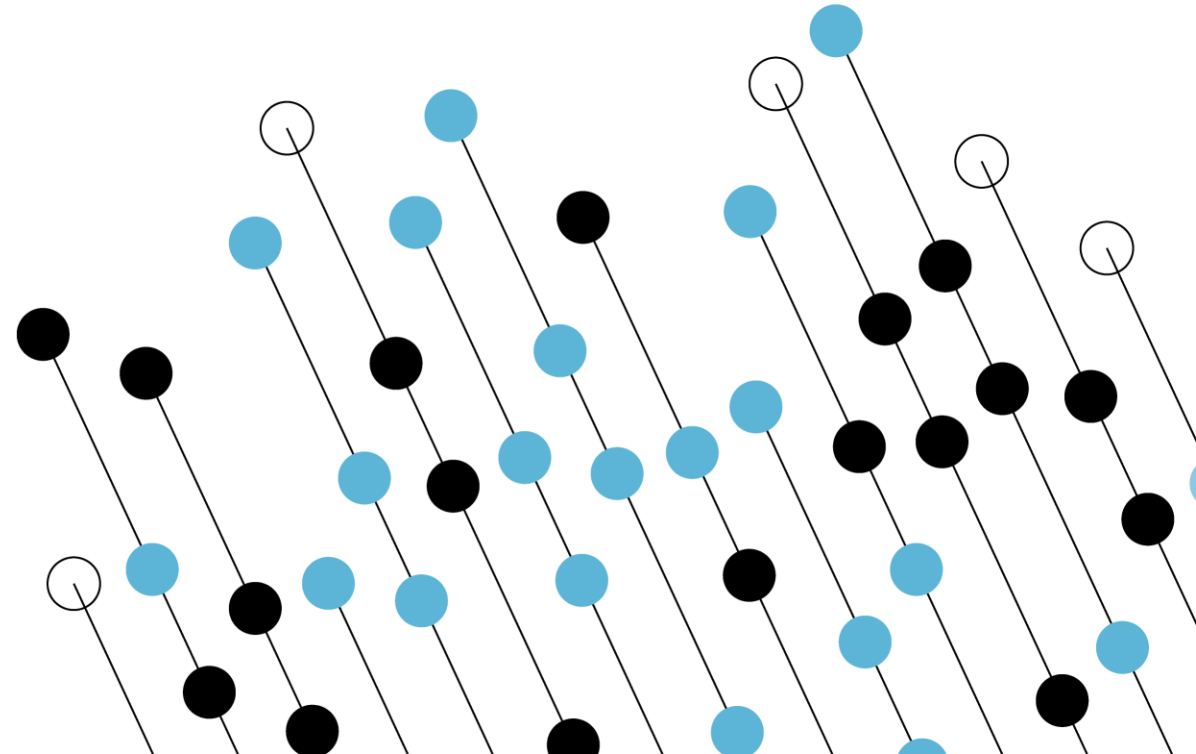


Introduction to RAID

- Provides additional capacity
- Striping offers performance improvements
- Fault tolerance
- Mirroring
- Parity
- Hardware vs. software RAID



Types of RAID



RAID 0

Striping

- Requires two or more disks.
- Data is stored in blocks across multiple drives.
- Improves read and write performance.
- Volume size is capacity of all disks, assuming disks are equal capacity!
- No fault tolerance.
- Disk failure can result in data loss.
- Data can be restored from backup.



RAID 1

Mirroring

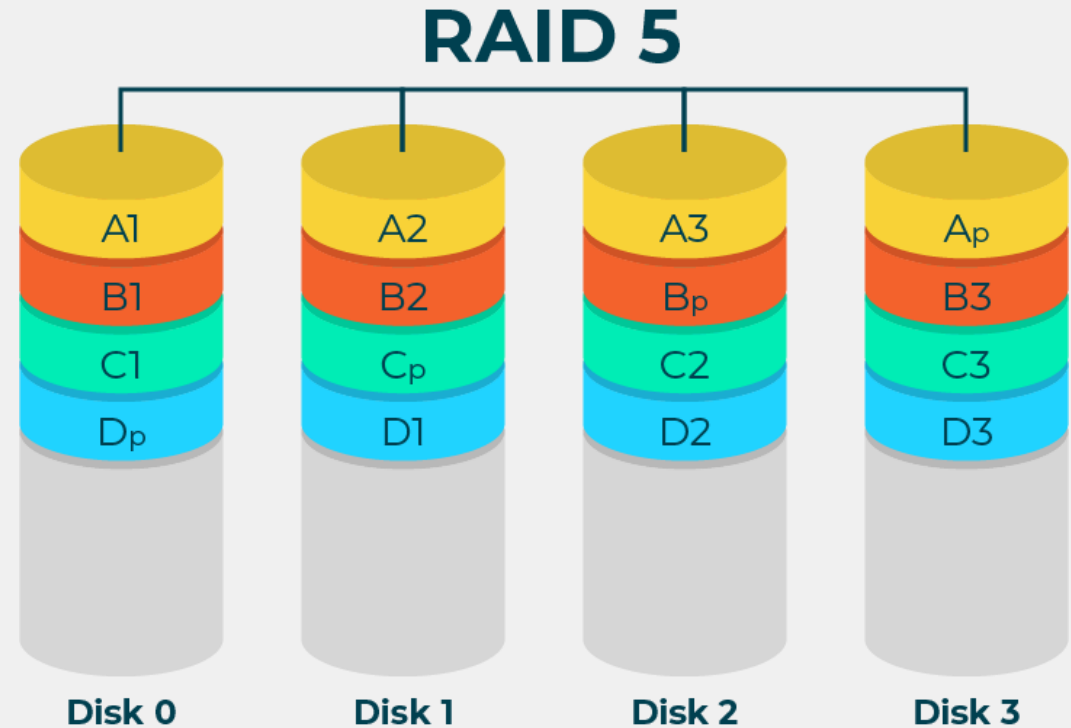
- Requires two or more disks.
 - 2-way mirror
 - 3-way mirror
- 50% disk utilisation.
- It tolerates $n-1$ disk failures.



RAID 5

Striping with distributed parity

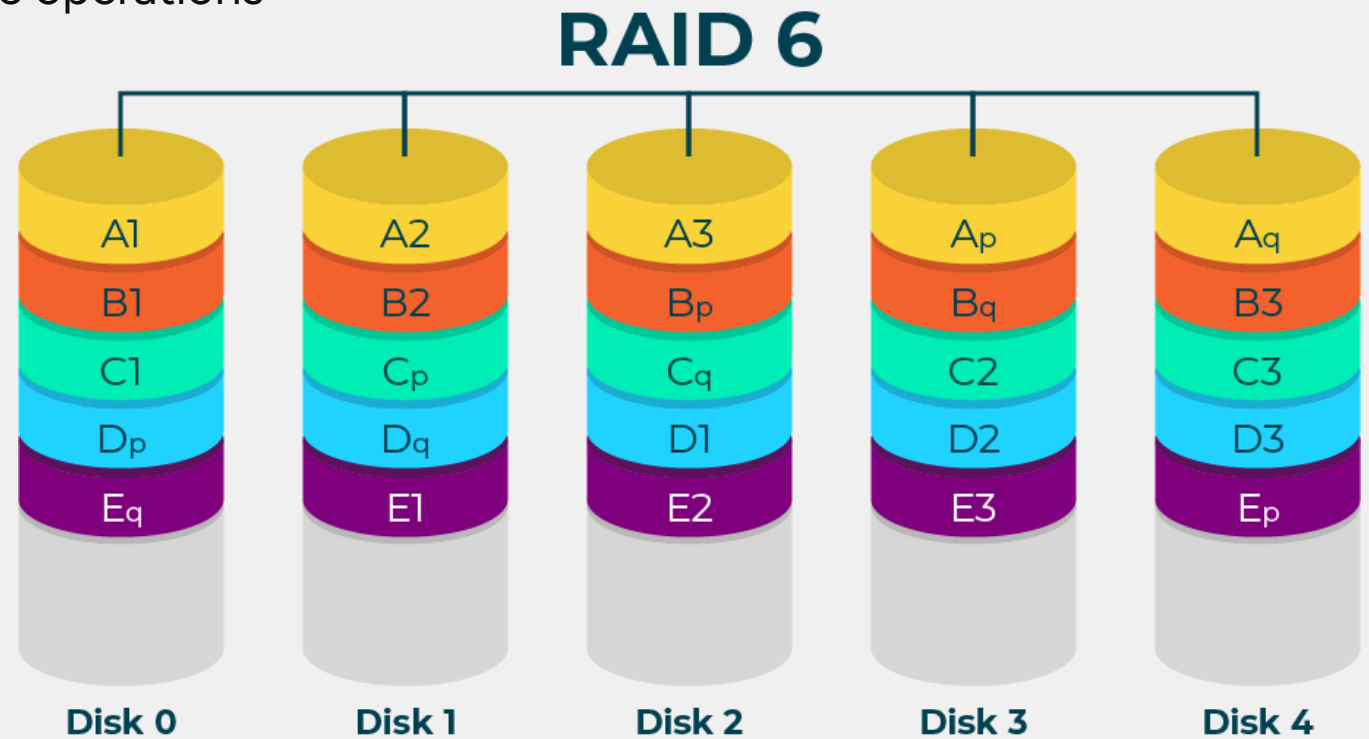
- Parity is generated using XOR.
- Requires three or more disks.
- Tolerates failure of one disk.
- Build data from parity with XOR calculations.



RAID 6

Striping with double parity

- There is a performance penalty for write operations due to extra parity data.
- Requires four or more disks.
- Tolerates failure of two disks.



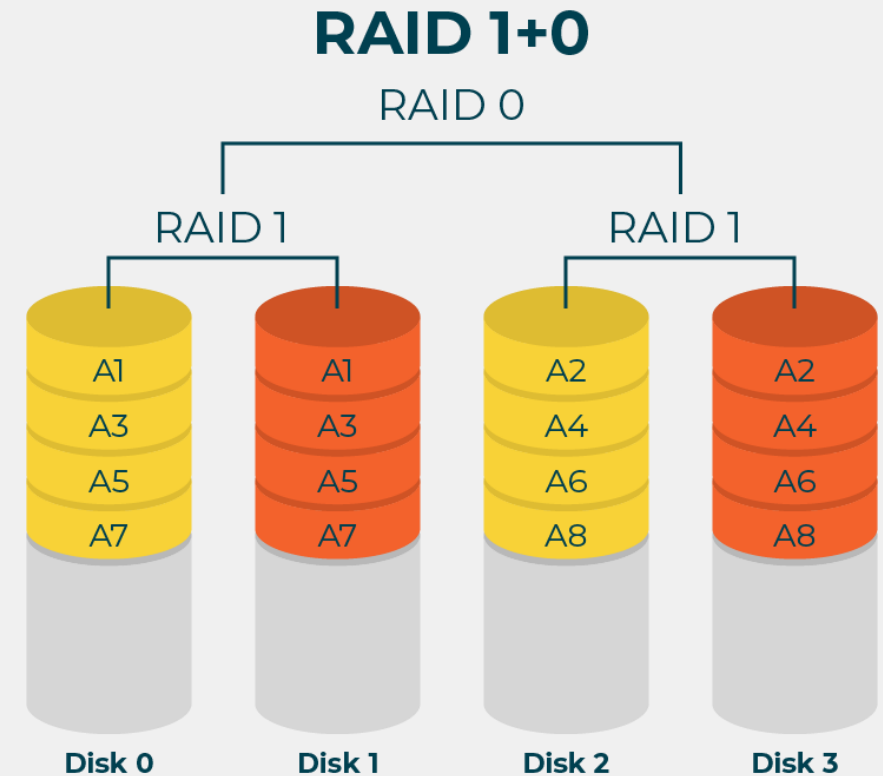
Nested RAID

Hybrid RAID:

- Provides additional fault tolerance, redundancy, or both.
- Is the perfect solution for IO intensive critical workloads.

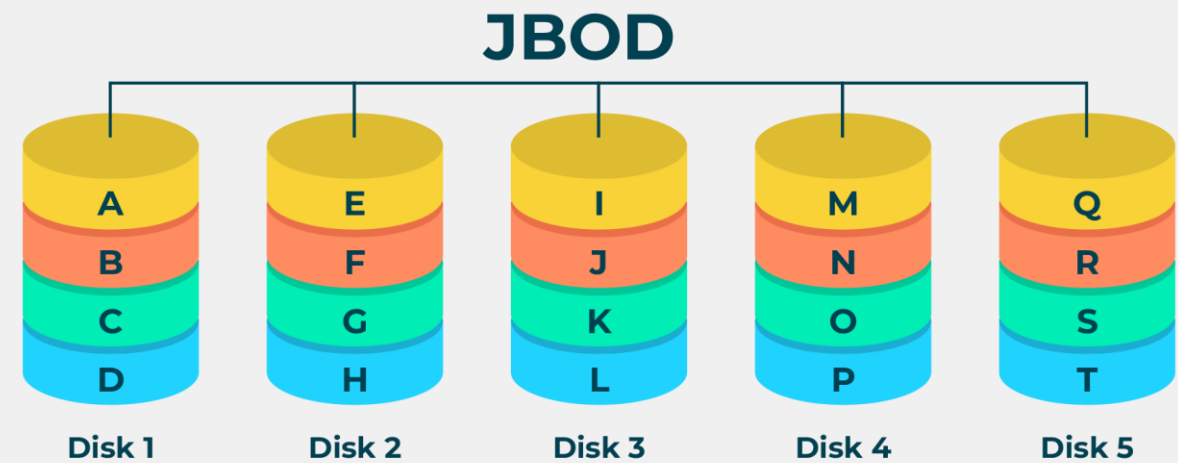
It commonly uses two levels:

- 1+0.
- 1+5.



Software RAID deployment

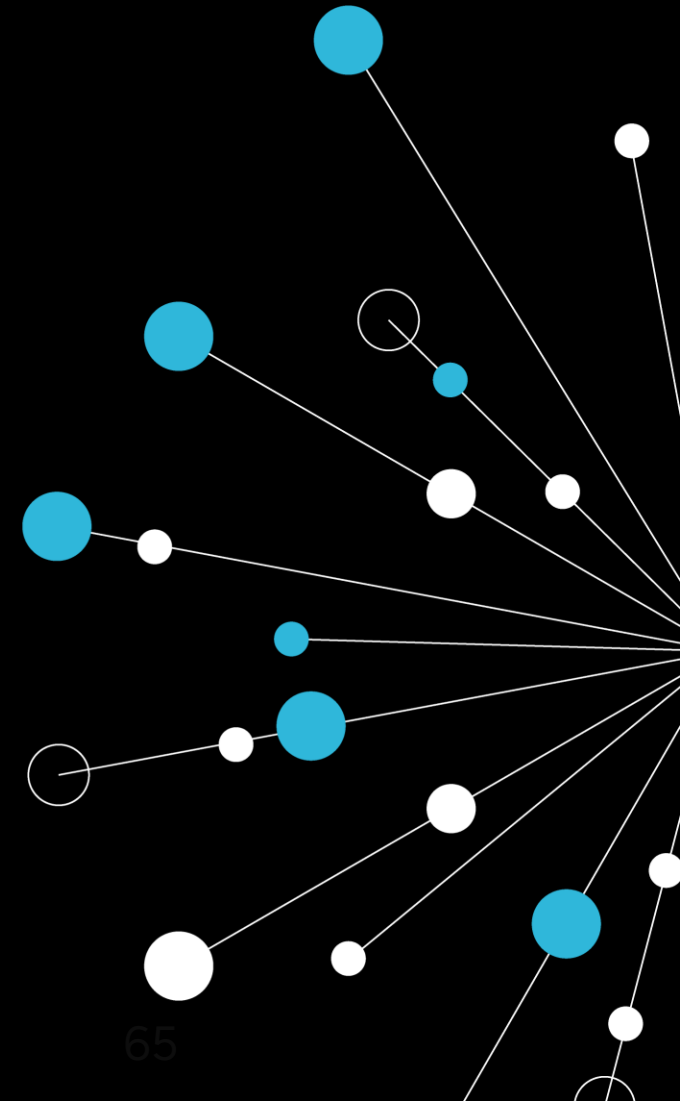
- Windows Dynamic Disks / Storage Spaces
- Linux
 - mdadm
 - Logical Volume Manager (LVM)
- ZFS
- JBOD
 - Just a bunch of disks
- SPAN / BIG
 - Spanned volumes
- MAID
 - Massive array of idle drives



RAID level	OS support
0	Windows / Linux / ZFS
1	Windows / Linux / ZFS
5	Windows / Linux / ZFS (RAID-Z)
6	Linux / ZFS (RAID-Z2)
1+0	Linux / ZFS

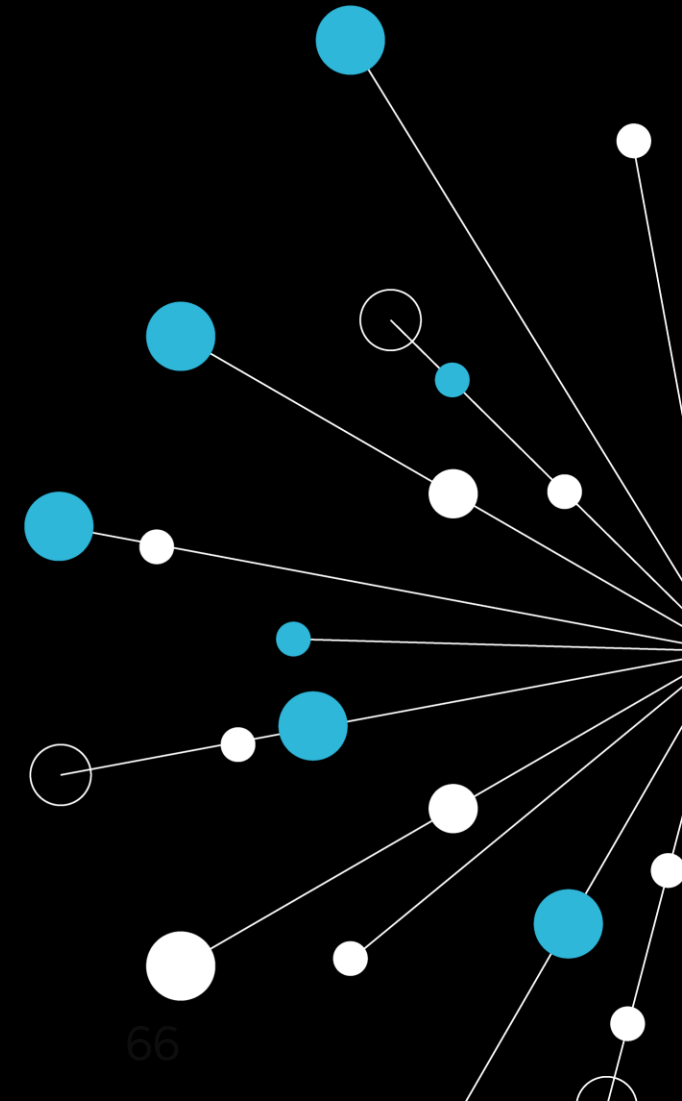
Activity 4: Recommend a RAID storage solution

Find the activity in your **Learner Activity Document** and read the instructions.



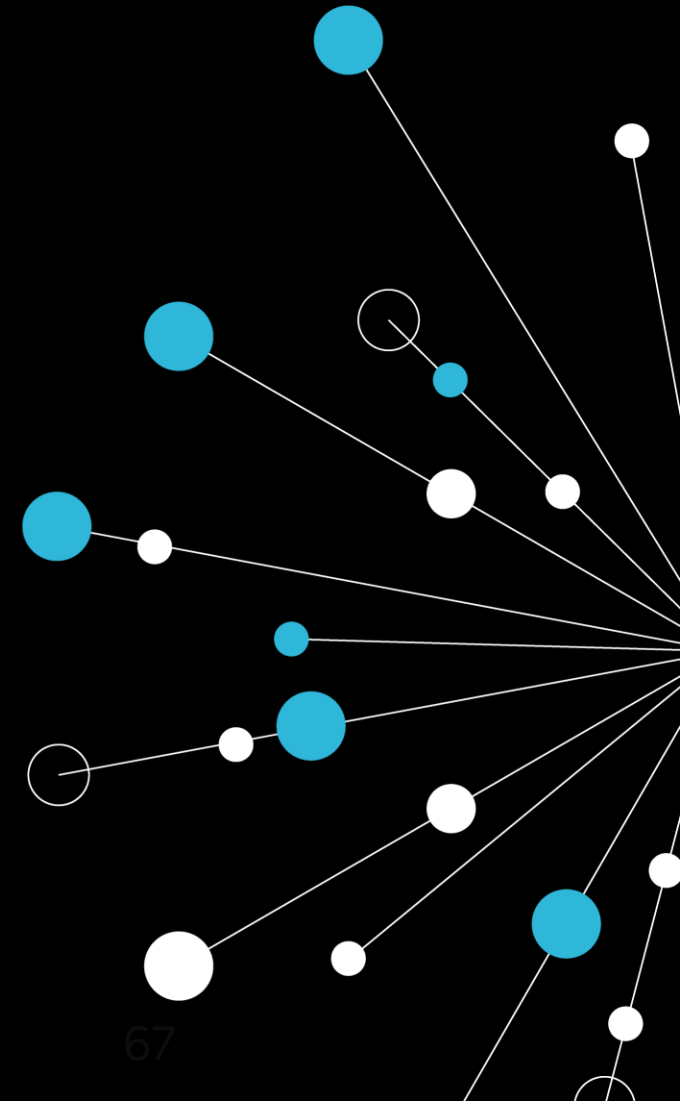
Activity 5: Create and verify RAID setup

Find the activity in your **Learner Activity Document** and read the instructions.

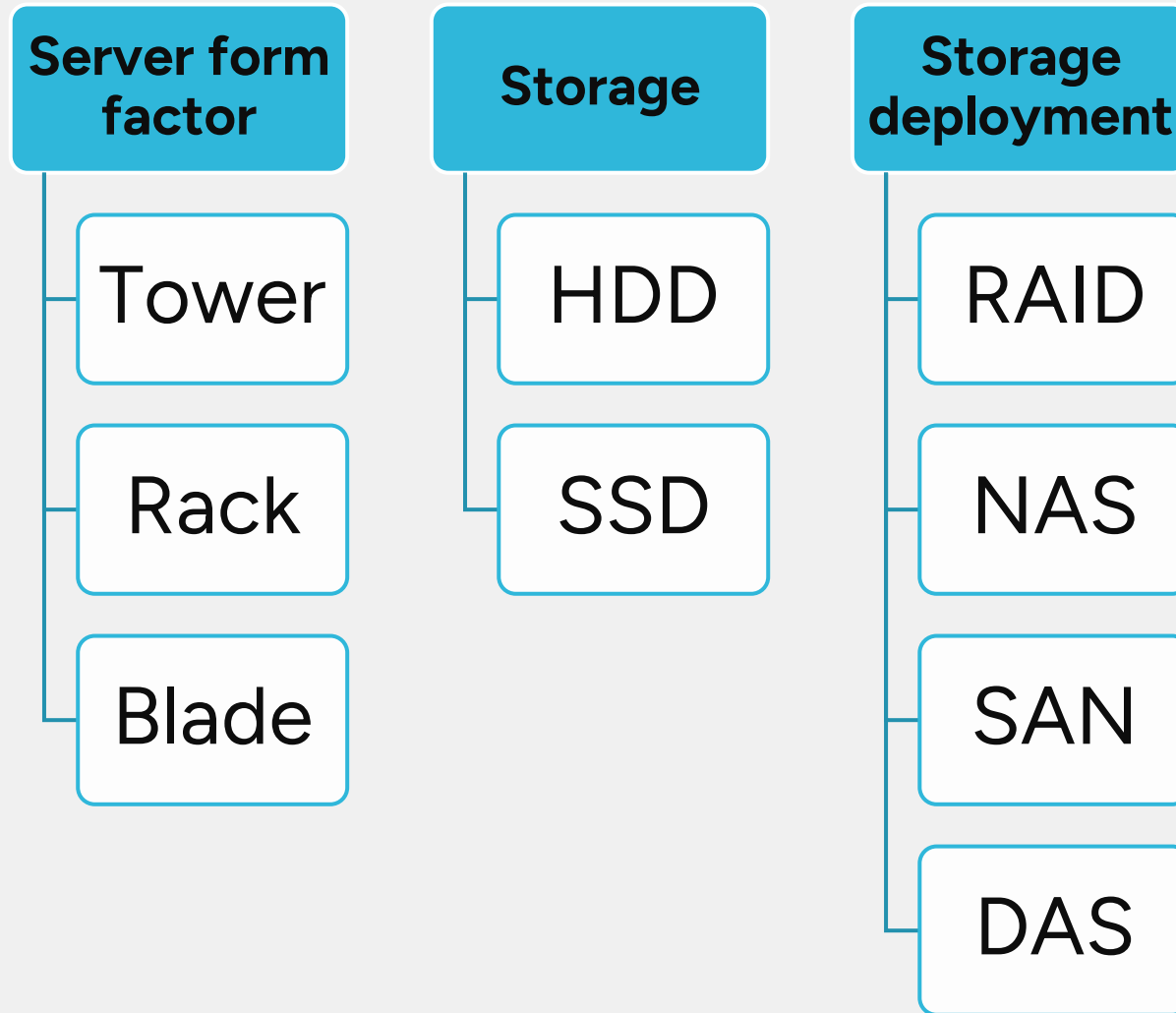


Activity 6: RAID multiple choice quiz

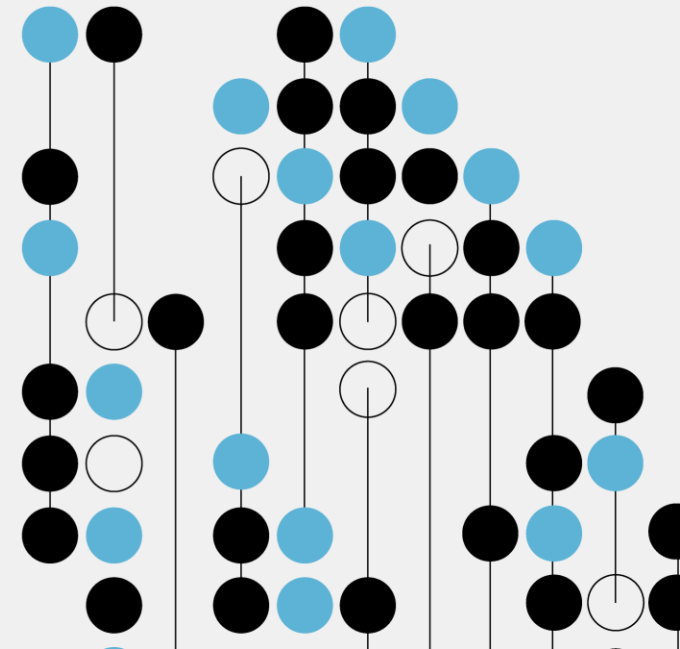
Find the activity in your **Learner Activity Document** and read the instructions.



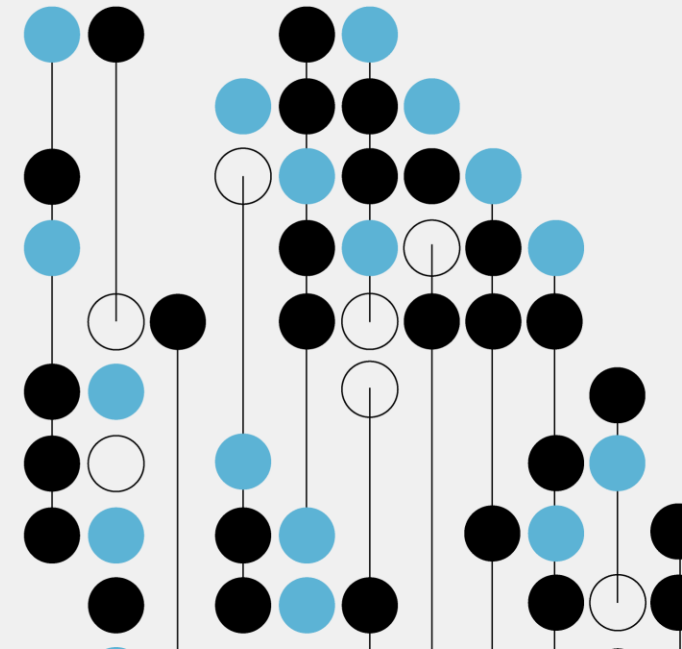
Day 1: Summary



End of Day 1



Day 2: Server OS install and administration

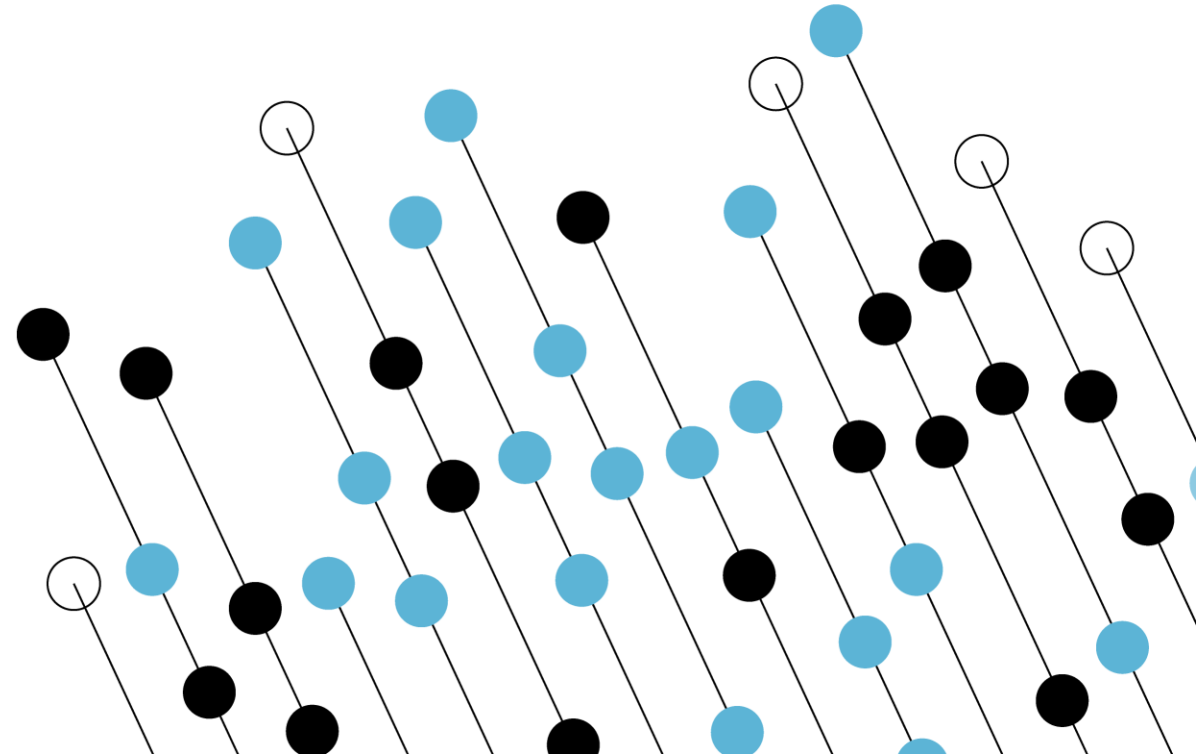


Objectives: Day 2

- Differentiate between Windows and Linux permissions.
- Deploy the right OS based on customer requirements.
- Identify key Windows and Linux Server network services.
- Differentiate how to configure Windows and Linux Server OS.



Windows Operating System



Windows Server: Discussion and Q&A

- What are the main features of Windows Server and how does it differ from desktop operating system?
- How does Active Directory enhance the management of users and resources?
- In what scenario would Windows Server core be preferred to Windows Server GUI?
- In what situations would a company prefer to use Windows Server over Linux Server?



Introduction to Windows Server OS

- Developed by Microsoft to support enterprise level network management.
- Provides GUI and Command-line based network administration.
- Used to centralised the management of network resources (users, files, services).
- Provides the ability to virtualise resources through the use of Hyper-V.
- Used to automate the deployments of updates and patches.



Core features of Windows Server

- **Active Directory:** Used to manage authentication and Authorisation.
- **Group Policy:** Apply allow or deny policies on users and network computers.
- **Server Manager:** Centralised management console for adding roles and features to the server.
- **Routing and Remote access:** Configure VPN or routing protocols.
- **Network Access Protection:** Used to manage health policy enforcement.
- **Firewall:** Create rules for packet filtering.
- **Windows Defender:** Advanced antivirus and antimalware protection.



Windows Server OS Versions

Windows Server versions:

- 2012 R2 / 2016 / 2019 / 2022
- Windows Server on Azure for cloud-based solution

Editions:

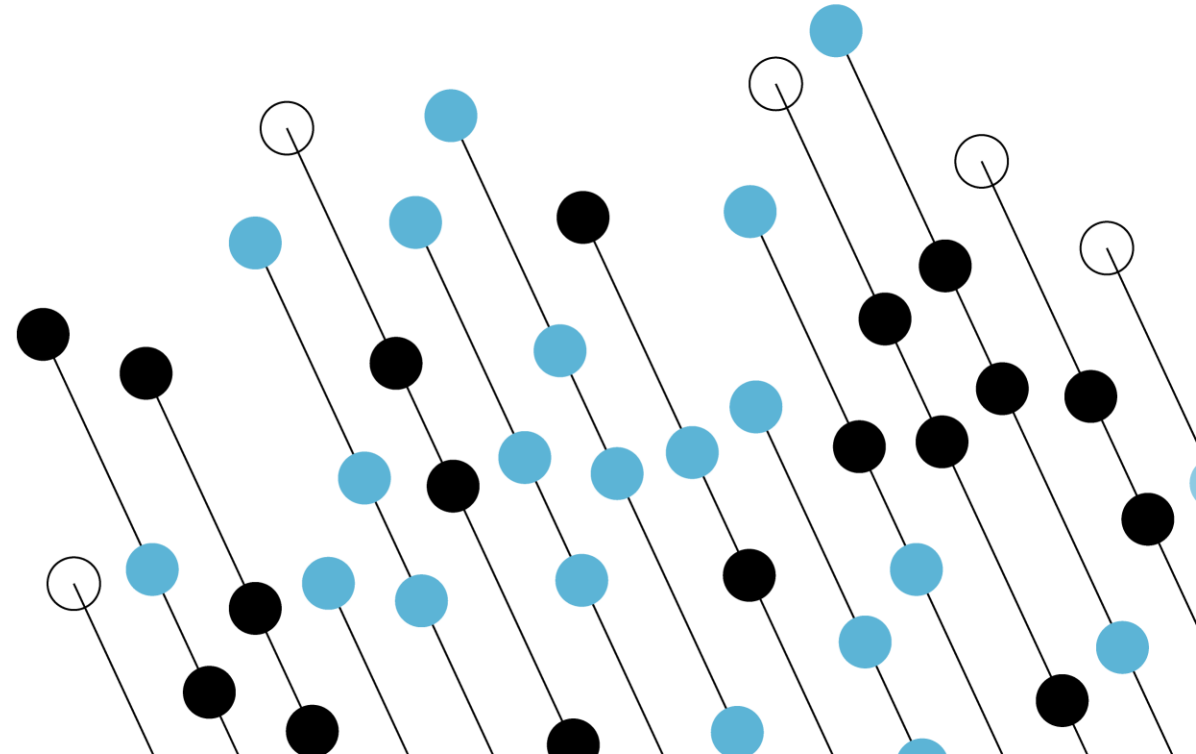
- Standard – Small to medium size businesses
- Datacentre – Features for virtualisation and Cloud integration
- Essentials – For small businesses (Max 25 users)

Installation options:

- GUI
- Core
- Hyper-V



Linux Operating System



Linux Server: Discussion and Q&A

- What are the key differences between Linux Server and Windows Server?
- How does Linux system performance and management differs from Windows Server?
- What are the key challenges of using Linux servers in production network?
- What are the security benefits of using Linux compared to Windows Server?
- What needs to be considered when making Linux deployment over Windows Server OS?



Introduction to Linux Operating System

- An open-source and community driven operating system.
- Versatile and can be used across multiple devices in respective of the underlying hardware.
- Mostly CLI based depending on the distribution.
- Has better security feature due to its default password hash salting.
- Has very good performance due to its lightweight hardware requirements.
- Less prone to malware and virus infections due to its security architecture and permission-based security.



Linux Operating System distros

- **Red Hat Enterprise Linux:** Enterprise grade distro with support
- **Debian:** Very stable distro with enterprise level security and networking abilities
- **SUSE Linux Enterprise Server:** Enterprise grade with Cloud support
- **Ubuntu Server:** Very user friendly with long term support (LTS)
- **CentOS Linux Server:** A long standing and widely used in enterprise

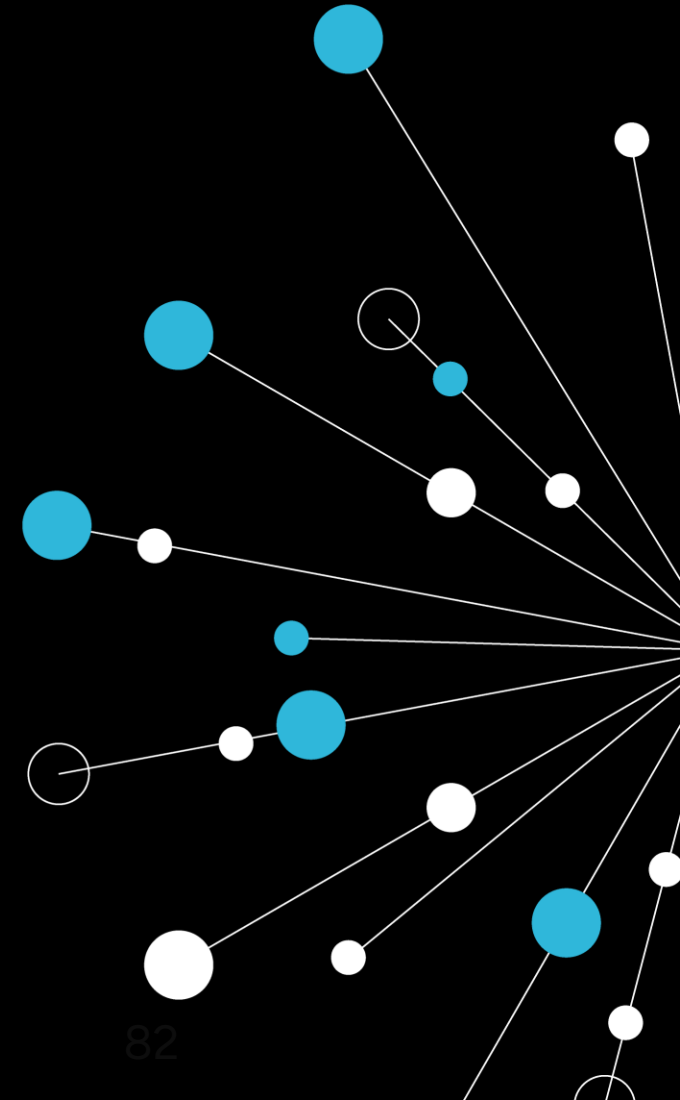


Key differences between Windows and Linux Server

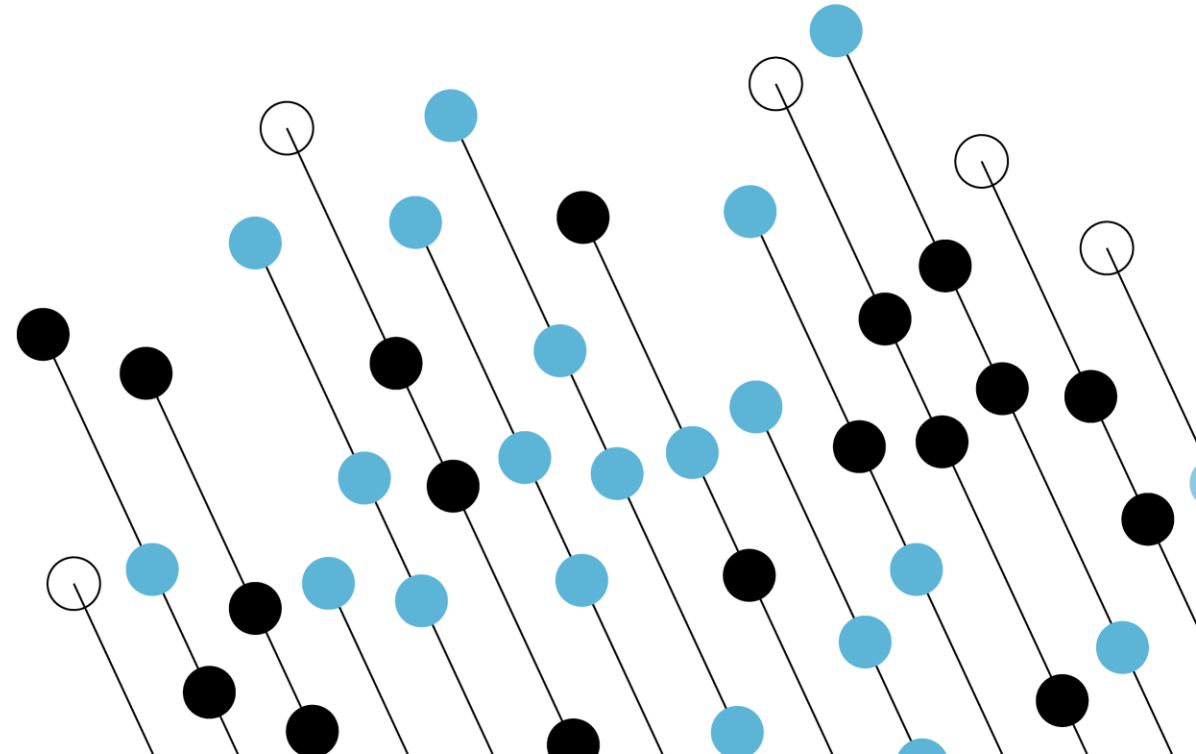
Feature	Windows Server	Linux Server
License	Commercial (paid)	Mostly free (except with support)
GUI	Full GUI (with the exception of Core)	Mostly CLI (GUI support)
Customisation	Limited	Highly customisable
Security	Built in Windows Defender and Firewall	Iptable firewall and minimal attack surface
Use cases	Centralised administration and services provision	Mostly service provision (web hosting, cloud hosting, and security services)

Activity 7: Recommend Windows vs Linux Server

Find the activity in your **Learner Activity Document** and read the instructions.



Operating System pre- installation requirements



OS pre-installation planning: Discussion and Q&A

- What are the key factors to be considered before server installation?
- Which file system should be used when security is paramount?
- How do we ensure data backup and recover measures are in place before installation?
- Should virtualisation be considered for OS installation instead of standard physical hardware install? Why?
- What are the best practices for backing up data before an OS installation or upgrade?
- What best practices should be followed to ensure adequate network and security settings during installation?



Pre-installation planning

- Define server role and budget.
- Identify system specification.
 - Cost
 - Reliability
 - Support
- Identify performance requirements.
 - Average
 - Peak loads
- Identify fault tolerance and backup requirements.
- Schedule installation and inform stakeholders.



General configuration consideration

Disk preparation

- Disk type (RAID level)
- Filesystem type (NTFS, ReiserFS)

Environment

- Physical security
- Cooling requirements

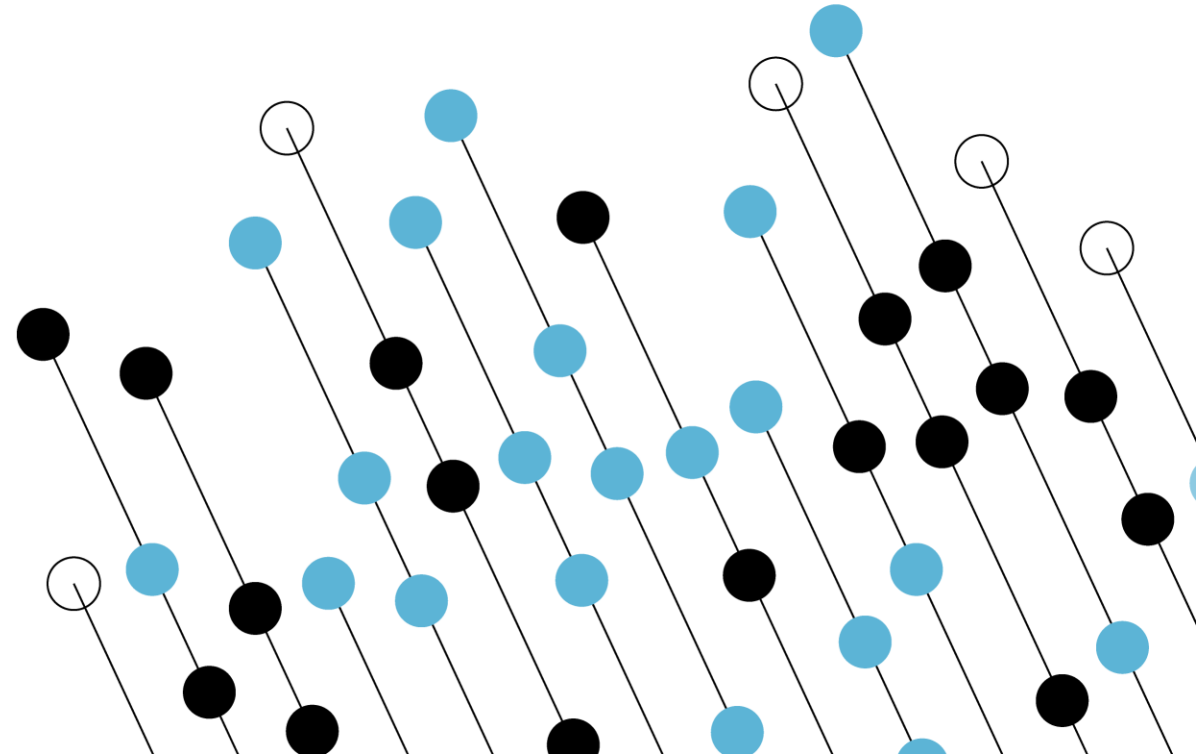
Network access

- Protocol support
- Client support

Network configuration

- Address information
- Network name / domain suffix
- Host name

Server install



Server installation options

Local media

- Use DVD
- USB
- Unattended installation
 - Use of answer file

Cross network

- Network share containing setup files
- Network booting (PXE)
- TFTP



Server installation tools: Discussion and Q&A

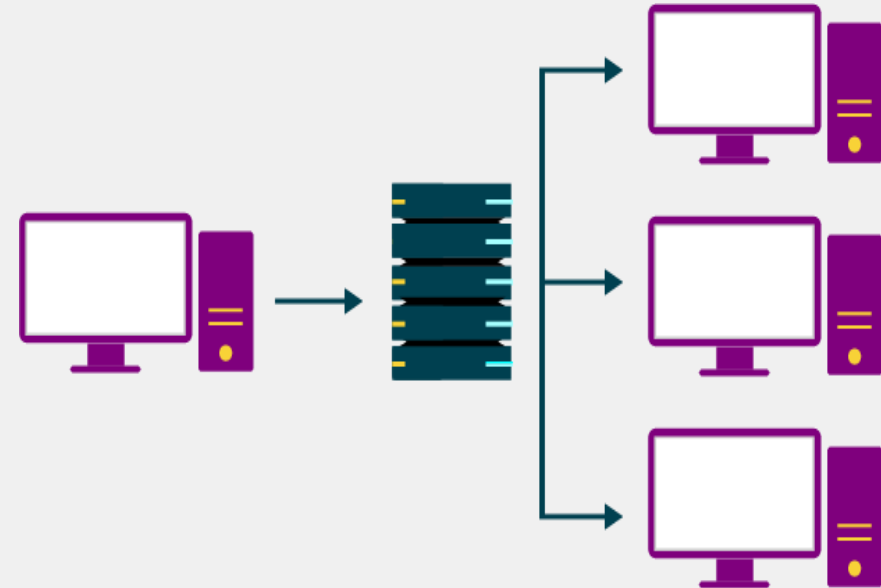
- What are the common tools that can be used for server deployment?
- Are there any examples as to how these tools can be used to automate installation for deployment efficiency?
- What factors should be considered when selecting server deployment tool for a large or small-scale deployment?
- What should be considered when creating pre-configured server image for deployment?



Installation tools

Image-based installation

- Windows System Image Manager (SIM)
- Drive cloning
 - Sysprep
- Windows Deployment Service (WDS)
- Microsoft Deployment Toolkit (MDT)
- **Third party imaging tools**
 - Symantec Ghost
 - Altiris Deployment Solution
 - Macrium Reflect



Post-installation configuration: Discussion and Q&A

- What post-installation steps should be considered after server install?
- Why is deploying patches and updates critical after installation?
- What measures should be taken when configuring server roles (DNS server, file server, domain controller)?
- Why monitoring important in post-installation configuration and what tools can be used?



Post-installation configuration: 1/2

What post-install would you consider?

- **Server configuration tasks**
 - Enable Remote Desktop (optional)
 - Set time zone
 - Change computer name, add domain suffix, or join the domain
 - Add any required features (.Net Framework, SNMP, etc.)
- **Network configuration**
 - IP address settings
 - Configure shared resources
- **Power settings**



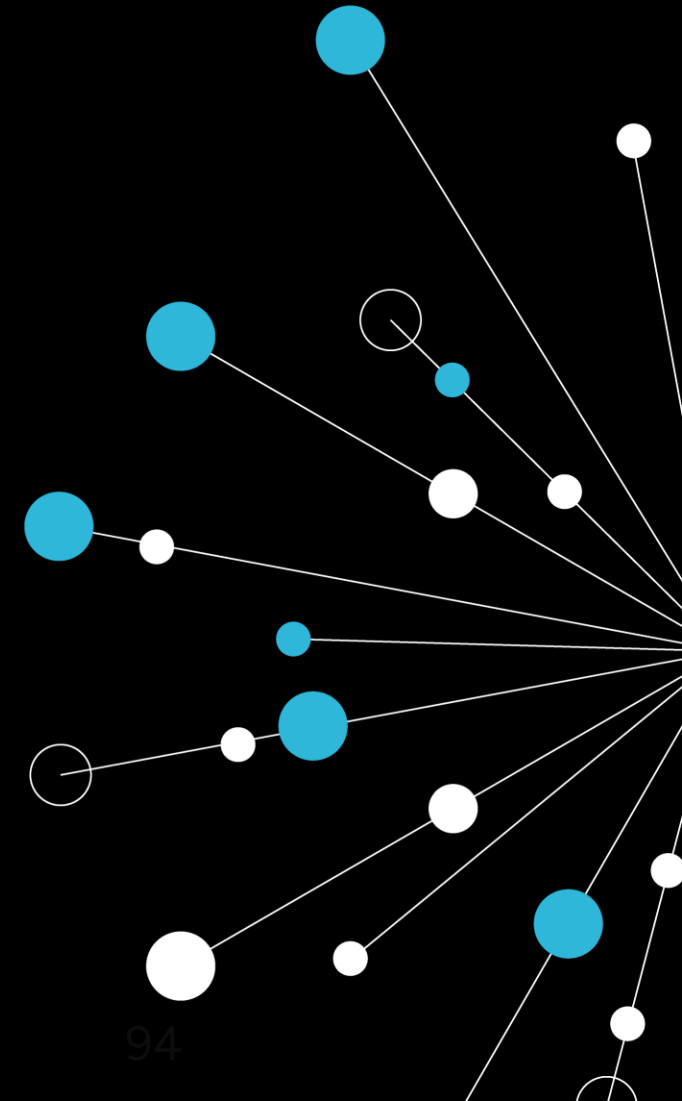
Post-installation configuration: 2/2

- **Security settings**
 - Firewall
 - User accounts
- **Windows updates**
 - Automatic Updates
 - WSUS
 - SCCM

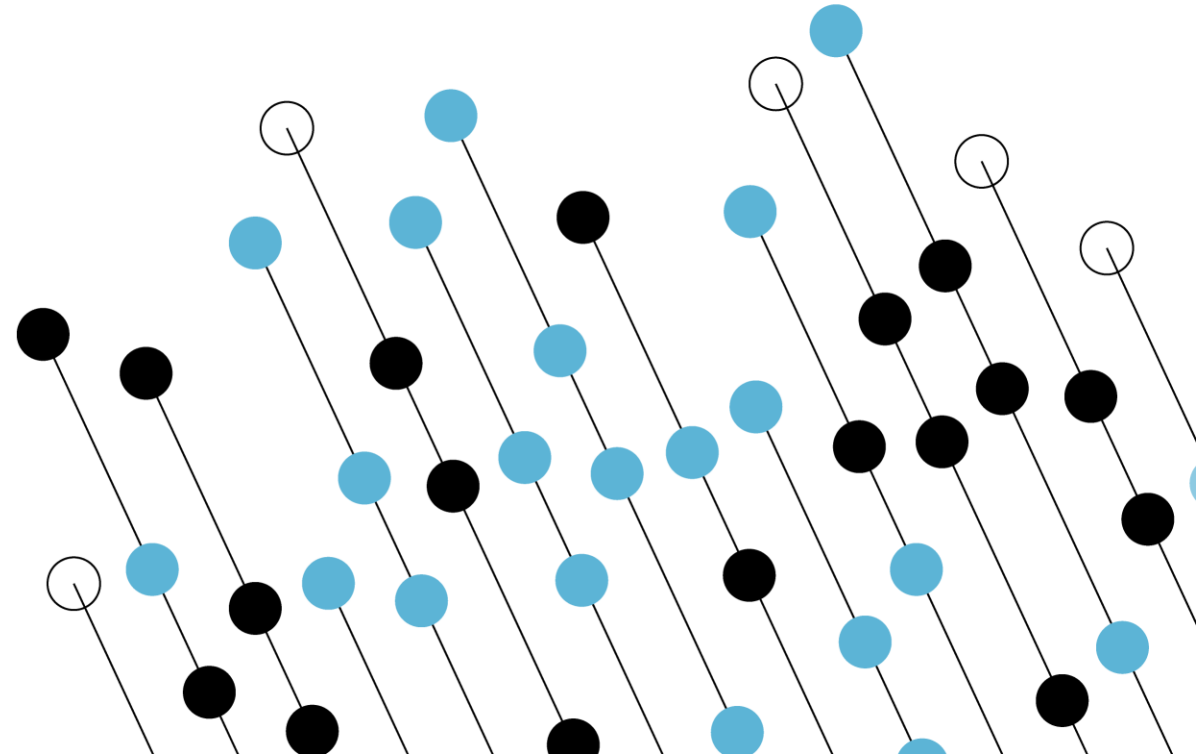


Activity 8: Windows and Linux Server Install

Find the activity in your **Learner Activity Document** and read the instructions.



Windows Operating Systems administration



Objectives: Windows Operating Systems administration

- Describe directory services role in Windows Server OS.
- Explain how Lightweight Directory Access Protocol (LDAP) is used by AD.
- Describe the structure of X500.
- Describe windows Print Services.
- Explain some basic Linux administration tools.



Directory services

- List of network users and resources
- Used to store and manage subjects and objects on the network
- Access Control Lists (authorization)
- Lightweight Directory Access Protocol (LDAP)
- X.500
 - Defines development of directory
 - Distinguished Name
 - Attribute=Value pairs
 - Hierarchical



Active directory

- **Domain Controller (DC)**
- Create domains, trees, forests and sites
- Two-way, automatic, transitive trusts
- Organisational Unit (OU)
- AD root namespace (in DNS)
 - For example, AD.QA.com
- Use OUs for object types
- Separate admin and user accounts
- Document
- Naming computer and user objects



User, group, and role-based management

User management:

- Assign privileges directly to user accounts.
- Unmanageable if number of users is large.

Group management:

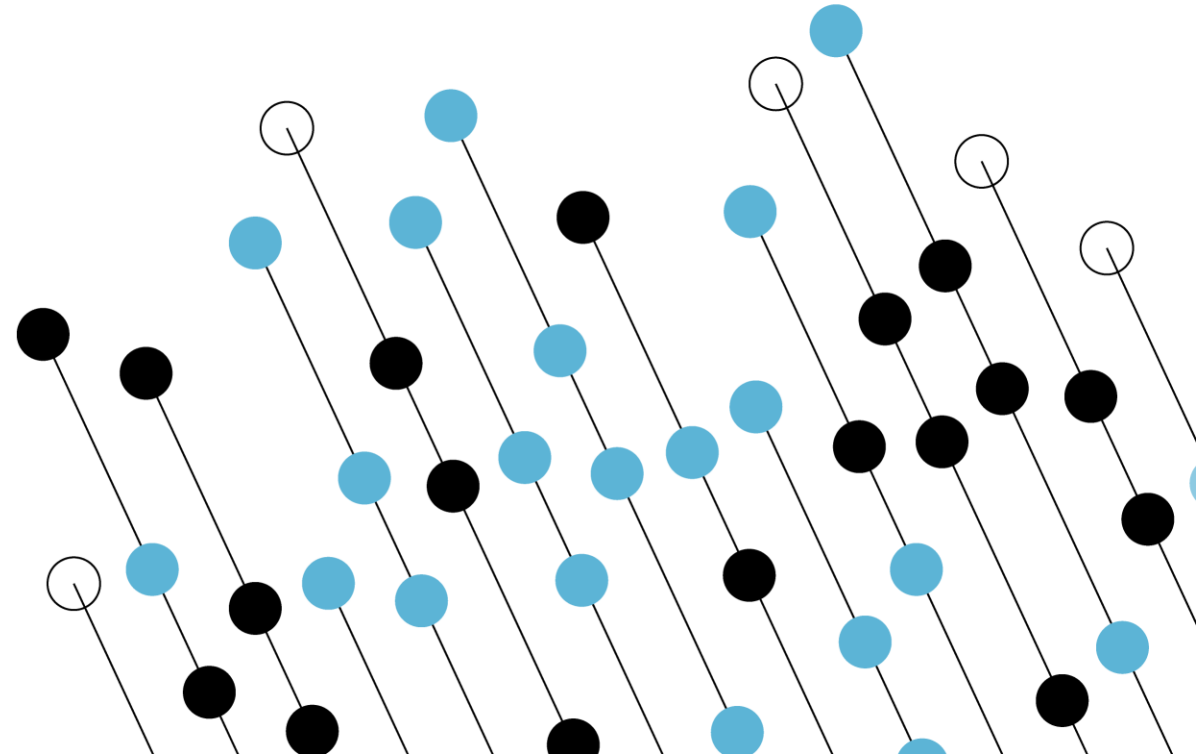
- Assign permissions to security groups and assign users to relevant groups.
- Issues with users inheriting multiple permissions.

Role-based management:

- Groups reflect system roles only.
- Membership of roles is more selective.
- User should only have permissions from one role at a time.



Windows printing services



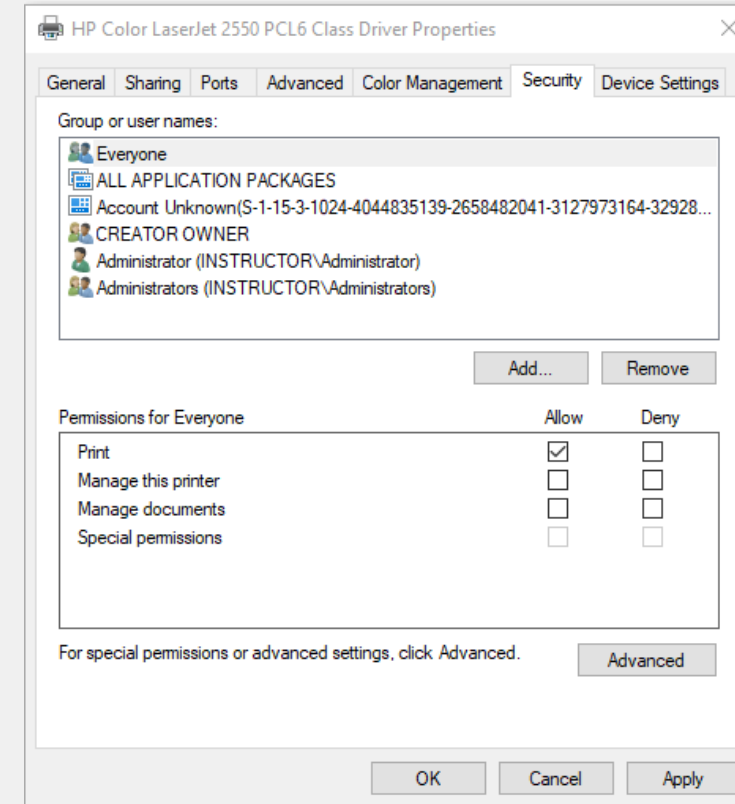
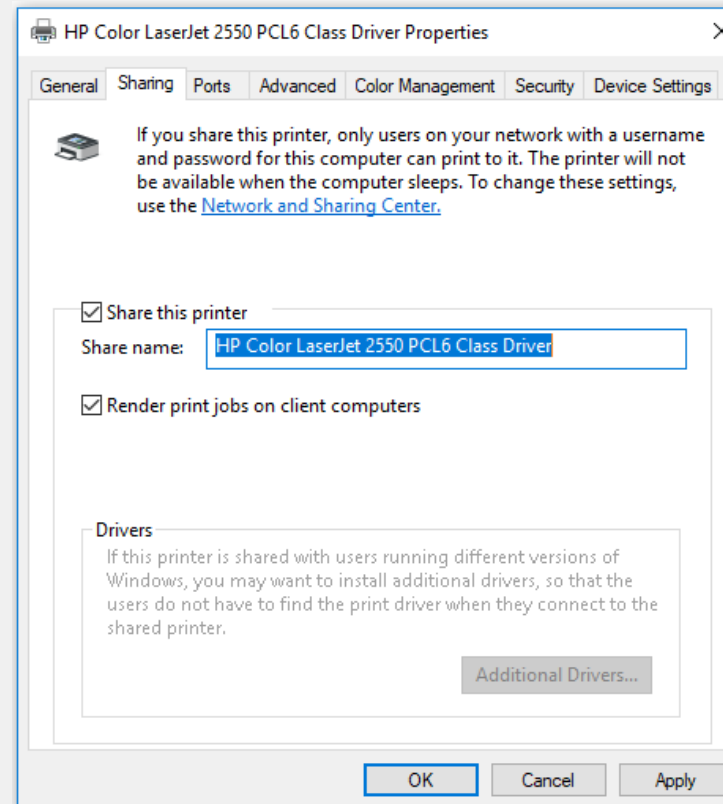
Print server role

- Print devices connected to network
- Centralised management of shared printers
- Print drivers installed on server
- Shared over the network



Managing print server

- Install printer on server and share.
- Connected devices download driver from server.
- Configure print permissions.
 - Print
 - Manage this printer
 - Manage documents



Print Server configuration

Supports multiple protocols

- SMB – Server message block
- LPR - Line Printer Remote protocol
- LPD - Linux Printer Daemon for Linux compatibility.

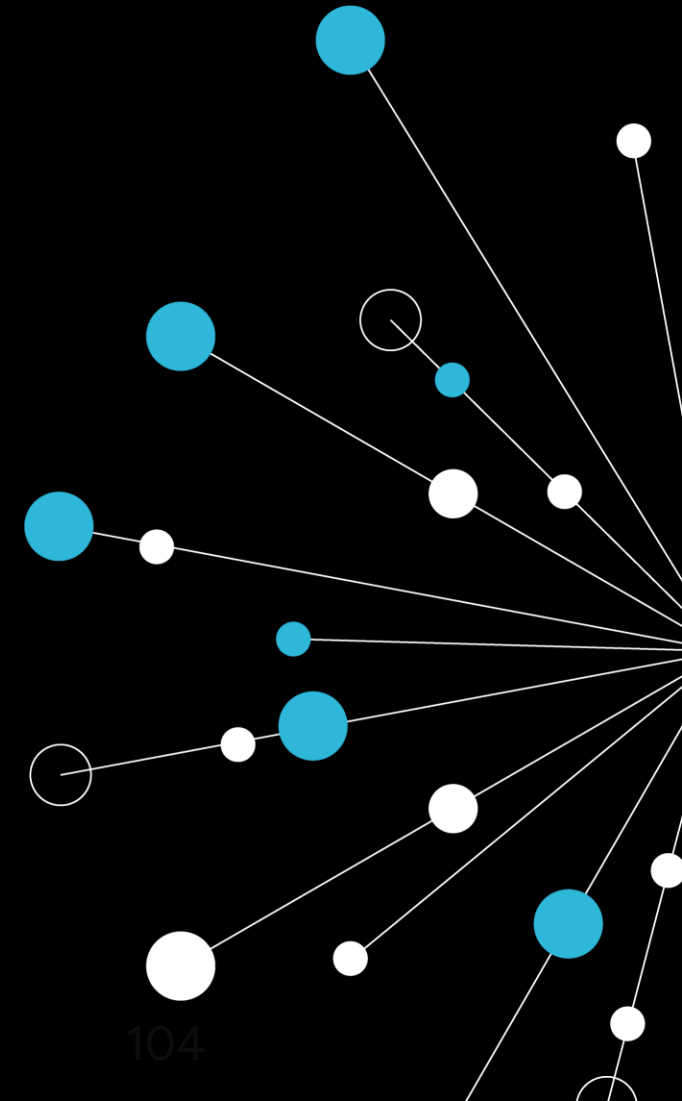
Internet printing

- Publish URL for printer on Intranet / Internet
- Easy configuration for users
- Internet Printing Protocol (IPP)

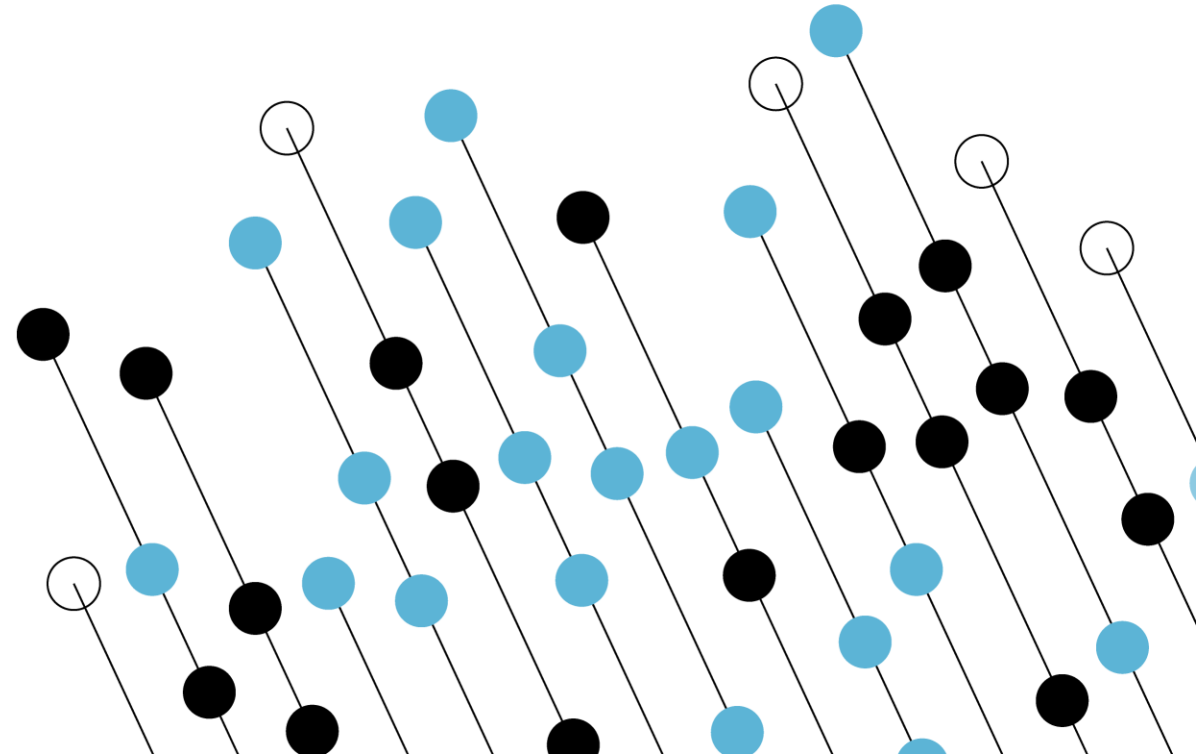


Activity 9: Install Active Directory Domain Services

Find the activity in your **Learner Activity Document** and read the instructions.



Linux basic services administration



Linux File Shares

Network File System (NFS)

- NFS client and server software is available for Windows.

Samba (Server Message Block implementation)

- Better compatibility with Windows clients.



RAM and CPU management

Free shows information about current RAM usage

- Default display is in kilobytes.
- **-m** displays in megabytes: **-b** for bytes and **-g** for gigabytes.
- **-t** displays totals for physical memory and swap space.
- **-s** (seconds) number displays a new, updated output at regular intervals.

lscpu displays CPU architecture information.

Option	Description
-e	Display the CPU information in human readable format
-s	Gather CPU data for a different Linux machine

Hard Drives management

Linux uses **fsck** to check the disk for errors:

- `fsck [options] [device]`
- **-p** option for automatic repair

Linux also uses:

- **parted** to list partitions and partition disks.
- **fdisk** to partition and format disks.



View running processes

ps displays a snapshot of current processes and detailed information.

By default, displays processes with the same effective user ID as the currently logged in user.

-a shows all running processes.

- Use | less to get one page at a time then type q to quit when done.



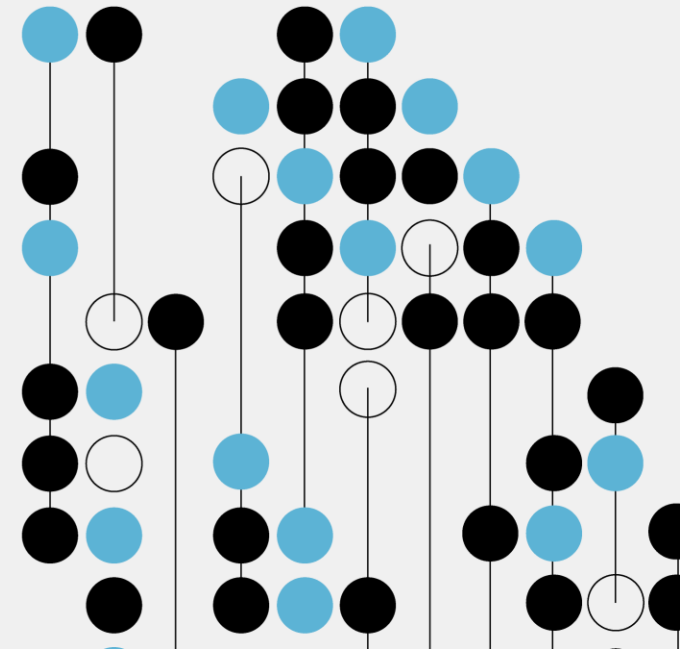
Viewing processes in real time

Top provides a dynamic real-time view of a running system. Once top has been run, the list can be ordered.

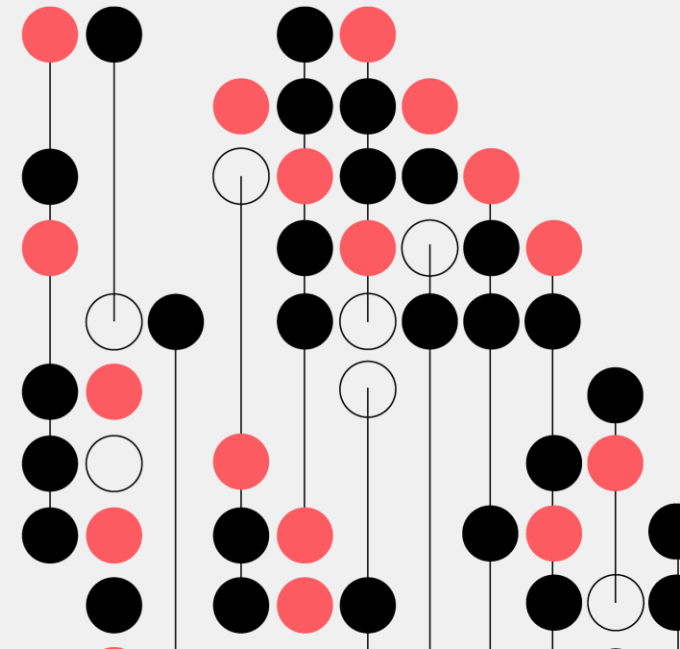
- **F** = fields management
- **M** = sorts by memory usage
- **P** = sorts by CPU usage
- **N** = sorts by Process ID
- **T** = sorts by running time
- **R** = reverses the order
- **x** = bolds values in the sort column
- **b** = highlights column background of sort column



End of Day 2



Day 3: System and network automation



Objectives: Day 3

- Describe the importance of network / system automation.
- List at least one automation tool.
- Explain the impact of automation on business processes.
- Describe different approaches to network automation.
- Explore the role of AI in network automation.
- Recommend best practices for system / network automation.



Network automation: Discussion and Q&A

- What is network automation, and why is it becoming a critical part of managing network environment?
- Are there any challenges and risks associated with network automation?
- Can you give examples of tasks that can be automated to simplify network administration?
- What approaches should be taken when automating network tasks?
- What are some benefits of implementing network automation in business network?
- Can automating improve system security and compliance? How?
- How can critical failures be avoided while using automation?



Introduction to network automation

- Automates the setup and maintenance of network resources.
- Deploys network resources based on predefined policies.
- Provides the means to continuously monitor network devices and sends alerts for anomalies.
- Automates updates deployments.
- Resources can be scaled to demand during peak usage.
- Automate response to alert to minimise downtime.



Advantages and disadvantages of network automation

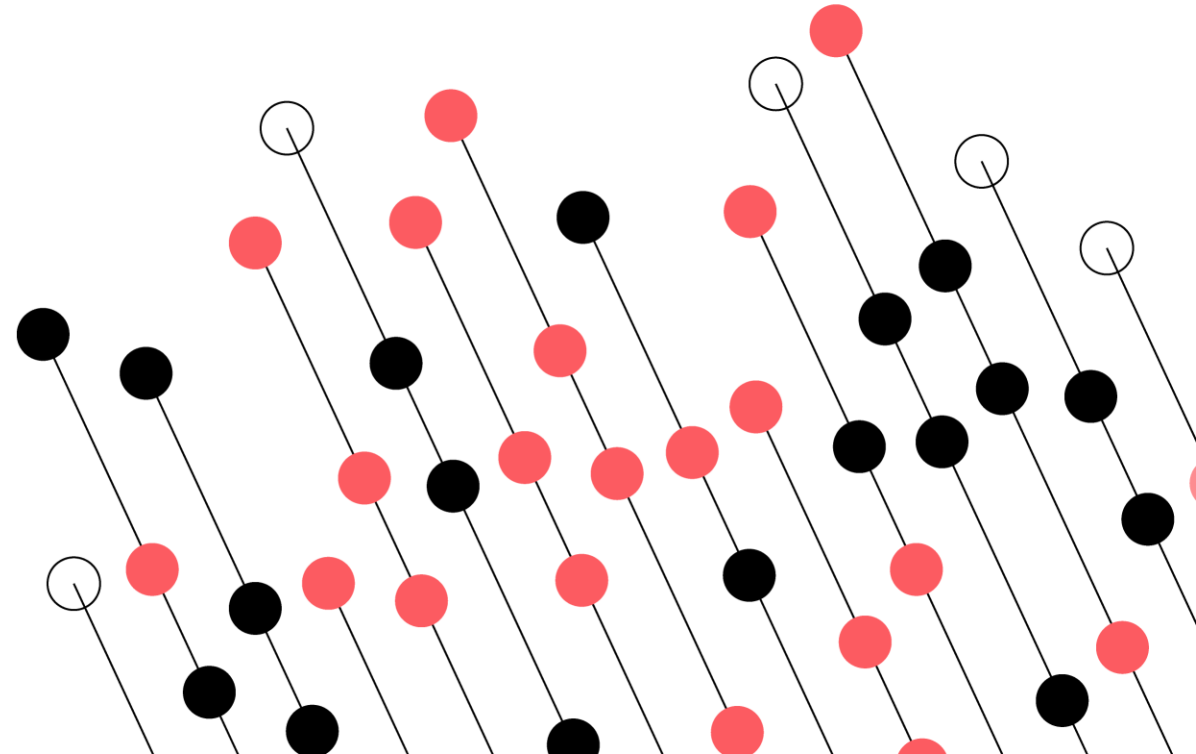
Advantages

- Network / system efficiency
- Speed of application deployments
- Consistency across network / systems
- Identify network / system issues
- Centralised administration
- Ease of application deployment

Disadvantages

- Complexity of implementation
- Difficult for automated tasks to adapt
- Security vulnerabilities if not properly maintained and managed

Application of network automation in network environment



Application of network automation: Discussion and Q&A

- What network tasks would you prioritise for automation (services, data sharing)?
- What tools or script would you use if you were to add multiple users to Active Directory and deploy group policies and software updates across multiple devices?
- What measure would you take if an automated task caused unexpected downtime? What measure should be in place to mitigate such risk?
- What are the primary components of SNMP and how do they interact with each other?
- Can you explain the differences between the SNMP versions (v1, v2, and v3) and their respective security features?

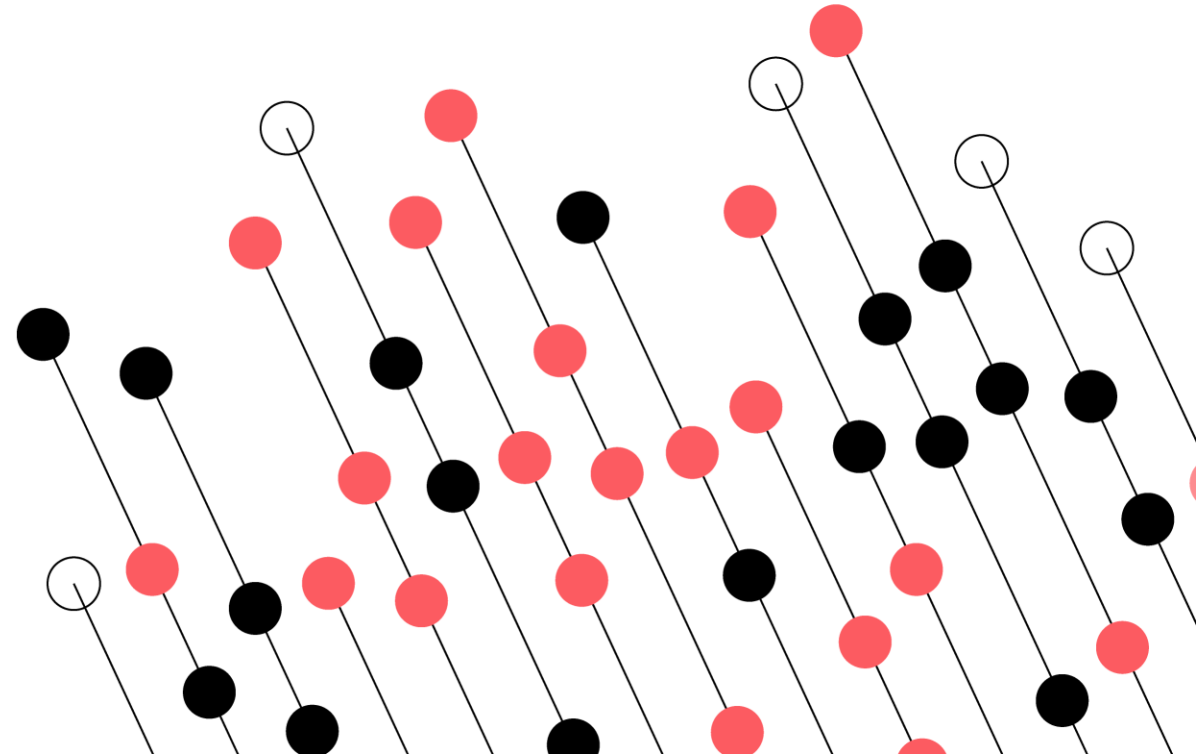


Application of network automation

- Identify network connectivity issues
- Resource allocation
- Service deployment
- Monitoring systems / network
- To locate network devices (inventory)
- Resource scalability
- Update and patch management
- Application connectivity checks
- Virtualisation (VM) deployments



Network automation using SNMP



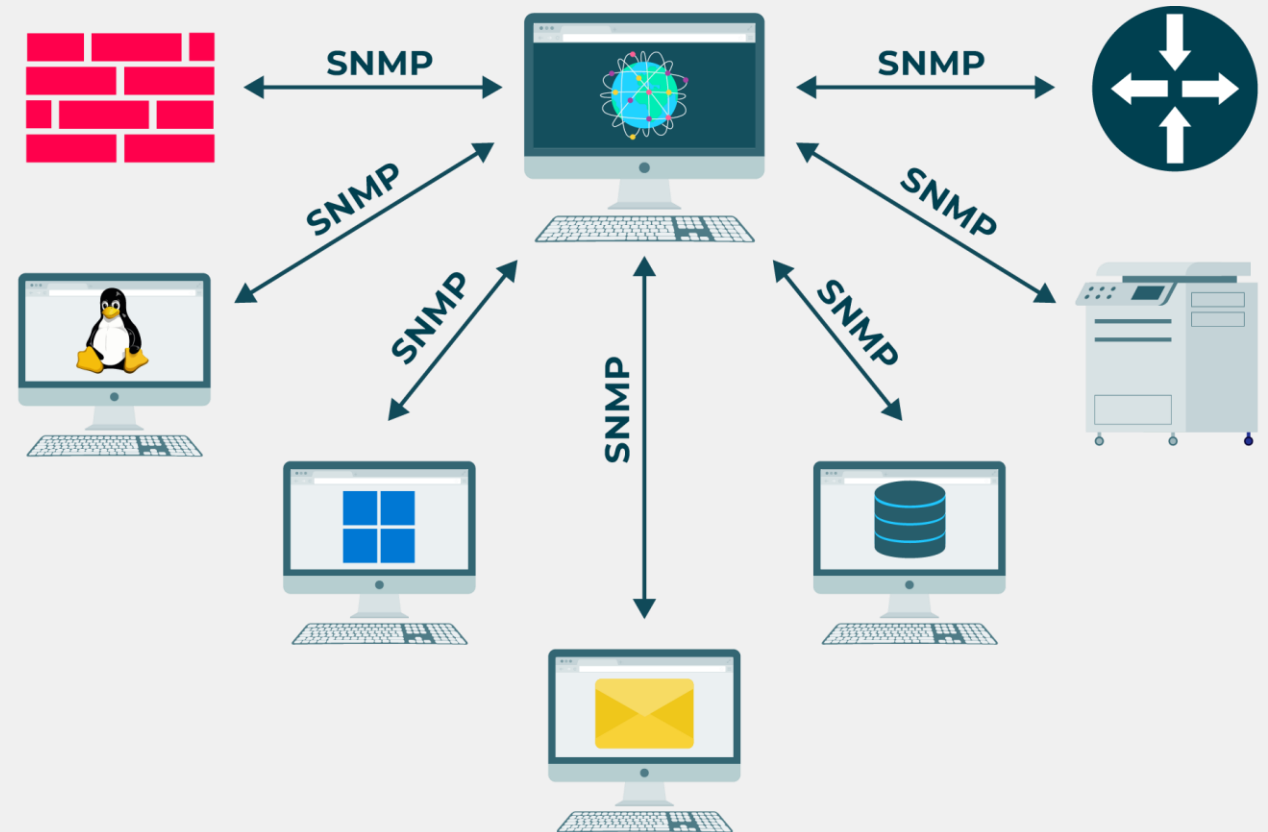
Simple Network Management Protocol (SNMP)

SNMP is an internet standard that provides a simple method for remotely managing or gathering statistics from network devices such as:

- Switch.
- Router.
- Firewall.

Instances monitored can include:

- Hardware and software configuration data.
- Computer performance and usage statistics.
- Computer event and error messages.
- Program and application usage statistics.



SNMP security

SNMPv1 and v2 use plaintext transmissions.

- Access Control List (ACL)

SNMP v3:

- Cryptographic functions
- Authentication and encryption
- Supports AES and 3-DES

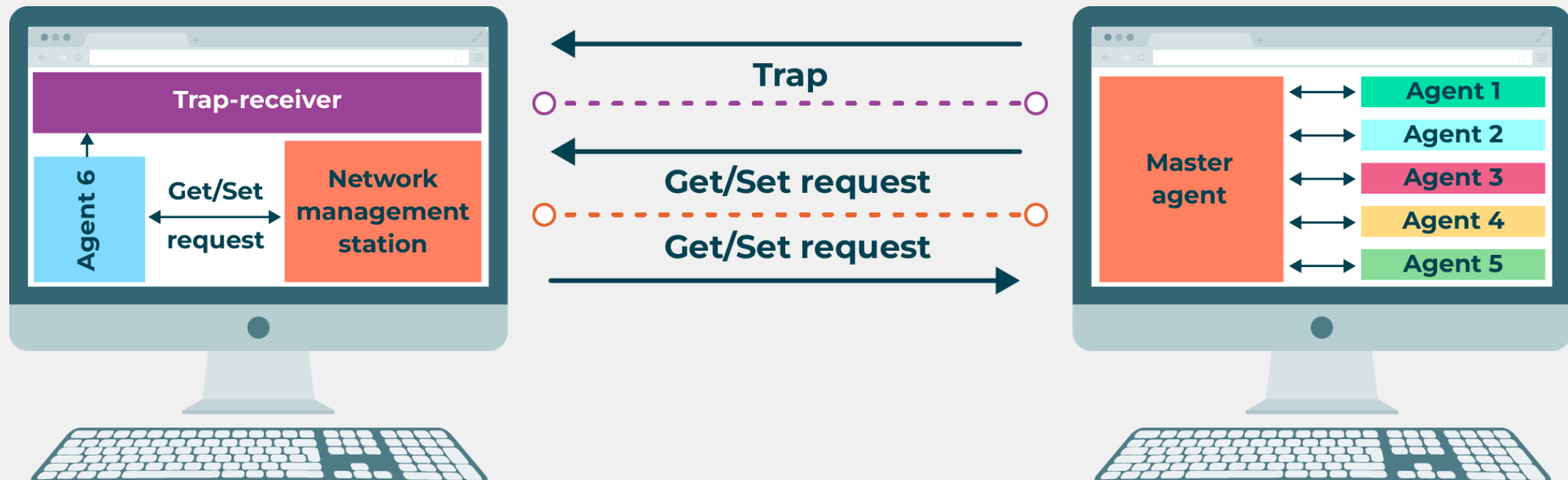
SNMP components:

- NMS (requests, compiles, and stores status information)
- Agent (monitors the device on which it is installed)
- MIB (framework dictating the format of data communicated between the network manager and each device agent)



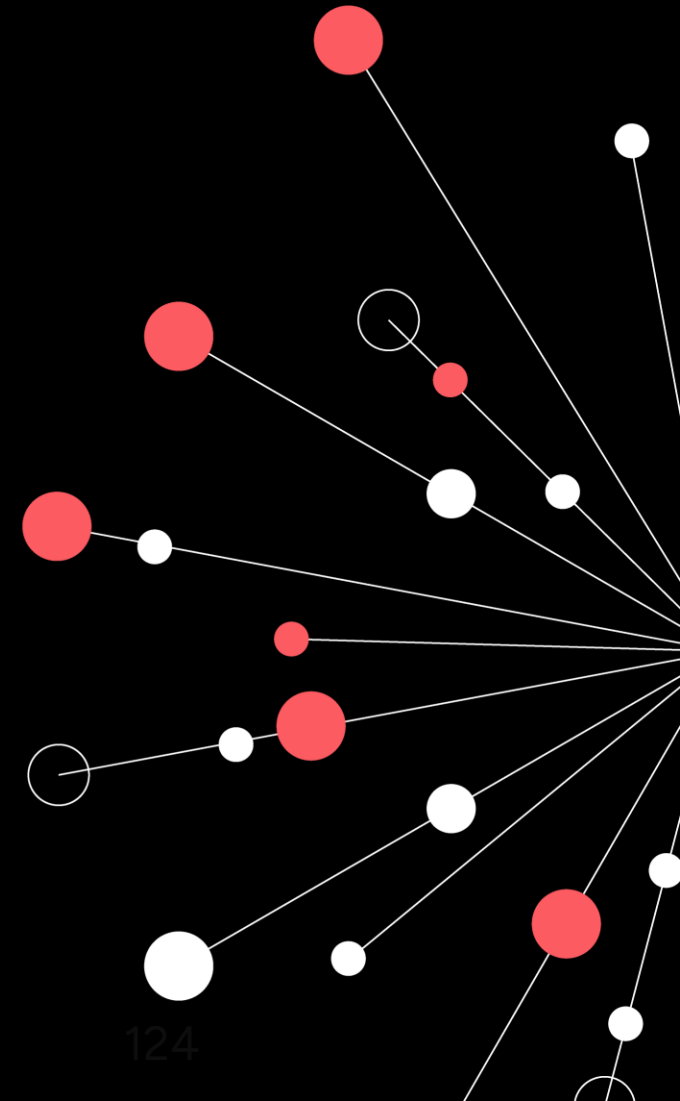
SNMP Interaction

- **Trap:** An alert sent from a device to SNMP manager (failures / maintenance)
- **Get:** An information request sent to a specific agent on a specific OID
- **GetNext:** Requests next information segment from ordered list of OIDs
- **Set:** Instructs an agent to alter a setting on the device it monitors
- **Walk:** Tests connectivity and query a network device for information

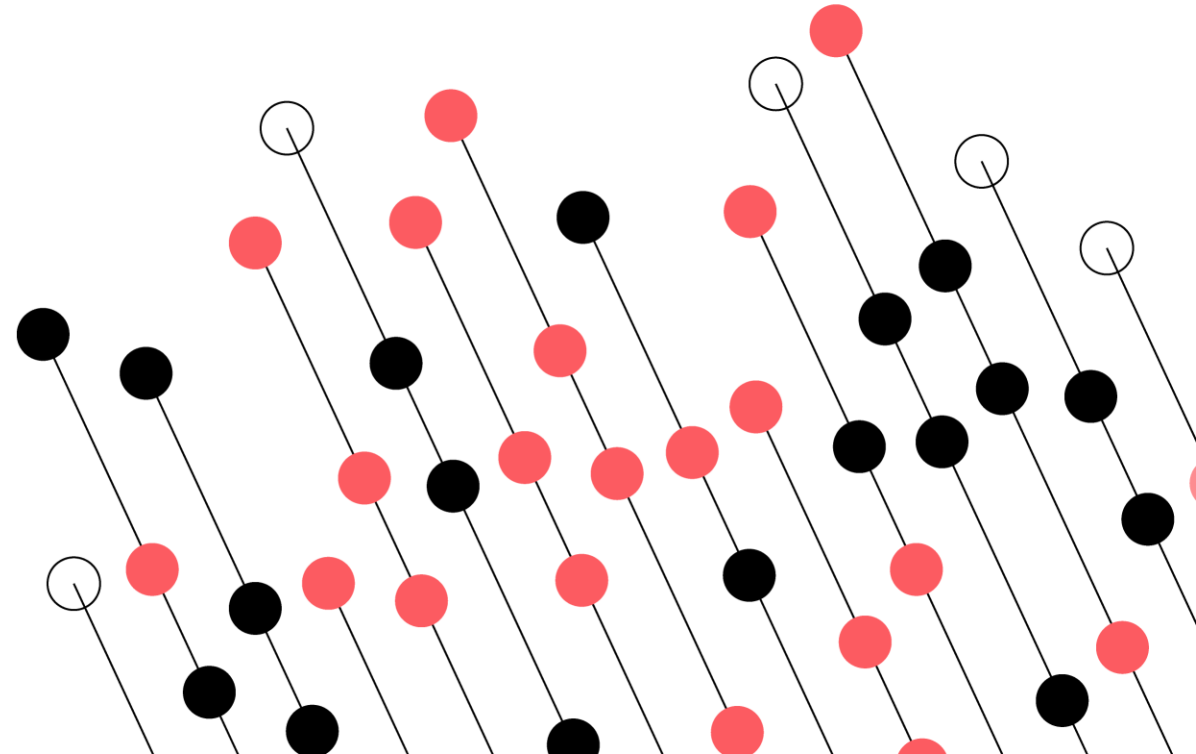


Activity 10: SNMP configuration

Find the activity in your **Learner Activity Document** and read the instructions.



System upgrade and testing



Objectives: System upgrade and testing

- Understand the importance of system upgrades and their impact on business operations.
- Identify key tools and techniques for securely operating and testing networks.
- Explain the benefits of regular system upgrades.
- Identify the risks associated with outdated network systems.
- Develop a structured approach for upgrading systems to minimise downtime.
- Verify the success of system upgrades through testing.
- Record and maintain the outcomes of system upgrades.



Understanding system upgrades: Discussion and Q&A

- What are system upgrades, and what are some key factors for initiating system upgrades?
- How do hardware upgrades differ from software upgrades? What should be considered for each scenario before upgrade?
- What upgrade scenario could lead to major business disruptions?
- How do business objectives and policies impact system upgrade?
- What risks are associated with system upgrade and how can downtime be minimised as a result of wrong upgrade?
- How does zero-trust architecture relate to system upgrades?



System upgrade

Measures businesses take to replace or update include:

- Hardware.
- Software.
- Firmware.

These updates are done due to:

- Performance requirements.
- Security.
- Functionality.

Upgrades can be minor or major



Reasons for system upgrades

Reasons for upgrade can include:

- Performance improvements.
- Security enhancement.
- Compatibility.
- To rectify bug issues.
- To take advantage of new system features.
- To comply with business policies.
- Cost efficiency.



Planning and preparing for system upgrade

- Identify scope of upgrade.
- Identify toolkit needed for the upgrade (WSUS, etc).
- Identify compatible software and drivers for the upgrade.
- Cross reference and document bugs and known issues.
- Identify tests and benchmark for upgrade.
- Set up a test system to avoid downtime and failure in services.
- Upgrade software.
- Carry out test against upgrade and benchmark the upgrade code.
- Identify backup requirement.



Securely operating and testing networks: Discussion and Q&A

- What are the best practices for testing before deploying system upgrade?
- What types of testing should be carried out after an upgrade?
- What measures should be taken to revert the system if an upgrade causes critical failures?
- What type of security testing should be carried out to verify OS upgrade?
- What aspect of disaster recovery planning should be incorporated with system upgrade?



Types of system testing after upgrade

Testing is carried out to validate system upgrades and to evaluate if upgrade objectives were met.

There are different types of testing to validate system upgrade and configurations:

- **Acceptance testing:** End users carry out a beta testing of the system.
- **System testing:** The newly deployed system is tested against the defined upgrade objectives.
- **Integration testing:** This testing is carried out where a component has been deployed as part of an existing system.
- **Unit testing:** Modular testing is carried with this type of testing where each upgraded component is tested individually as part of the system.



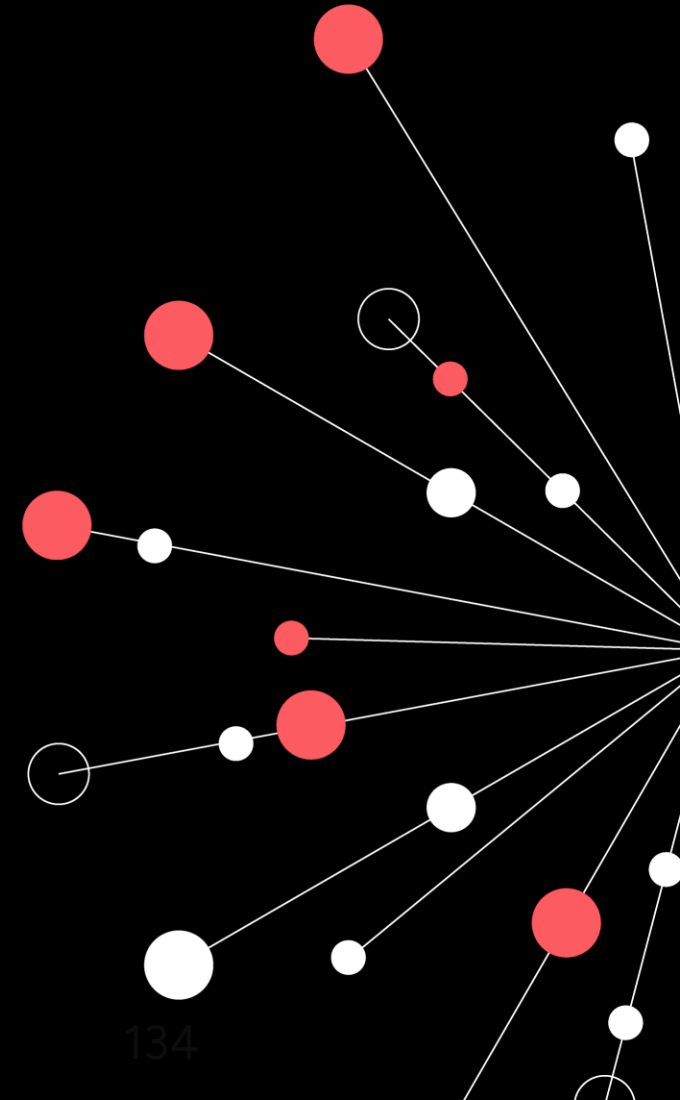
Summary and case studies: Reflection

- You are currently in the process of upgrading your business routers and switches. What measures would you take to minimise downtime and functionality?
- Your business needs to upgrade Windows Server OS and Windows Client OS. What pre-upgrade and post-upgrade actions should you take? What would consider on behalf of your users?
- You have deployed an upgrade, and it has unexpectedly caused performance issues. How would you troubleshoot and resolve the issue?
- Deploying patches for a critical vulnerability could cause system / network disruptions. What approach would you take to prevent any issues occurring from the upgrade?

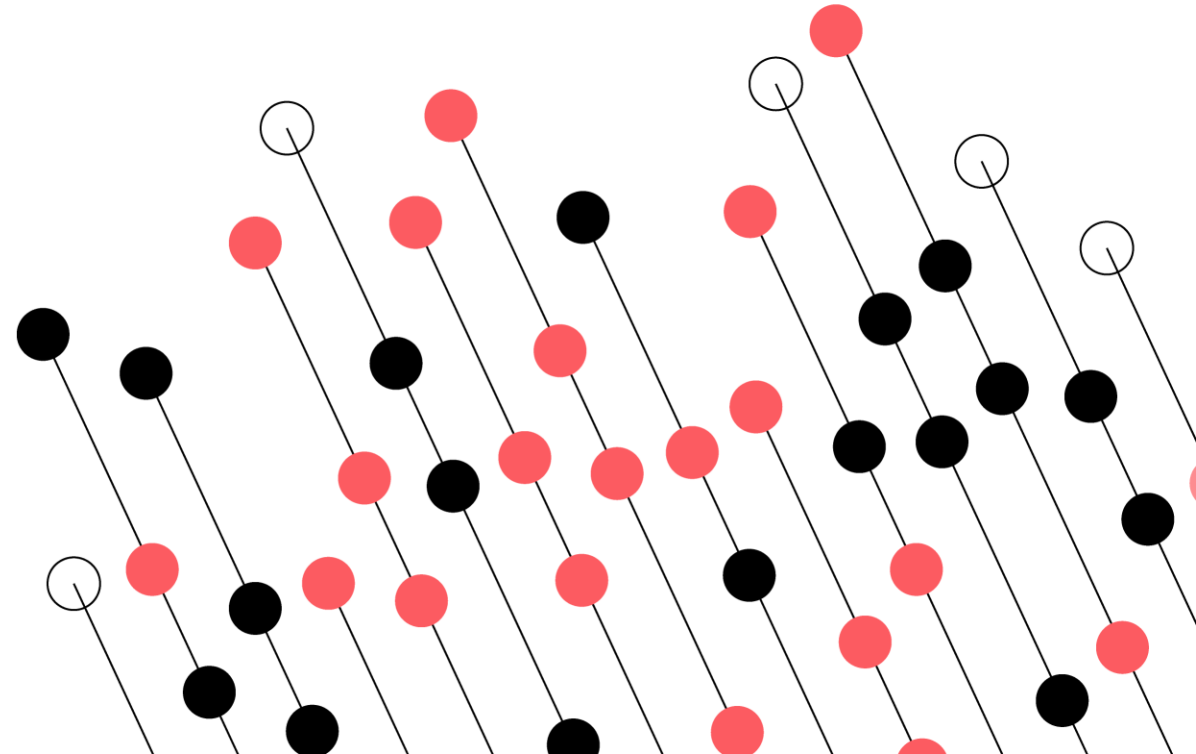


Activity 11: System upgrading and testing

Find the activity in your **Learner Activity Document** and read the instructions.



Different types of network maintenance



Objectives: Different types of network maintenance

- Describe different types of network maintenance.
- List best practices for implementing maintenance strategies.
- Evaluate the role of automation and monitoring in managing organisation network infrastructure.
- Explain some network maintenance challenges that businesses can face.



Understanding network maintenance: Discussion and Q&A

- What is network maintenance, and why is it essential for IT infrastructures?
- In what ways does network maintenance contribute to business continuity?
- Can business network / system function without a well-defined network maintenance strategy? Why or why not?
- Can anyone provide some examples of consequences of poor network maintenance?
- How do we balance network maintenance whilst ensuring system / high performance and resilience?
- What are the differences between reactive, preventive, predictive, and proactive maintenance? Which one would you recommend and why?



Introduction to network maintenance

Network maintenance is required to prevent network failures.

Carrying out preventative maintenance can help identify issues with network components before they occur.

Network maintenance can include:

- System monitoring.
- Upgrading and updating.
- Troubleshooting issues.
- Capacity planning.
- Ensuring policy compliance.



Why do businesses undertake regular system / network maintenance?

- Support business to ensure system / network reliability.
- Maintain consistent uptime and prevent failures.
- Minimise the downtime to ensure business continuity.
- Enhance the security posture of systems / network.
- Minimise data loss or corruptions.
- Reduce cost of repairs or as a result of downtime.
- Optimise system / network utilisation.



Reactive maintenance

- Maintenance carried out after system / network failures
- A reactive approach
- Unplanned / unscheduled
- Can result in downtime and expensive system restore

Pros

- Low initial cost
- Simple to implement

Cons

- Might result in prolong downtime
- Can lead to extensive damage



Preventive maintenance

- Includes regular hardware inspection and maintenance.
- System schedule is put in place to apply corrective maintenance.
- The aim of preventative maintenance is to prevent serious issue from occurring.

Examples of preventative maintenance:

- **Clustering:** Adding additional servers can prevent a single point of failure and ensure availability of network services.
- **Scalability:** Having the right level of hardware and software to support the business operations can prevent system lag in the event of expansion.



Predictive maintenance

- Maintenance carried out based on data
- Used to predict system / network failures
- Usage of proactive tools to prevent future system issues

Pros

Minimise system downtime

Optimise regular maintenance schedules

Cons

- Monitoring and predictive tools can be costly
- Challenges in complex system data analytics to predict possible issue



Proactive maintenance

- A proactive measure in identifying system failure
- Can be very effective in identifying root cause of system issue
- Continuous monitoring and improvement strategy

Pros

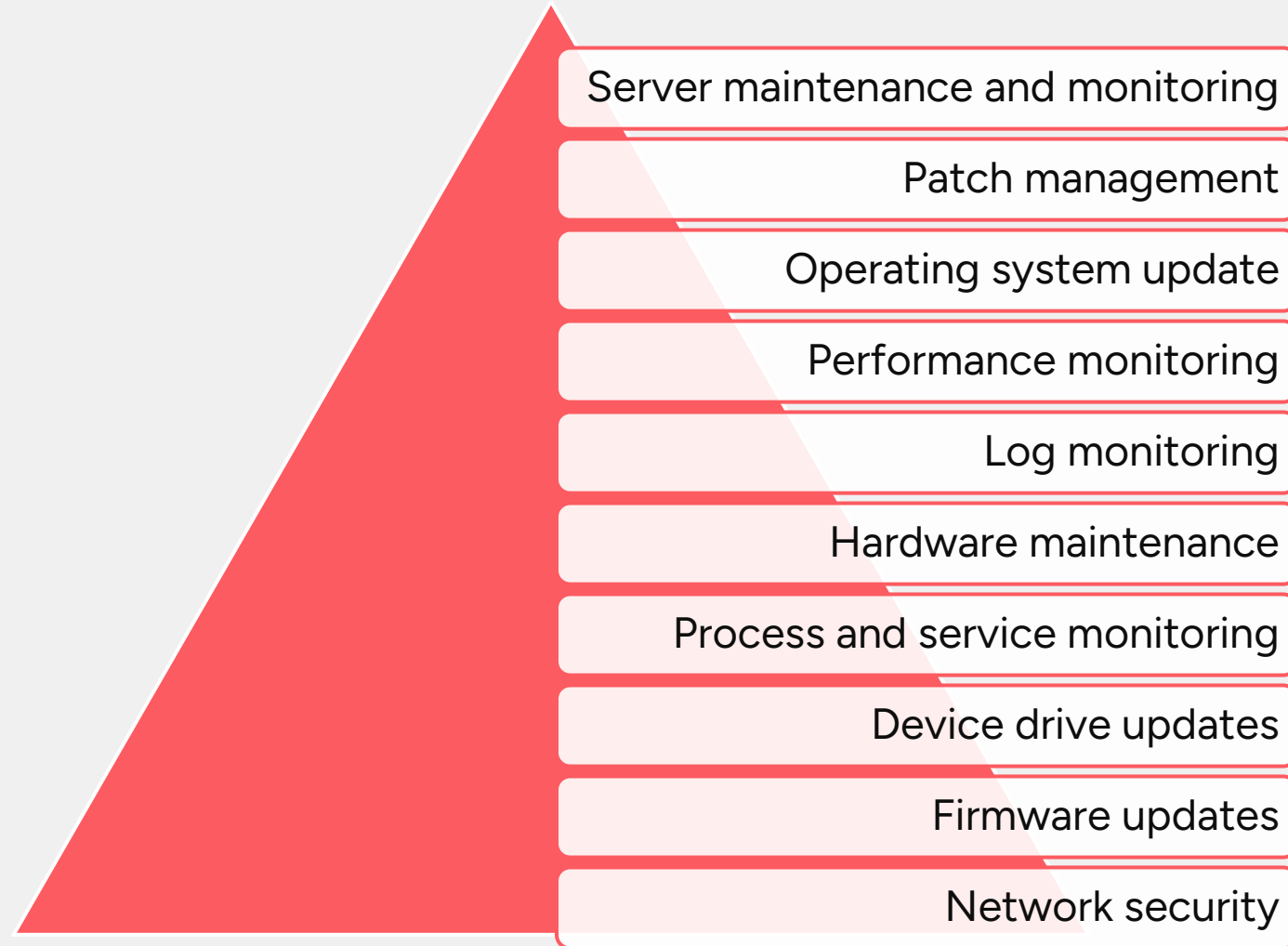
- Identify root cause to prevent reoccurring issue
- Provides a greater degree of system reliability
- Reduce maintenance cost

Cons

- Cost can be a deterring factor
- Implementation can be complex



Other types of network maintenance



Review: Discussions and Q&A

Based on what we have just covered:

- How do reactive and proactive maintenance differ in terms of deployment effectiveness?
- Can AI and machine learning be used to effectively predict network failure?
- Which type of network maintenance is most effective in ensuring 24/7 network availability? Why?
- Is it possible for businesses to rely solely on just one type of network maintenance, or should they use a combination?
- What measures should be taken when responding to network failures in a reactive maintenance approach?



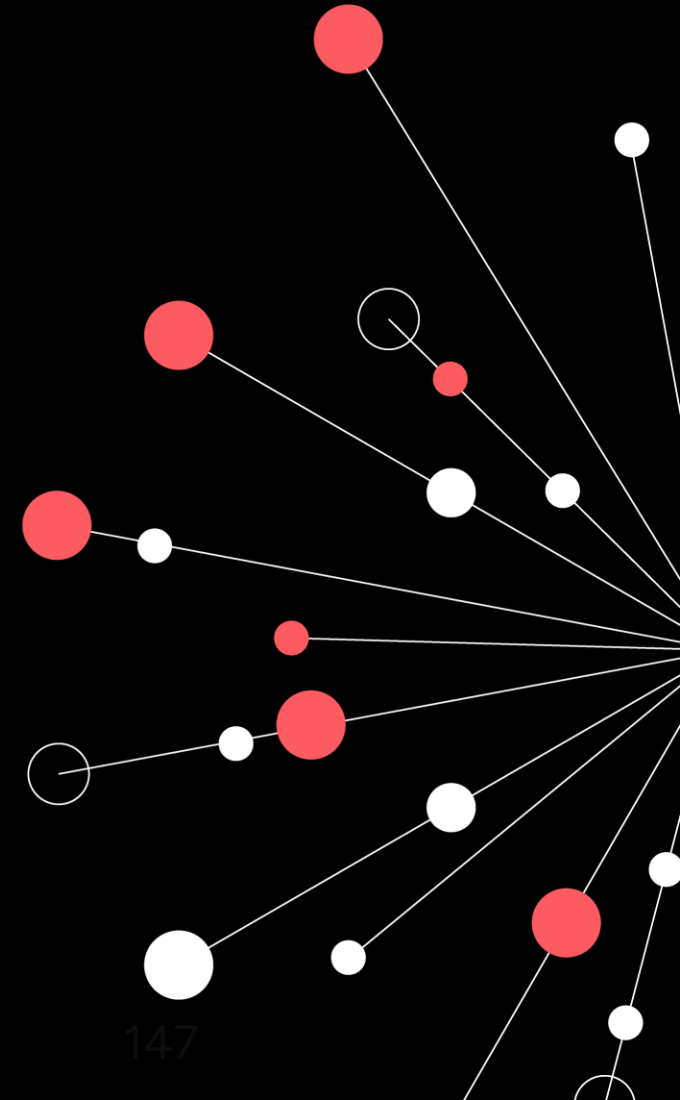
Summary and case studies: Reflection

- Your business constantly experiences network downtime. As a result, you have been delegated to select the right maintenance strategy. Which type of maintenance would you implement to resolve this and why?
- Your business recently deployed new network devices including firewalls, routers, and switches. What type of maintenance would you recommend for a long-term reliability?
- One of your router log shows unusual activity, but there are no clear signs of failure. How would you deal with the situation using a maintenance strategy?
- You have been delegated to create a new maintenance policy for your business. What components would you include and why?

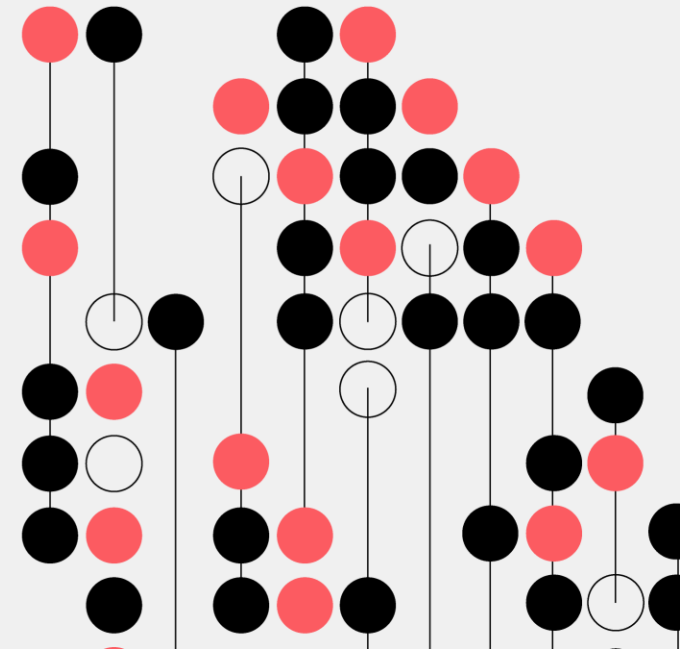


Activity 12: System performance improvement

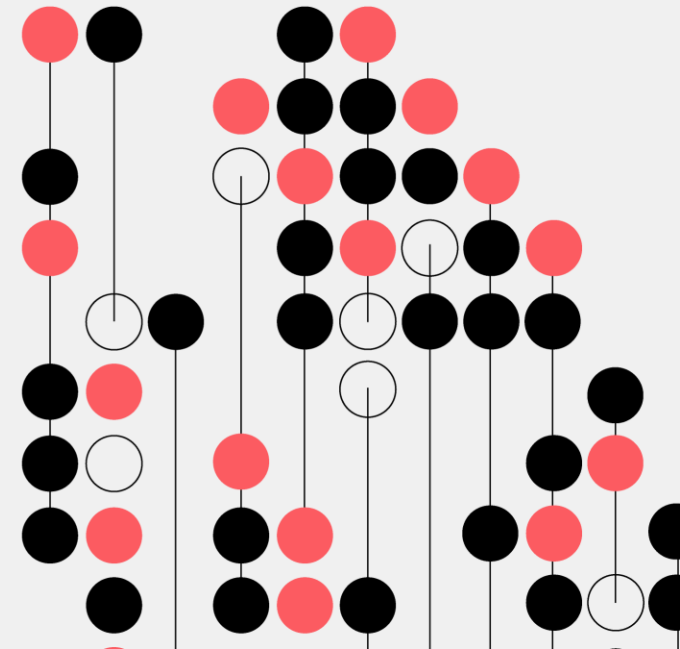
Find the activity in your **Learner Activity Document** and read the instructions.



End of Day 3



Day 4: Performance monitoring and analysis



Objectives: Day 4

- Explain the importance of system / network monitoring.
- Document network / system performance inline with business requirements.
- Describe different monitoring tools and techniques.
- Recommend at least one solution in optimising network / system performance.
- Describe how automation can be used for effective system performance monitoring.



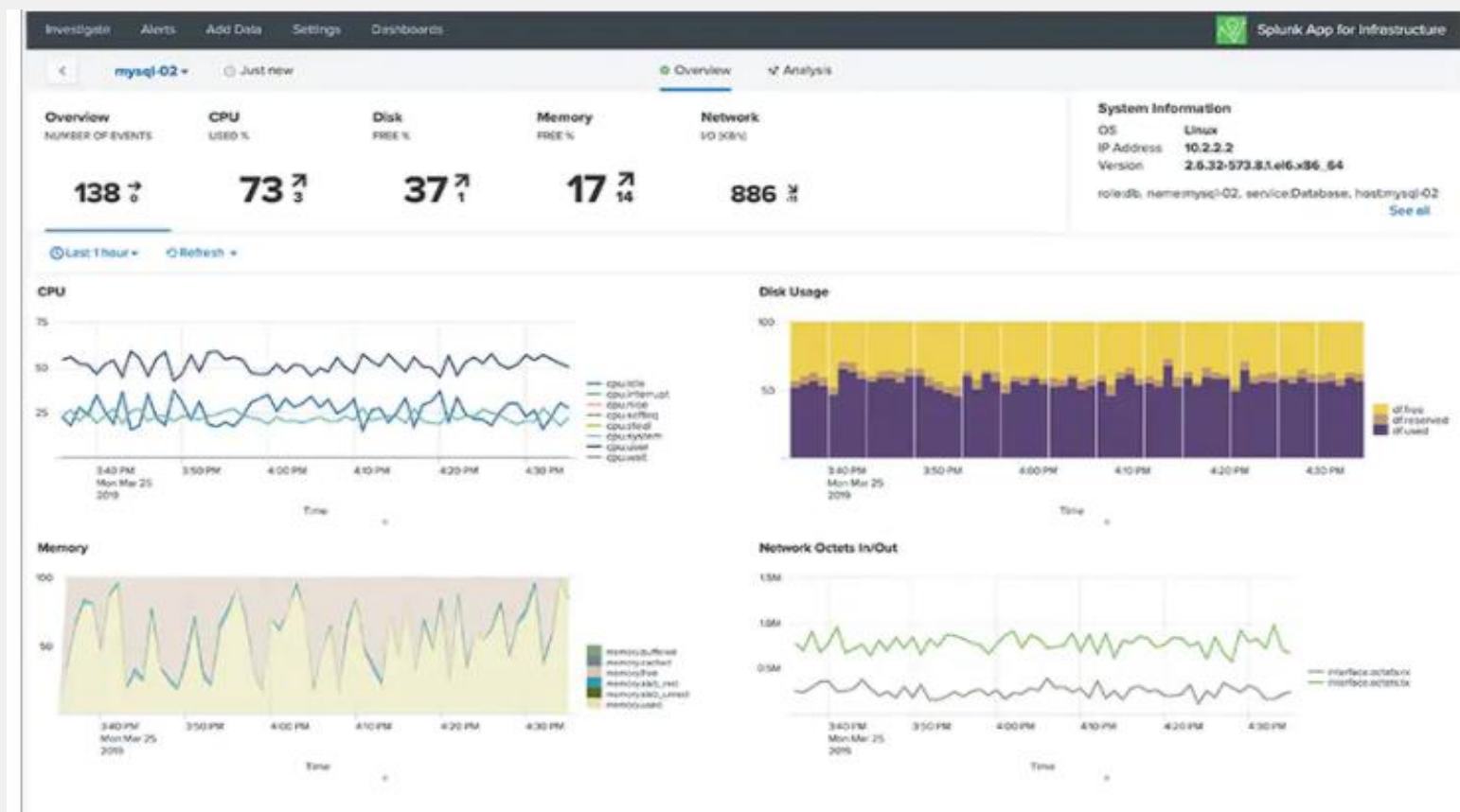
Network / system performance and monitoring: Discussion and Q&A

- Why is it critical to monitor network and system performance in a business network?
- What impact does poor performance in a network or system have on business processes?
- What are some key performance indicators (KPIs) for measuring network / system performance?
- What are the differences between real-time and historical performance analysis?
- What are the benefits of continuous system / network monitoring?



Introduction to performance monitoring

- Performance monitoring is used to determine if the system is handling input / out process as required.
- It involves system tracking and optimising system for smooth operations.
- Good practice is to create system baseline during peak time.
- Continuous monitoring can improve system reliability, security, and availability.

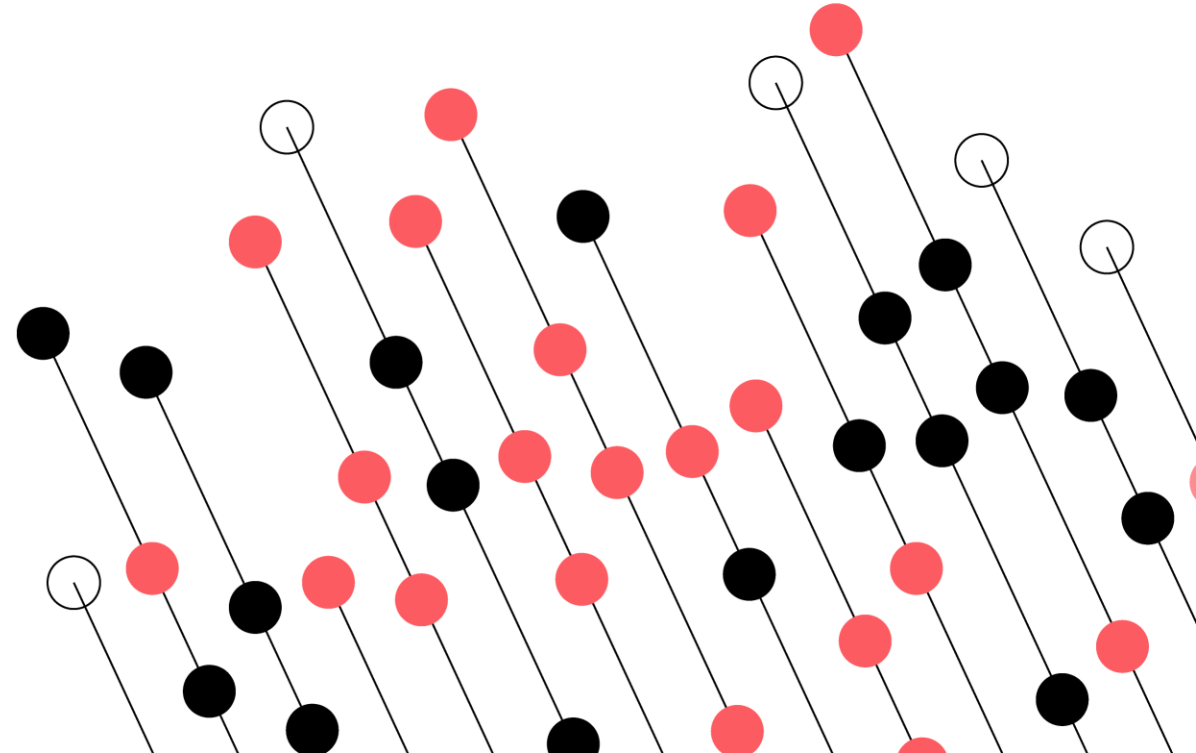


Why monitor system / network performance?

- Optimal performance and reliability of system
- Prevents downtime proactive detection
- Enhances security through identification of suspicious system activity
- Ensures seamless system and network operations



Tools and techniques for monitoring network / system performance



Tools for monitoring network / system performance:

Discussions and Q&A

- What monitoring tools have you used within your business network?
- Are you familiar with any network performance monitoring (NPM) tools (SolarWinds, Nagios, PRTG, Wireshark)?
 - Other commonly used system performance monitoring tools include Splunk and Microsoft Performance Monitor.
- What should be considered when selecting a performance monitoring tool for your business?



Tools for monitoring network / system performance

Tools include:

- **SolarWinds Network Performance Monitor:** Comprehensive network monitoring with real-time analytics.
- **ManageEngine OpManager.**
- **Integrated network management** with automated alerts.
- **Nagios:** Open-source monitoring for servers and network devices.
- **New Relic:** Application performance monitoring with detailed insights.
- **Datadog:** Cloud-based monitoring with AI-driven insights.



Techniques for identifying and analysing performance issues: Discussion and Q&A

- What are some common network performance issues that you have encountered and how was it identified and resolved?
- How does traffic analysis help in identifying performance bottlenecks? Can you provide a scenario of how bottleneck can occur?
- What role can log analysis play in detecting troubleshooting system performance?
- Do you think AI and machine learning can play a critical role in predictive performance analysis?



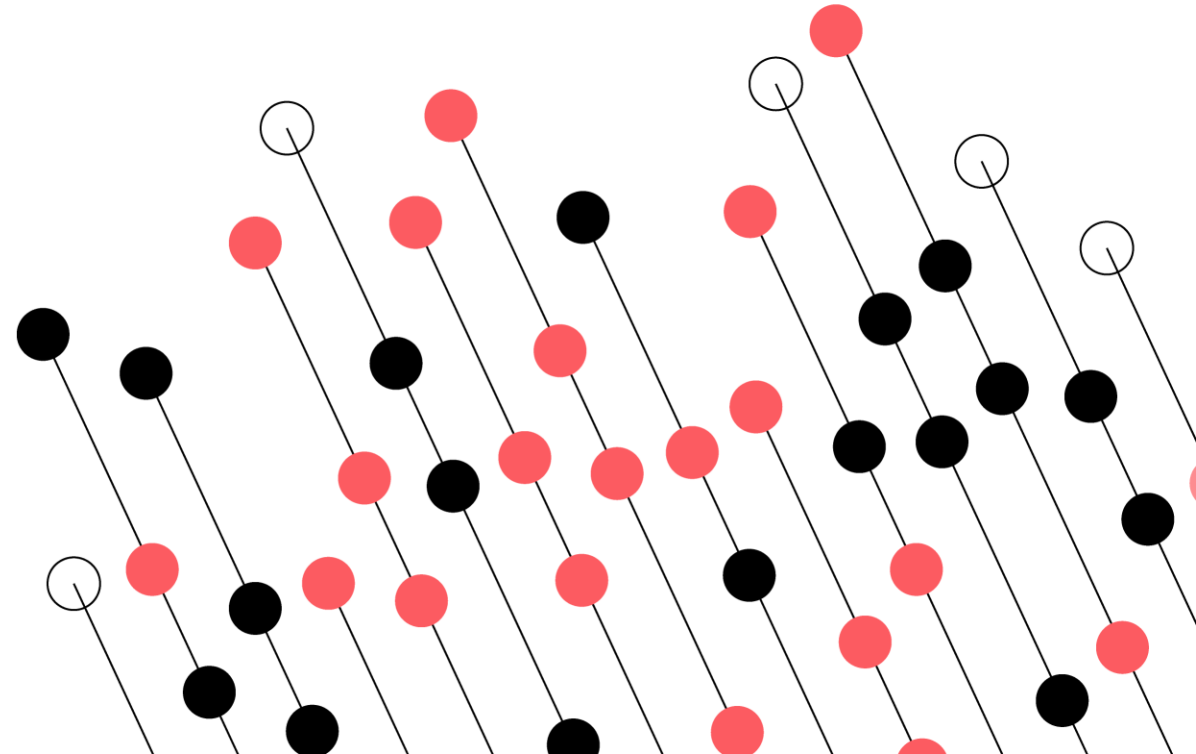
Techniques for identifying and analysing performance issues

- Identify the underlying causes of performance issues.
- Analyse system behaviour during peak and off-peak period.
- Detect areas of network / system congestions.
- Review logs for system crashes.

Various tools can be used to identify the points listed to the left:

- Perf
- Strace
- Wireshark
- Ping
- Traceroute
- Task manager
- Htop

Recording and analysing network / system performance



Recording and analysing network / system performance:

Discussion and Q&A

- Why is it critical to document and log performance data?
- What best practices should be in place for reviewing and maintaining network / system performance logs?
- How do automation of log analysis help with a more proactive response?
- What role does organisation policies and procedures play in performance monitoring?
- What role does performance logs play in business continuity and capacity planning?



Approaches to recording network / system performance

- Continuous real time monitoring and tracking of network traffic
- Collate system logs
- Log and retain historical log data
- Use agent-based log system to monitor and log performance
- SNMP polling
- Early detection through log analysis



Approaches to analysing network / system performance data

- Baseline comparative analysis
- The use of AI and machine learning to analyse log files
- Detect congestion through system / network profiling
- Analyse system behaviour compared to log data
- Compare ingress and egress traffic for anomalies
- Create network simulation using log data



Challenges to recording and analysing network / system performance

- High volume of log files
- False positive alerts
- Tools not able to scale with larger enterprise networks
- Privacy and security concerns with log tools
- Complexity of log collation from different sites
- Lack of encryption during transmission of log files
- Cost implication of using AI powered tools



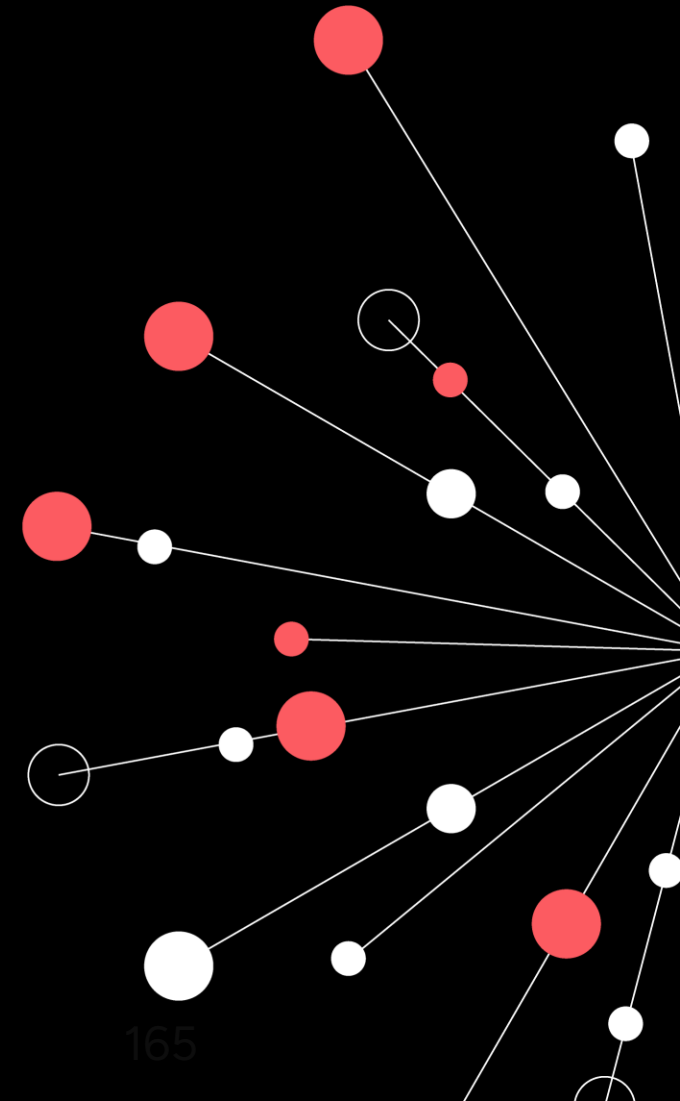
Summary and case studies: Reflection

- What tools and approach would you use to monitor and identify throughput issues?
- Your business has asked you to expand your network by adding an additional four segments, which will increase the network traffic significantly. What approach would you take to prevent network / system overload?
- Your business is currently experiencing performance issues during peak hours. What optimisation measures would you implement?
- Users are complaining of intermittent connections when they attempt to access your intranet. What approach would you take to identify the root cause. What tools would you use?
- In summary, what steps would you take to create a monitoring and response plan for your organisation?

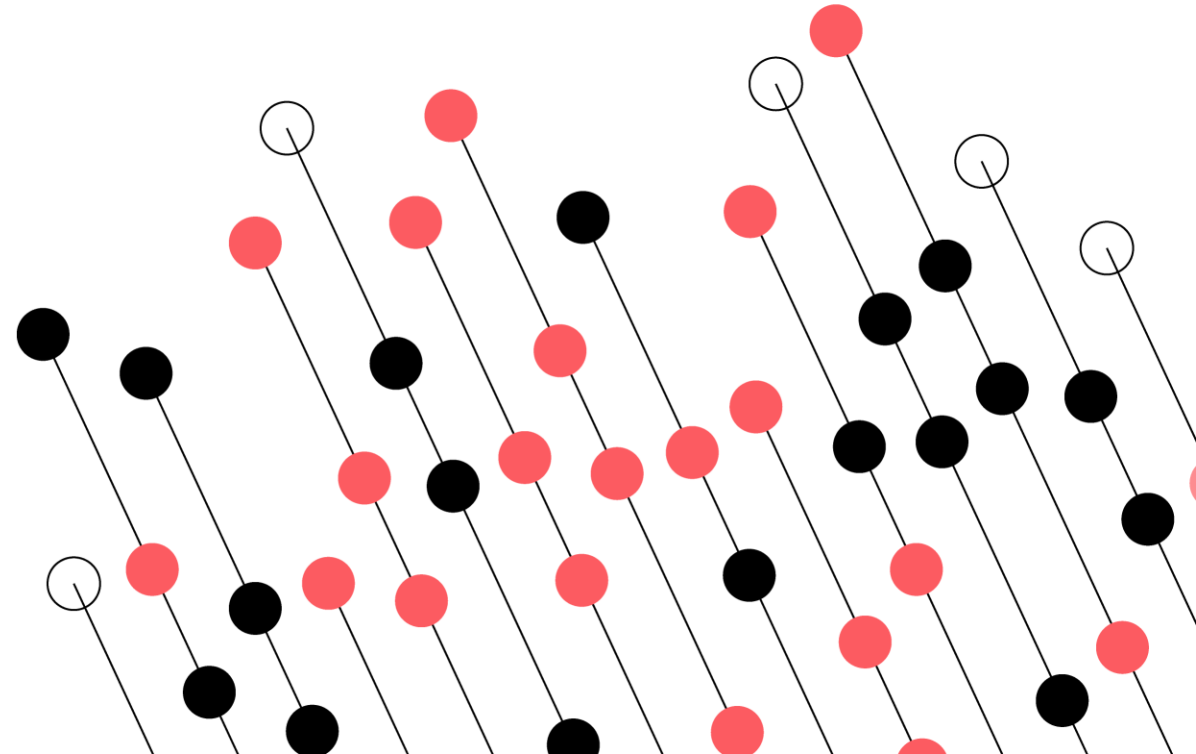


Activity 13: System performance monitoring

Find the activity in your **Learner Activity Document** and read the instructions.



Service Level Agreement (SLA)



Objectives: Service Level Agreement (SLA)

- Explain Service Level Agreements (SLAs) in relation to service provision.
- Differentiate between various types of SLA.
- Describe some challenges service providers faced when creating SLA.
- Recommend some best practices when creating, monitoring, and updating SLA.



Service Level Agreements (SLAs): Discussion and Q&A

- What is an SLA and what role does it play in IT service provision?
- What benefits does it provide for the service provider and customer?
- What are the main components of an SLA?
- How is SLA used to manage customer / service provider expectations?
- Why do businesses need clear and measurable SLA objectives?
- In what scenario would a customer be liable when there is a breach of SLA contract?
- What key performance metrics should be included as part of an SLAs?
- In what ways does SLA aid in service quality assurance?
- What is the best way to align SLA with IT strategy?



What are Service Level Agreements (SLAs)?

- A contract between service provider and end users.
- Specifies the level of service provision.
- Used to agree on an acceptable level of service provision.
- Matrix used to measure quality of services provided.
- Penalty implications when agreed service levels aren't achieved.

Examples of where SLA is used include:

- Cloud service provision.
- IT management services.
- In-house IT delivery services.



Types of SLA

There are three types of SLA which can be used based on an enterprise's requirements:

- Customer-based SLA.
- Service-based SLA.
- Multi-level SLA.



Customer-based SLAs: Discussion and Q&A

- What is customer-based SLA and how does it differ from other SLA types?
- Can you provide some example of customer-based SLA application?
- If you were to create a customer-based SLA, what would you include and how would you ensure it will meet customer expectations?
- What are some challenges business face when creating SLA for different customers?
- What is the best approach when creating SLA? Should customers be allowed input SLA terms? Why or why not?



Customer-based SLA

Applicable in an enterprise when dealing with individual customers, based on their level of support requirements.

Customer based SLA example:

An end user with support level SLA would include several services with the same contract. In other words:

- Troubleshooting network problems.
- Install and configure operating systems.
- Remote support.
- Application automation.



Service-based SLAs: Discussion and Q&A

- What is a service-based SLA, and can you provide some examples?
- What are the advantages and disadvantages of having a specific service SLA?
- In what ways do service-based SLAs help with cost efficiency?
- Which industry is most likely to use service-based SLAs, and why?
- What are the main key metrics used in service-based SLAs?



Service-based SLA

- The same type of service level is provided to an organisation.
- Convenient if all users require the same level of service.
- Additional charges are incurred if services provided are not within the contract scope.

Service based SLA example:

A business with service SLA may require server and network security services which is included in the SLA. But the business will be billed additionally if users require services outside the scope of the SLA (in other words, network installation and troubleshooting which is outside the SLA).



Multi-level SLAs: Discussion and Q&A

- How do multi-level SLAs differ from customer-based and service-based SLAs?
- Can you provide some examples where a multi-level SLA can be implemented?
- Can anyone tell us the three types of a multi-level SLA?
- In what scenario would you recommend a multi-level SLA?
- Should a multi-level SLA approach be the standard practice when buying services? Why or why not?



Multi-level SLA

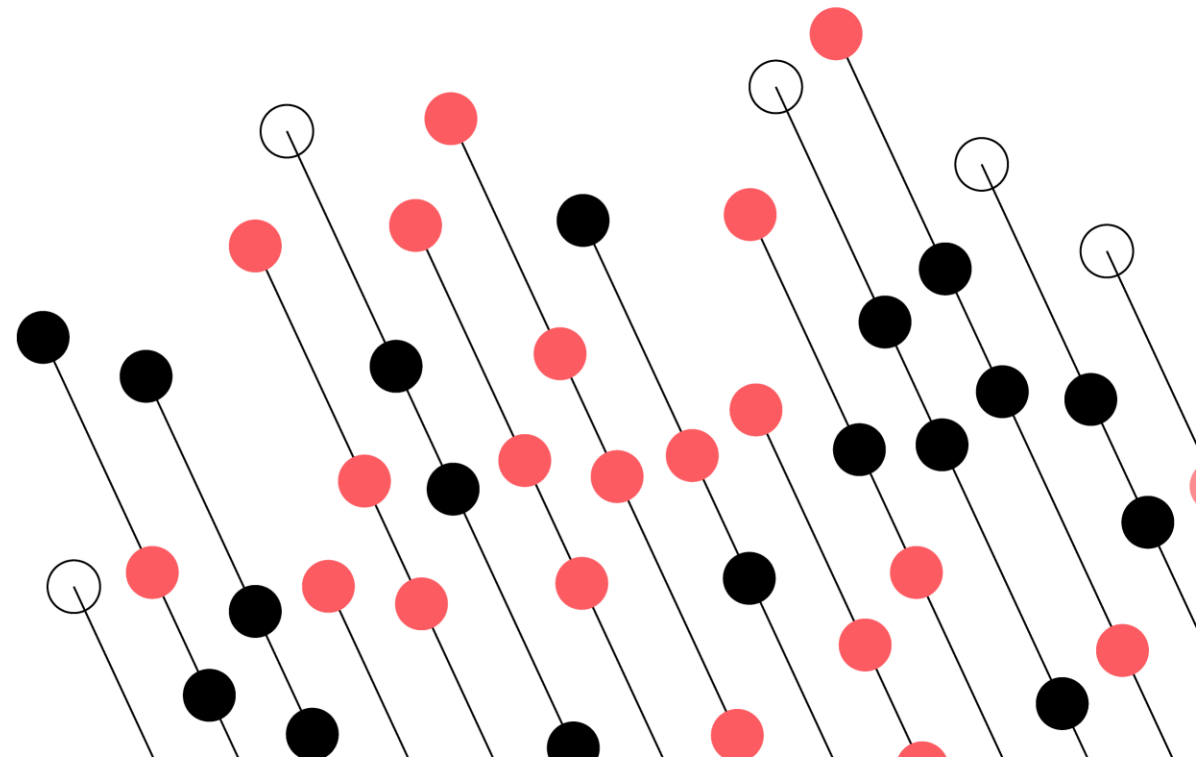
- This contract type is defined to cover three distinct areas.
- Each specific area addresses different sets of needs with the contract.
- This contract provide flexibility for the enterprise to integrate different aspect of the business needs into the same contract.

MLS sub-contract include the following levels:

- Corporate level.
- Customer level.
- Service level.



SLA challenges



SLA challenges: Discussion and Q&A

- What are some challenges businesses face when creating SLA?
- What measures should be taken to make sure SLA terms are realistic and achievable?
- What measures should be in place to balance customer requirements with service provider capabilities?
- Are there any legal or compliance challenges when drafting an SLA?
- What are the risks and implications when SLAs expectations are set too high or too low?
- What should businesses do when they encounter unexpected service disruptions not covered in an SLA?



Challenges of creating and implementing SLA

- Difficulty in in setting measurable metrics
- Inadequate response time
- Defining a realistic expectations between customer and service provider
- Meeting different client needs whilst maintaining a consistent SLA structure
- Legal and specific industry regulatory compliance requirements
- Updating SLA to meet constant evolving technologies and services
- Monitoring and measuring SLA performance
- Managing SLA violations (penalties, corrective actions)
- Adequate customer dispute resolution



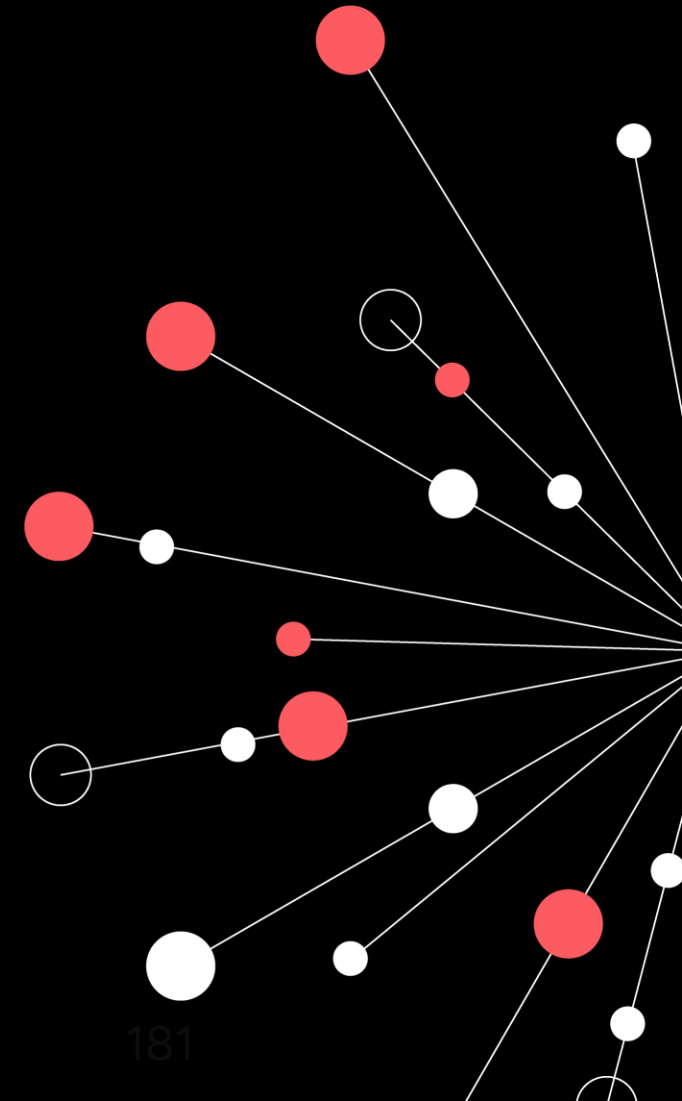
Summary and case studies: Reflection

- Your business guarantees 99.9% uptime, but customers are experiencing frequent downtime. How should your business respond?
- Your business has multiple customers who require different levels of support. Should you create customer-based or service-based SLA? Why?
- You have just discovered that your business hasn't specified any response time guarantee in their SLA. What are the potential risks to both parties (customer / business)?
- Your company's SLA metrics are outdated and do not reflect current service capabilities. What steps should you take to address this issue?

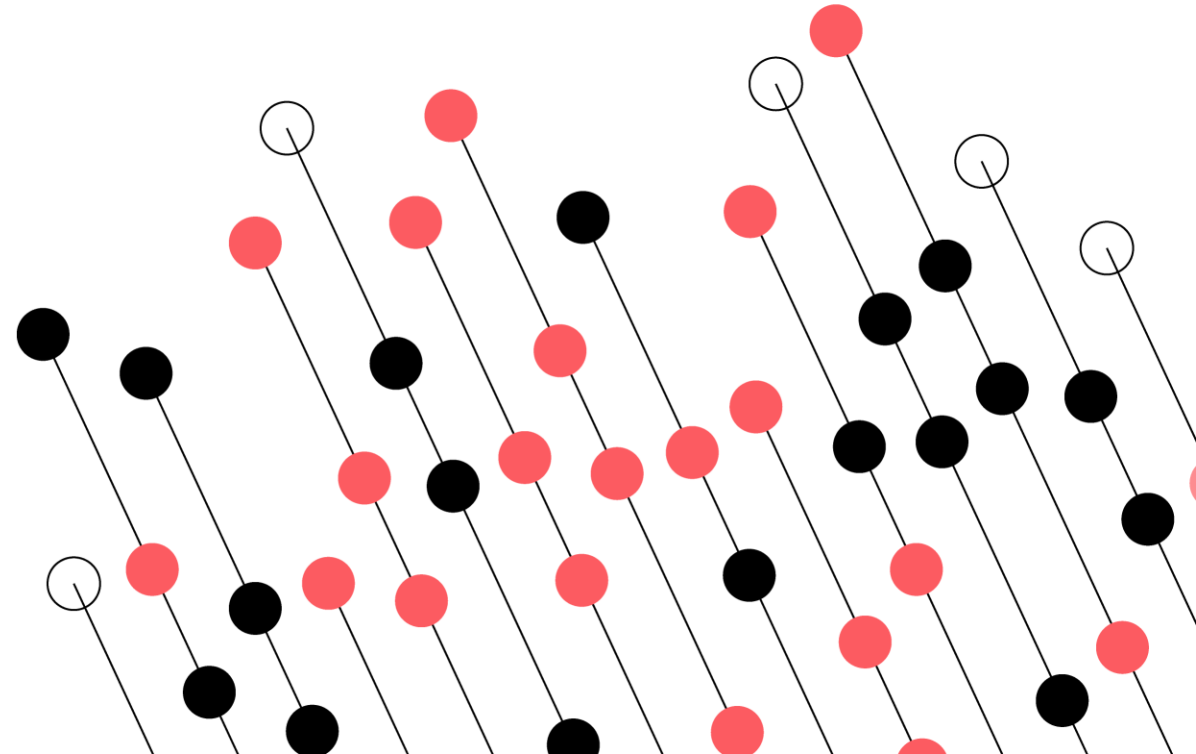


Activity 14: Service Level Agreement

Find the activity in your **Learner Activity Document** and read the instructions.



Organisational policies and procedures to ensure accurate recording and management of information



Organisational policies and procedures: Discussions and Q&A

- What are organisational policies and procedures, and why are they critical as part of information management?
- How do businesses use policies and procedures to ensure consistency and accuracy in information recording?
- Can you provide examples of policies and procedures used in your workplace?
- What key factors should be considered when creating and implementing policies to manage information effectively?
- How do business procedures contribute to accurate data recording and asset management?
- Why is it important to have clearly and concise policies for information handling?
- Can you give some examples of poorly defined policies and how they impact business operations?



Organisational policies and procedures

Policies:

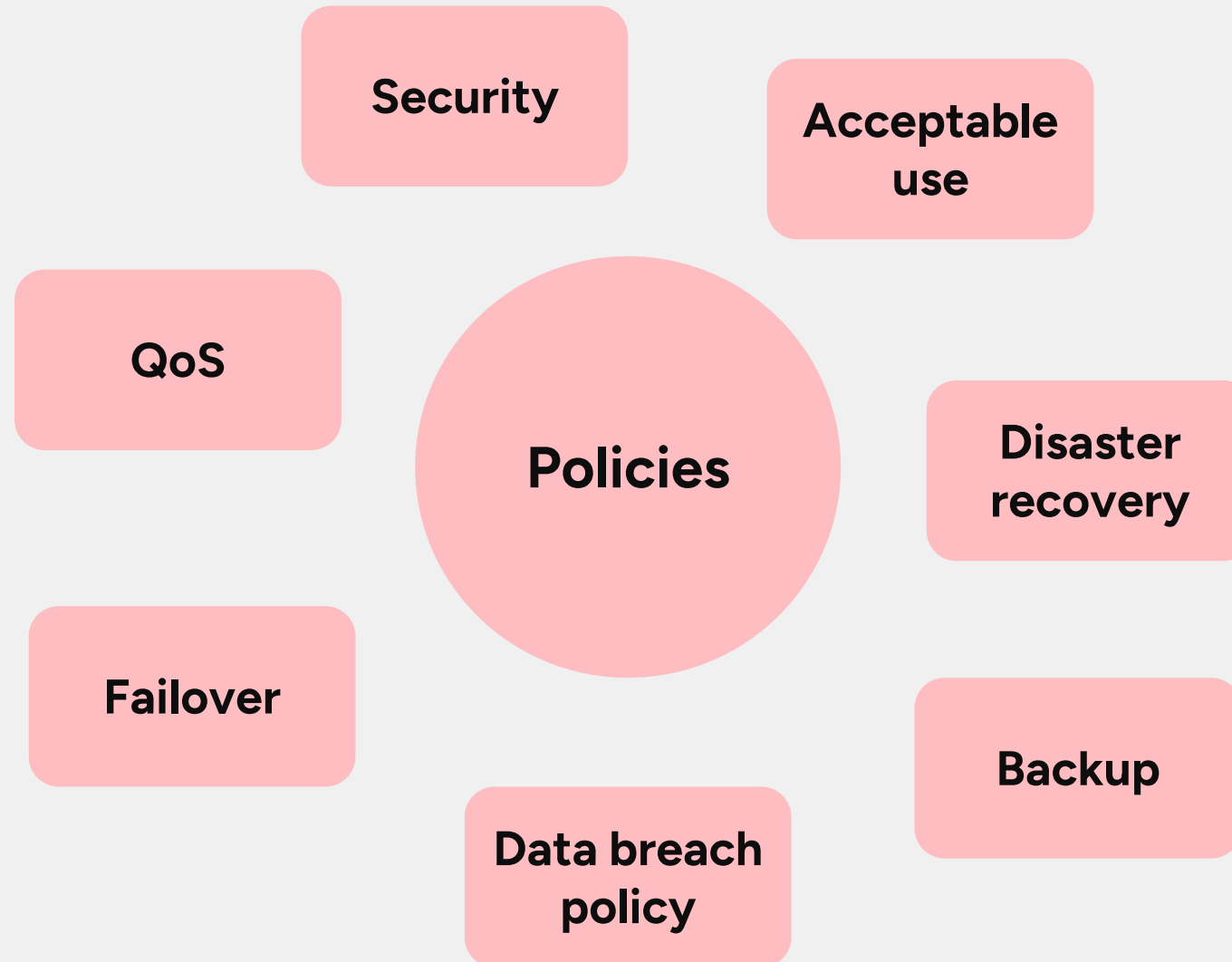
- Statements defined and produced by management.
- Written to comply with the business strategic objectives.
- Aids how organisation functions.
- Each policy will address a specific area of the business.

Procedures:

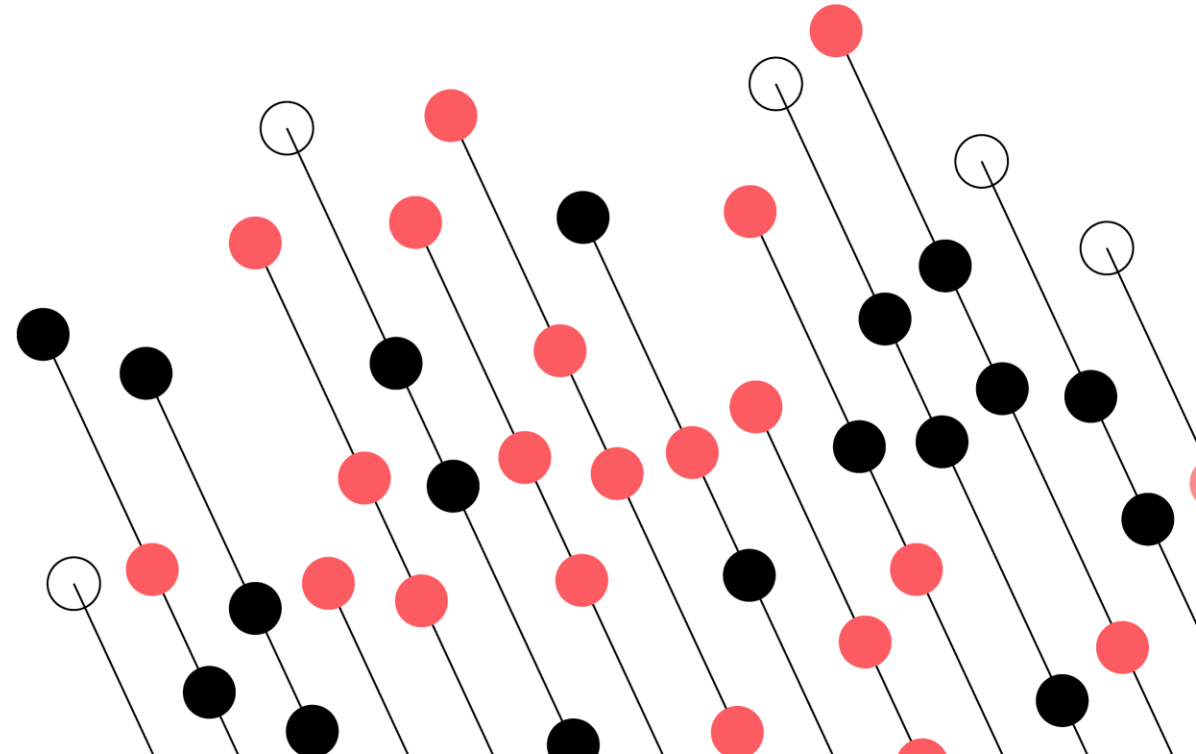
- A step-by-step instructions on how to carry specific task.
- Specify how to comply with policy.
- Well structured steps to align with organisation objectives.



Types of policies: Discussion and Q&A



Ensuring accurate recording and information management



How to ensuring accurate recording of information:

Discussion and Q&A

- What common mistakes can be made when recording information (customer, details, system logs) and how can these be prevented?
- What measures should be in place to ensure accuracy and integrity of records?
- What risks can businesses and customer face because of inaccurate information recording? What are the potential impact to customer and businesses?
- Can automation be used in improving business data accuracy? How?
- What training would you put in place for employees to improve data recording practice?
- What roll does version control play in managing records?
- What should be added in a policy to ensure that data recording and updates are consistent and accurate?
- What measures should be implemented to detect error detection during data entry?

How to ensuring accurate recording of information

- Have a standardised approach to documenting.
- Create predefined templates for information recording consistency.
- Use automation in collating and analysing data for accuracy.
- Regular validation and audit control to ensure integrity.
- Restrict access and modifications to authorised users only.
- Awareness user training on proper information recording and compliance.



Information management and compliance: Discussions and Q&A

- Are there any laws and regulations that provide guidelines on how businesses record and manage information?
- How do these regulations impact how businesses record and process data?
- What measures should be in place to ensure the security of data when it is **in transit, at rest, and in use**?
- What consequences do businesses face for non-compliance with information management regulations?
- What role do policies and procedures play in ensuring data confidentiality and security?
- If you were in charge of data recording, what measures would you put in place to ensure compliance with legal and regulatory requirements?
- How should businesses design their access control and data permissions to ensure data accuracy and integrity?



Information management and compliance

- Use regulatory frameworks for information management (GDPR, ISO, DPA).
- Keep logs and access audit for accountability.
- Ensure encryption for sensitive information and use principles of least privilege.
- Align information recording with business policies and procedures.
- Engage in regular audits to identify non-compliance.



Challenges in recording and managing information:

Discussions and Q&A

- What are some challenges businesses face in making sure data is accurately recorded and managed?
- What are the challenges when businesses handle large volumes of data while ensuring effective access control data accuracy?
- How does a hybrid approach to work impact information recording policies?
- What challenges can arise when ensuring data consistency across multiple systems?
- How does data classification help in effective data management? How does this remove some challenges in managing data?



Challenges in recording and managing information

- Incorrect data entry which can lead to compliance risks.
- Parsing through huge amount of data can be overwhelming.
- Protecting data distributed across different systems.
- Inconsistent data gathering and maintenance standard.
- The need to comply with different regulations or industry specific regulations.



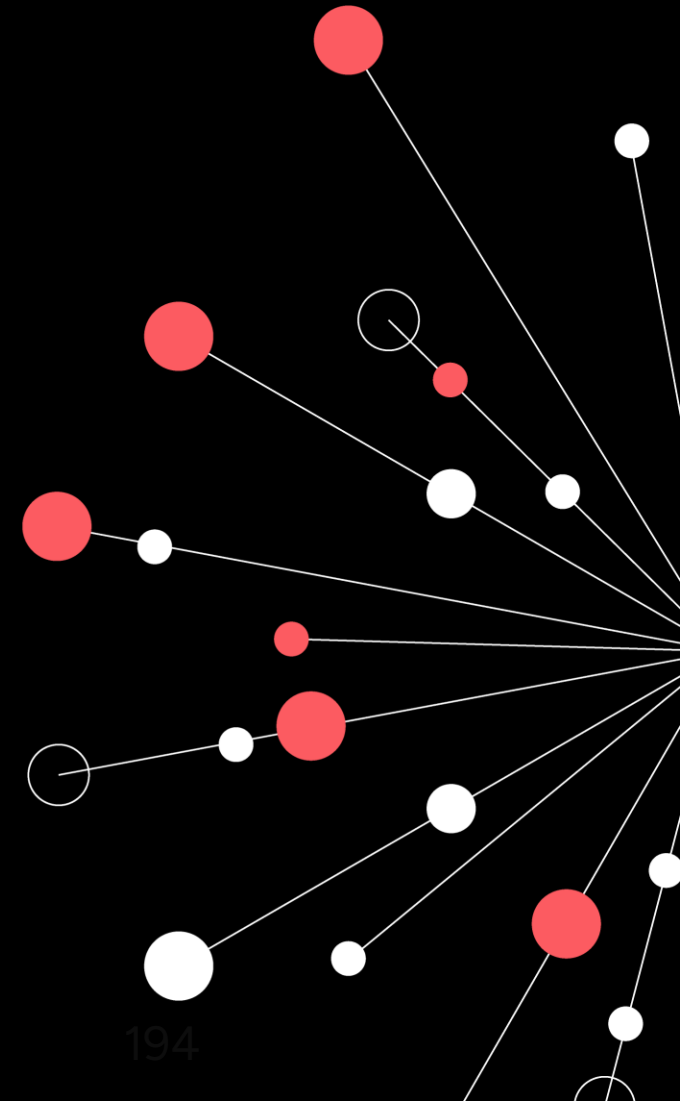
Summary and case studies: Reflection

- You have discovered that your business employees are not following procedures for accurate data entry, which is causing critical errors. What measures should be implemented to resolve this issue?
- Your business is transitioning from an on-prem to cloud-based information recording system. How should you update the business policies to reflect these changes?
- One of the first-line support agents has accidentally deleted critical record. What mitigating strategy should have been in place to prevent this from occurring and how should the business respond?

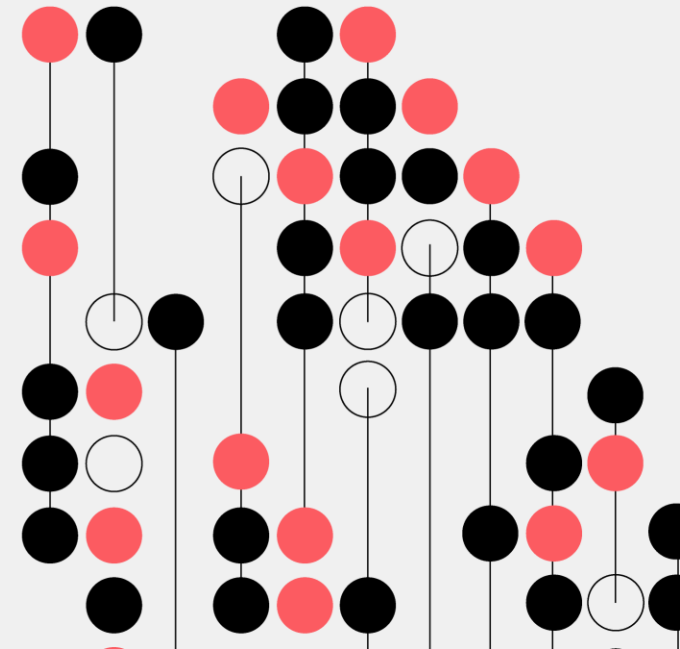


Activity 15: Policy and procedure

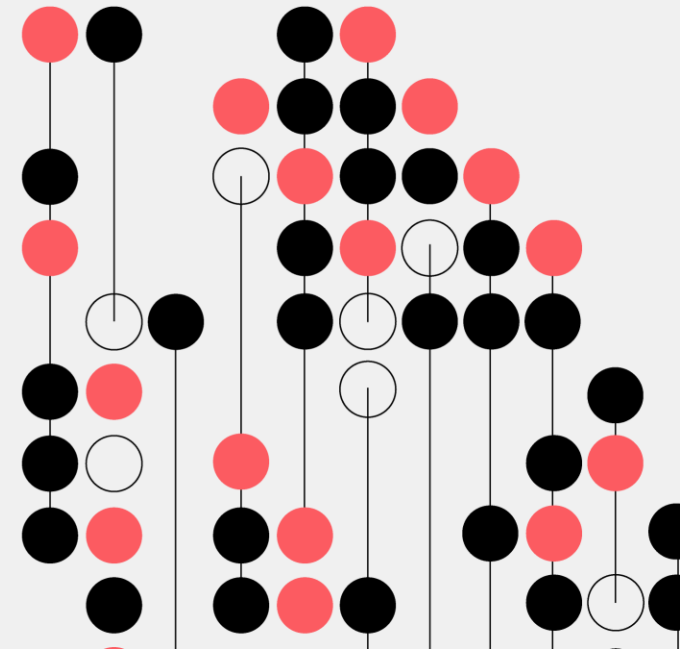
Find the activity in your **Learner Activity Document** and read the instructions.



End of Day 4



Day 5: Business continuity



Objectives: Day 5

- Describe business continuity and disaster recovery principles.
- Recommend best practices for system configuration preservation.
- Explain the role of network engineers in disaster planning.
- Analyse a given disaster recovery scenario and recommend appropriate solutions.



Business continuity planning: Discussion and Q&A

- What is business continuity, and why is it for businesses to have continuity planning in place?
- How does business continuity planning (BCP) minimise downtime during disruptions?
- What are some factors to consider in putting effective continuity plan in place?
- What is the difference between business continuity and disaster recovery?
- Why is it critical for businesses to test their business continuity on a regular basis?
- What types of events can cause disruptions and impact business continuity?

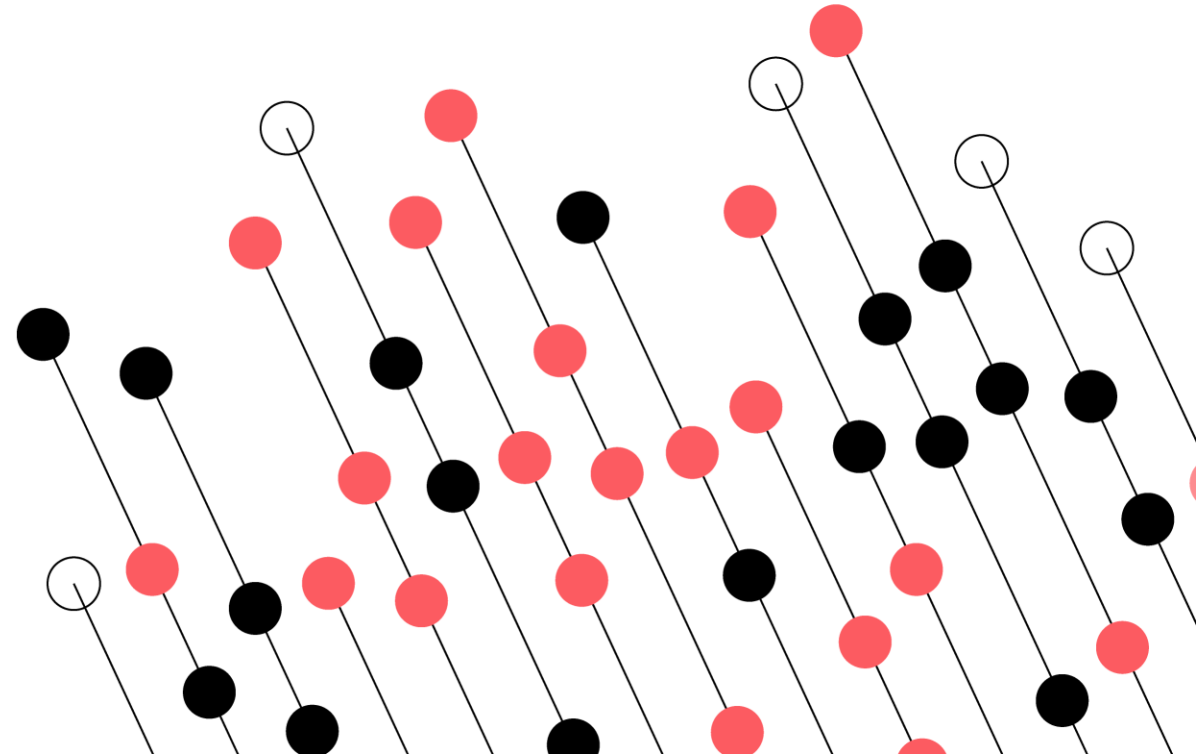


Business continuity plan (BCP)

- The business continuity plan (BCP) considers all aspects that are affected by a disaster (systems, personnel, and facilities).
- It identifies and prioritises the services that are needed, particularly the telecommunications and IT functions.
- Identify the process, functions, and hardware devices that should be addressed in the DRP.
- BCP deals with identifying the impact of any disaster.
- It ensures that there is a viable recovery plan.



Disaster recovery



Disaster recovery planning (DRP): Discussion and Q&A

- What is disaster recovery and how does it help businesses recover from a disaster?
- What are the key factors to consider as part of a disaster recovery plan?
- What are the differences between cold, warm, and hot disaster recovery sites?
- How do businesses decide on the best recovery sites? What criteria would they use in selecting a recovery site?
- Why are backups critical part of disaster recovery planning?
- What implication does cost have in disaster recovery planning?
- Should businesses prioritise cloud solution as part of their disaster recovery plan? Why?
- What role does BIA play in disaster recovery planning?



What is disaster recovery planning?

- Disaster recovery is a measure put in place to protect against disaster.
- Measures go beyond just back and recovery.
- Identify critical systems that needs to be duplicated.
- Business continuity is the primary objective.
- Identify sequence for recovery in the event of a disaster.



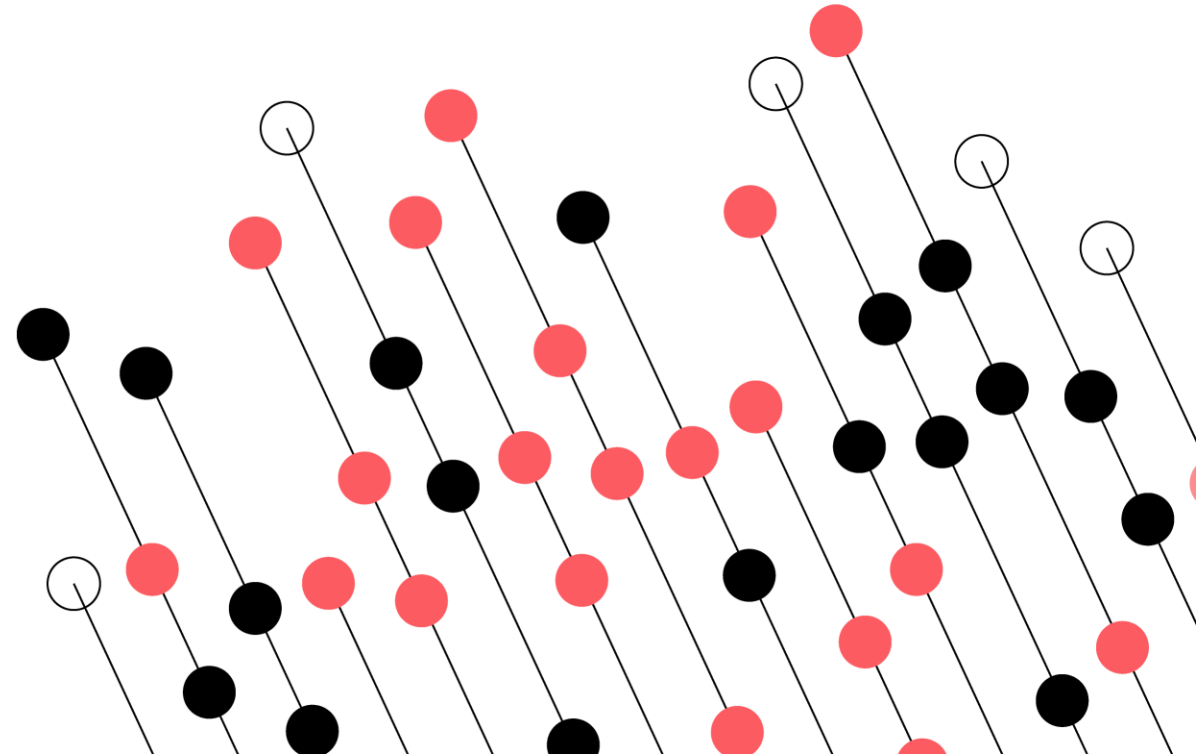
Factors that can lead to system / network failures

There are some factors that can cause system / network failure, including:

- File corruption.
- Malicious intent.
- Storage failures.
- User errors.
- Lack of security.



Network engineer role in disaster recovery

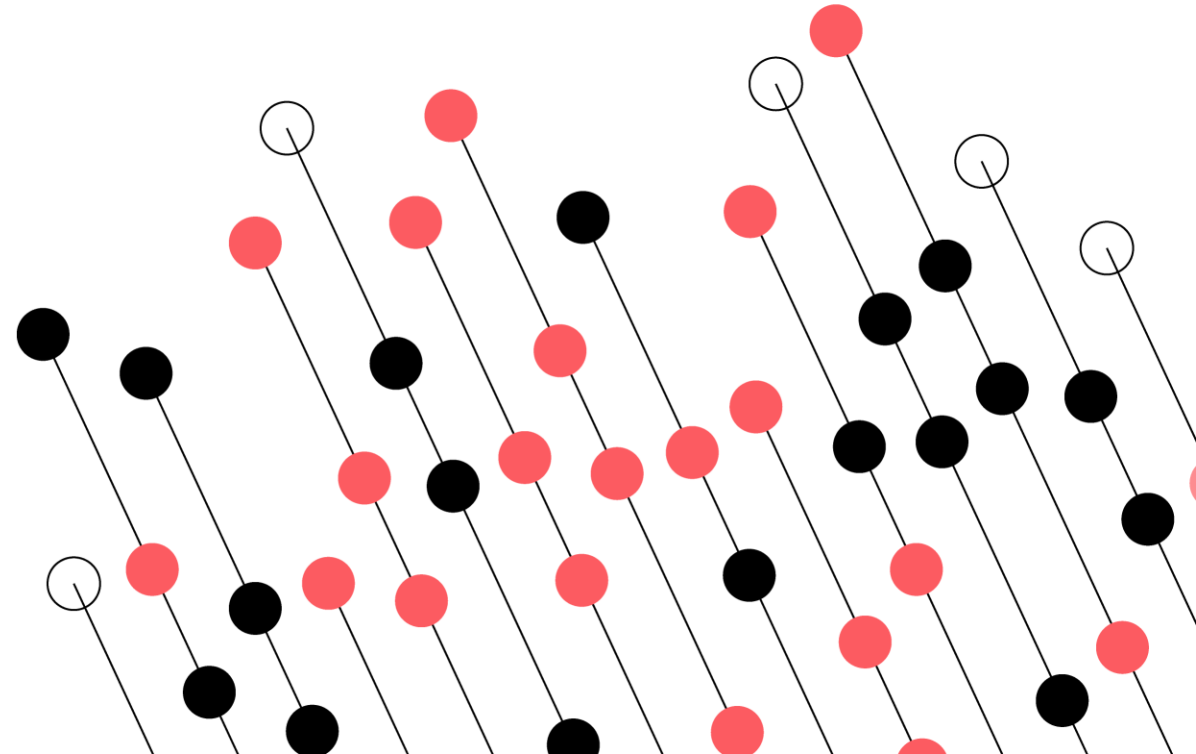


Network engineer role in disaster recovery: Discussion and Q&A

- What are the responsibilities of a network engineer in disaster recovery planning?
- What measures do network engineers take to minimise system / network downtime during failures?
- What proactive measures should be in place for monitoring systems / network health to prevent failures?
- What tools can network engineers use to detect and respond to network failures?
- What role should a network engineers play in creating and testing disaster recovery and business continuity plans?
- What skills are required by network engineer to be able to participate recovery operations?
- What role does incident response play in disaster recovery ?



Predicting disaster impact



Business Impact Analysis (BIA)

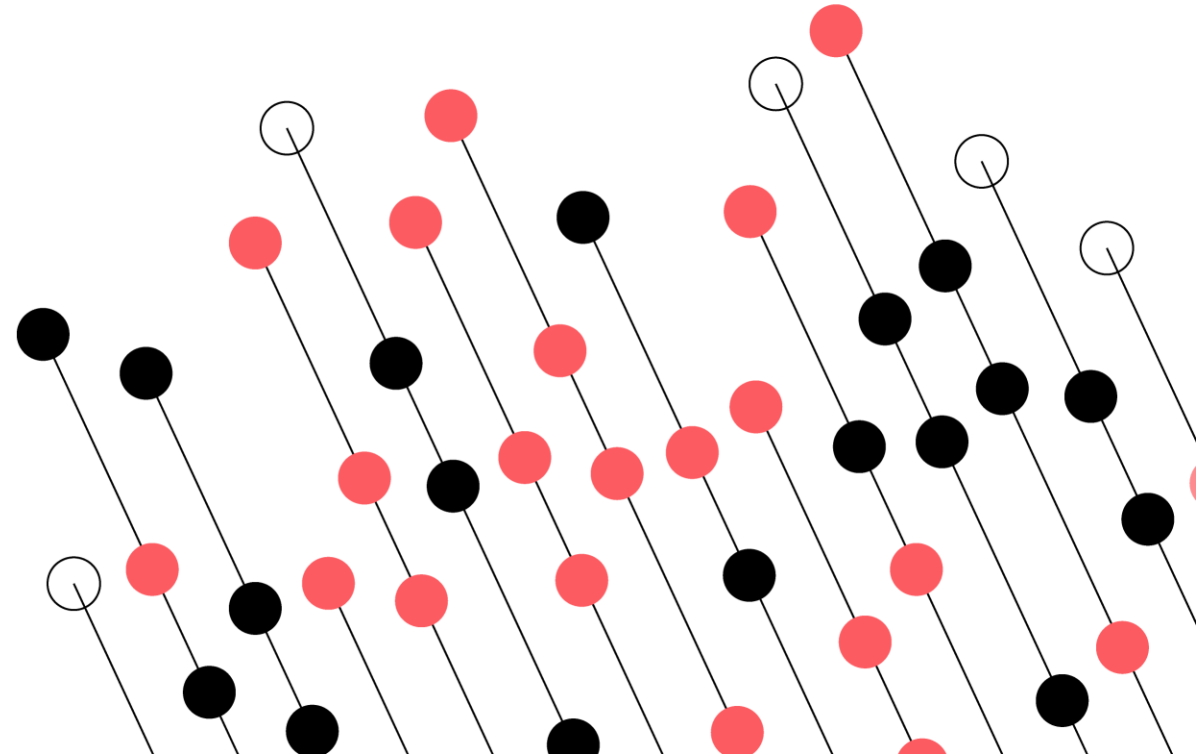
- BIA is part of business continuity and disaster recovery.
- A thorough BIA will help business units understand the impact of a disaster.
- The resulting document that is produced lists the critical and necessary business functions and their criticality.
- Questions are around:
 - Who is affected.
 - What is affected.
 - The severity of impact.



Business continuity and disaster recover measurement matrix

Business Impact Analysis (BIA)	Recovery Point Objective (RPO)	Recovery Time Objective (RTO)	Maximum Tolerable Period Of Disruption (MTPOD)	Mean Time Between Failure (MTBF)
• Risk management function • How could the disasters that are highlighted impact business function?	• Maximum acceptable level of data loss measured in time • Age of the files or data in backup storage required to resume normal operations	• Maximum time allowed for disaster recovery to take place	• The point at which the system / data has degraded to the point where it is not recoverable	• Method to measure how reliable a device is • (MTTR)

Disaster recovery sites



Disaster recovery sites

Hot sites:

- Are a mirror of the production environment.
- Provide real time failover strategy.
- Have the least minimum transition downtime.
- May be in a different geographical region.

Warm sites:

- Include some necessary equipment required for recovery.
- Require some additional configuration before a complete recovery.
- Are suitable when the business can handle some recovery downtime.

Cold sites:

- Require more configurations for the site to go live.
- Are not ideal where recovery downtime can't be tolerated.



Processes for disaster recovery planning

- Form a recovery team with multiple job roles and responsibilities.
- Carry out risk and cost assessment (residual risks).
- Ensure the DRP includes alternate sites, asset inventory, backup procedures, and other critical information.
- Train staff in disaster recovery and change management.
- Identify notification channels.
- Consider maintaining alternate recovery sites to quickly restore operations when the main site is compromised.

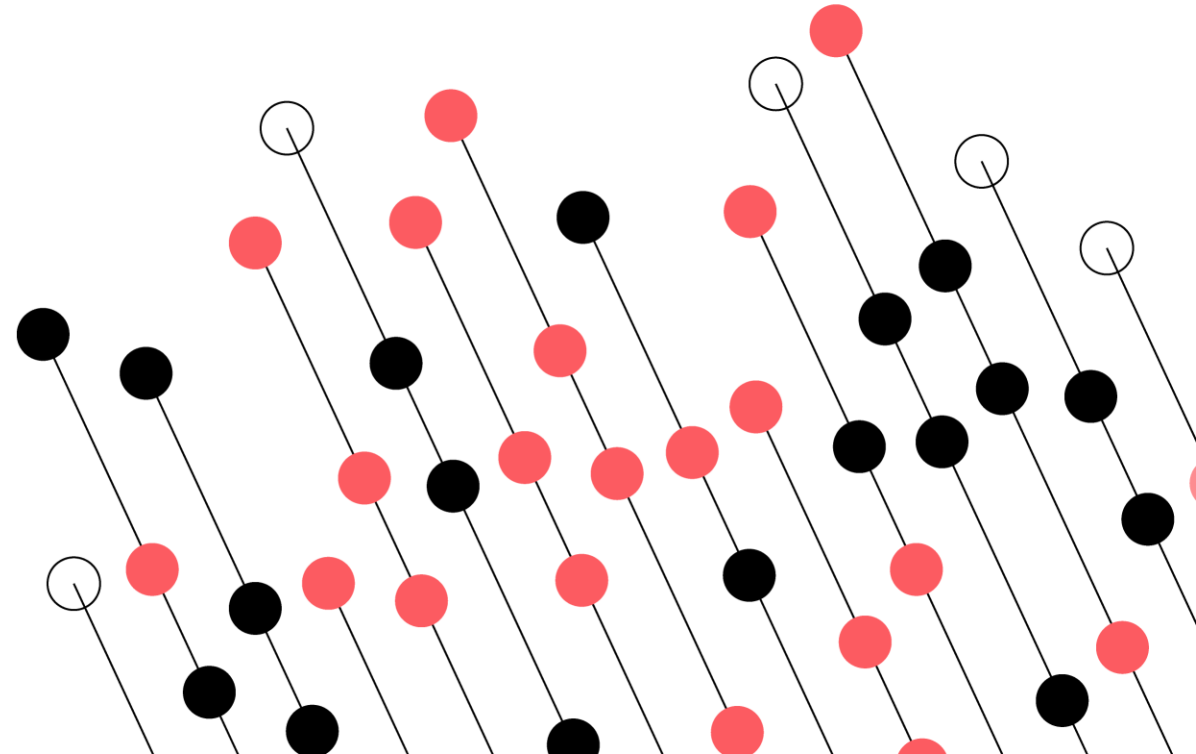


DRP procedures

- **Scope:** This include the potential causes of damages to systems.
- **Evaluation:** An in-depth analysis to identify the impact of damages to the system and how to mitigate and have a recovery action in place.
- **Staging:** Ensure equipment and staff are adequate during the recovery process.
- **Restore:** Include steps on how to recover and restore systems and applications to their functioning state.
- **Check:** A continuous process of reviewing the recovery process and make any adjustment where necessary.



Testing disaster recovery



Testing DRP

Why?

- Determine if plan can work
- Determine if plan meets organisational requirements (RPO / RTO)
- Provides feedback to improve plan
- Helps staff in emergency situations

When?

- Regularly
 - Yearly / once every 2-3 years
 - More often is better
- In response to significant change

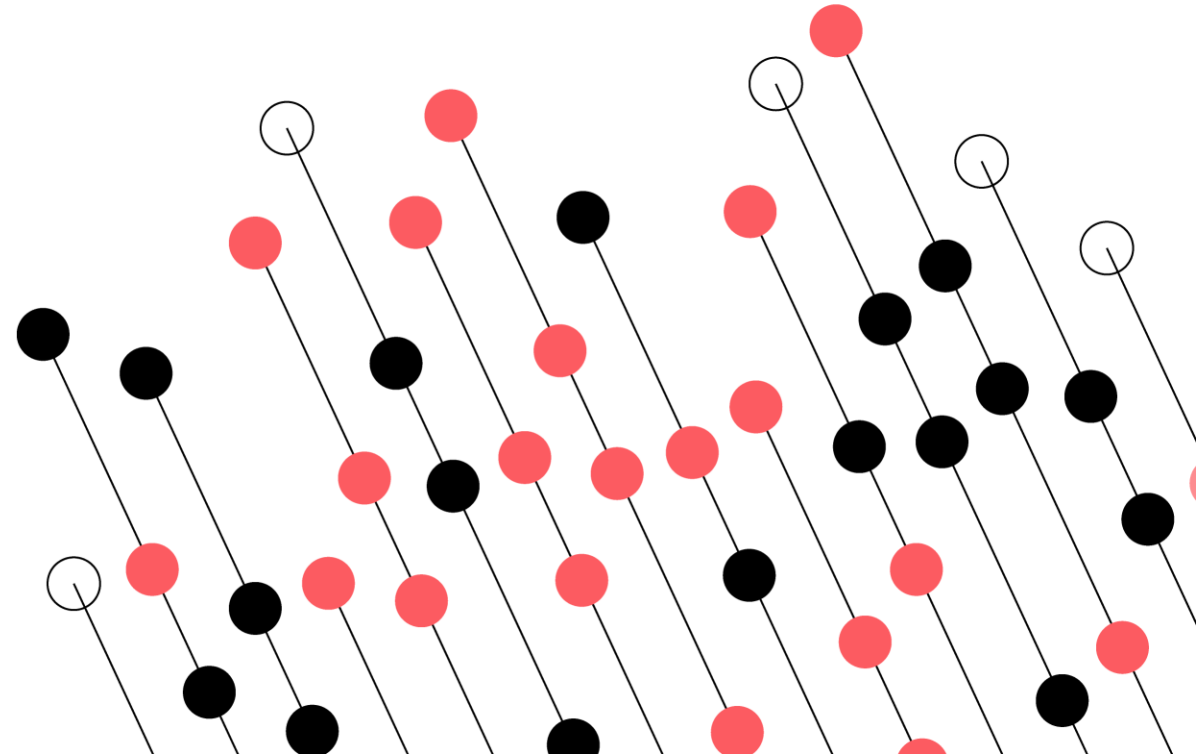


Types of DRP testing

- **Paper test:** Individuals read and annotate recovery plans.
- **Walkthrough test:** Groups walk through plans to identify issues and changes.
- **Simulation:** Groups go through a simulated disaster to identify whether emergency response plans are adequate.
- **Parallel test:** Recovery systems are built / set up and tested to see if they can perform actual business transactions to support key processes. Primary systems still carry the full production workload.



System and configuration preservation



Preserving system configuration through backups:

Discussion and Q&A

- What role does system configuration preservation play in disaster recovery?
- What other measures should be put in place to prevent data loss and corruption during system failure?
- What best practices should be in place for backing up and restoring network configurations / data?
- Would you recommend automation and configuration management in preserving system configurations? Why?
- What measures can businesses take to ensure that system configurations remain secure and up-to-date?
- What issues can businesses face in preserving system configurations across multiple systems?



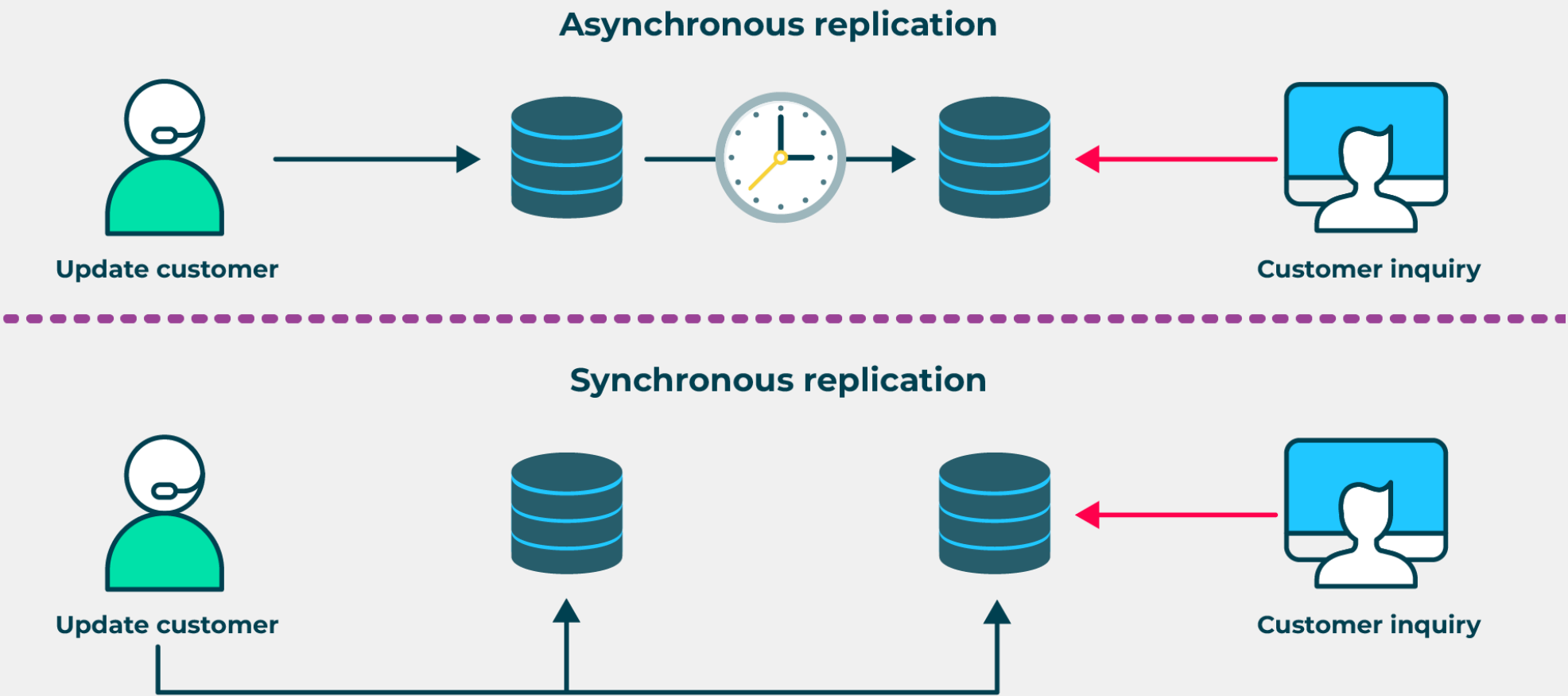
Replication and backup

- Replication and backup are primary measures are put in place to protect data.
- Measures make it easy to recover after storage failure.
- Backup is slightly different from replication.
- Backup stores a copy of data in a different location.
- Replication on the other hand stores multiple copies of data across a distributed system / locations.
- These measures can be put in place to preserve data or system configurations.



Replication methods

Replication of data is done in real time using any of these methods:



Replication types

Disk to disk replication:

- Writes data from one storage to another.
- Is similar to normal backup.
- Happens in real time.

Server based replication includes these types:

- Replication is done on the same server but to different volume.
- Have a disk cluster replicate to another cluster.
- Server to server replication in a synchronous or asynchronous method (Site to Site).



Backup methods

Data can be backed up in several ways:

- Full backup
- Incremental
- Differential
- Snapshot
- Selective backup
- Bare metal backup
- Open file backup



Types of backup strategies

Full

- Backs up all data.
- Easy to restore and manage.

Incremental

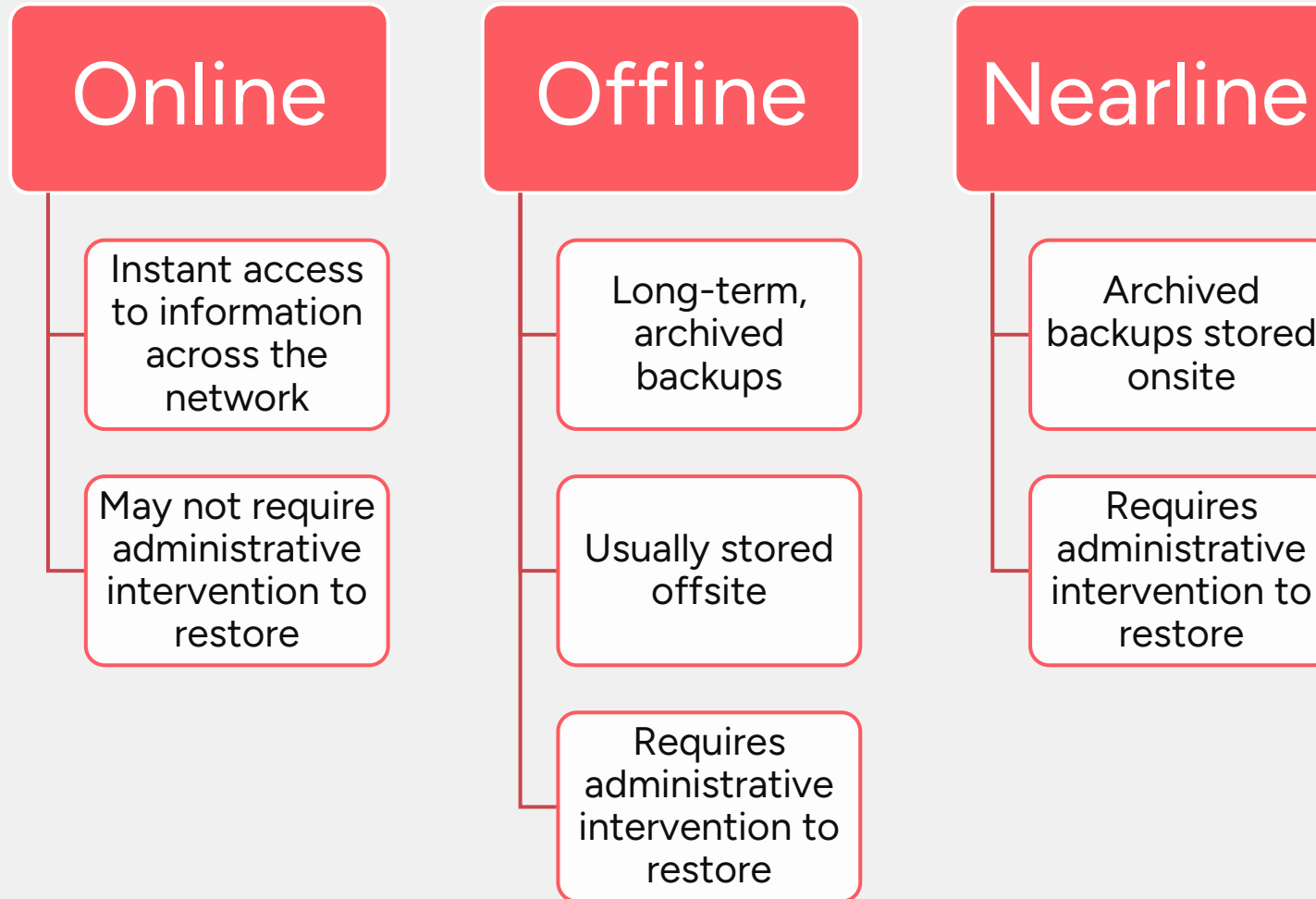
- Backs up changes since last backup.
- Quicker to back up.
- Harder / longer to restore.

Differential

- Backs up changes since last full backup.
- Quicker to backup full image.
- Easier to restore than incremental.

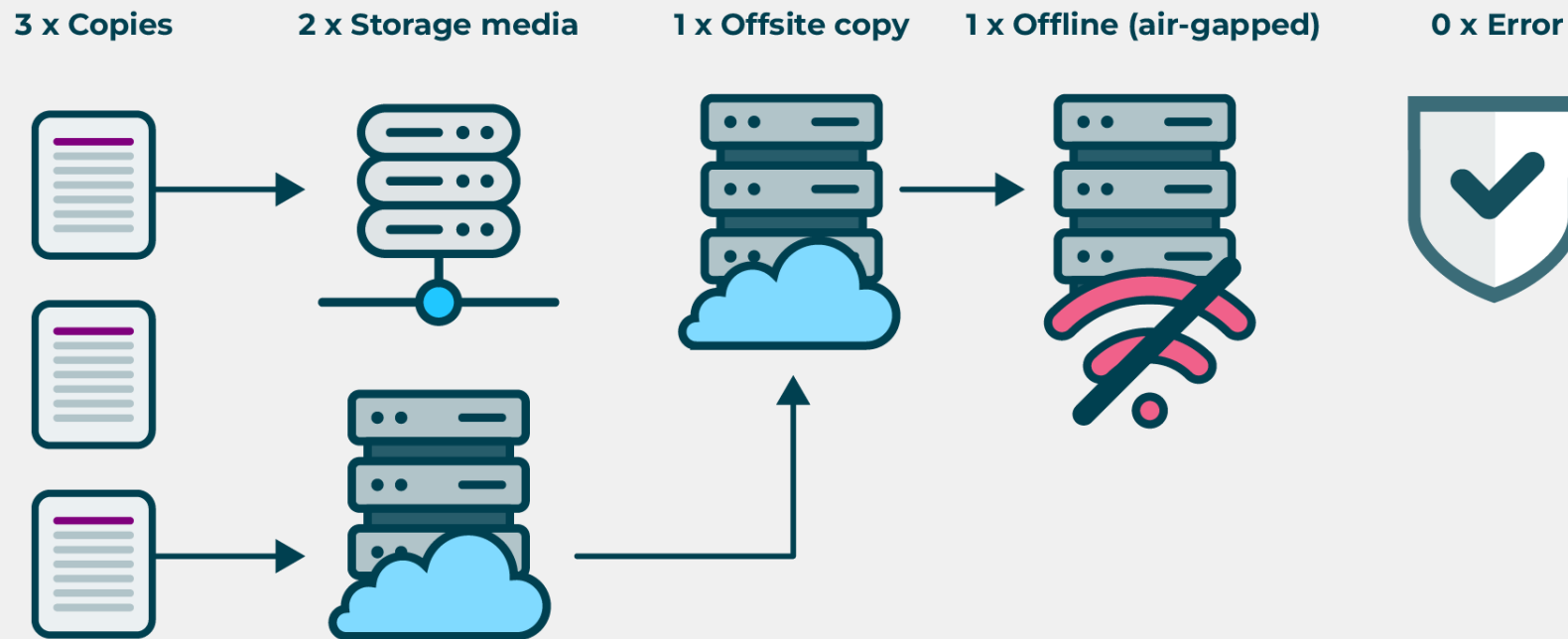


Types of backup location

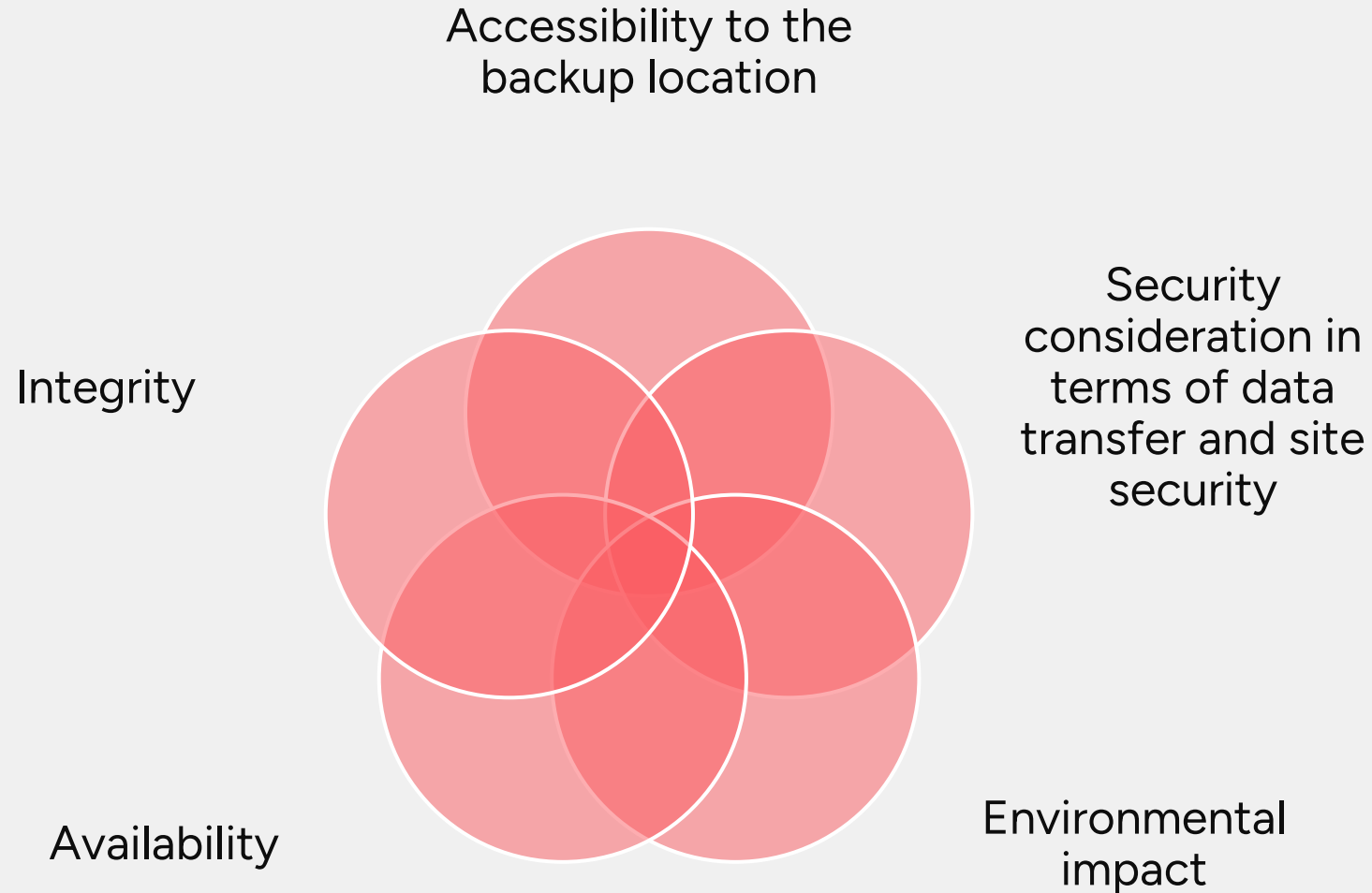


3-2-1 backup rule

- Keep at least **three** copies of the data.
- Store the backups on **two** different media types.
- Keep at least **one** copy of the data offsite.



Organisation backup requirements



Backup and restore testing

Why?

- Validate integrity of backup.
- Prevent permanent loss of data.

How?

- Create documented test plan.
- Carry out routine test restores.
- Test frequently and regularly.
- Automate where possible.
- Check accuracy of restored files.
- Keep redundant backups.

Challenges in business continuity and disaster recovery planning: Discussion and Q&A

- What do you think are some challenges businesses will face when designing disaster recovery plans?
- Why do some businesses DRP fail after a system / network failure?
- What different constraints can businesses face when creating disaster recovery and business continuity planning?
- What are the risk implications of not having adequate DRP / BCP testing strategy in place?
- How would businesses address cybersecurity incidents when planning DR and BCP
- Is it possible to achieve a zero downtime in businesses? How?



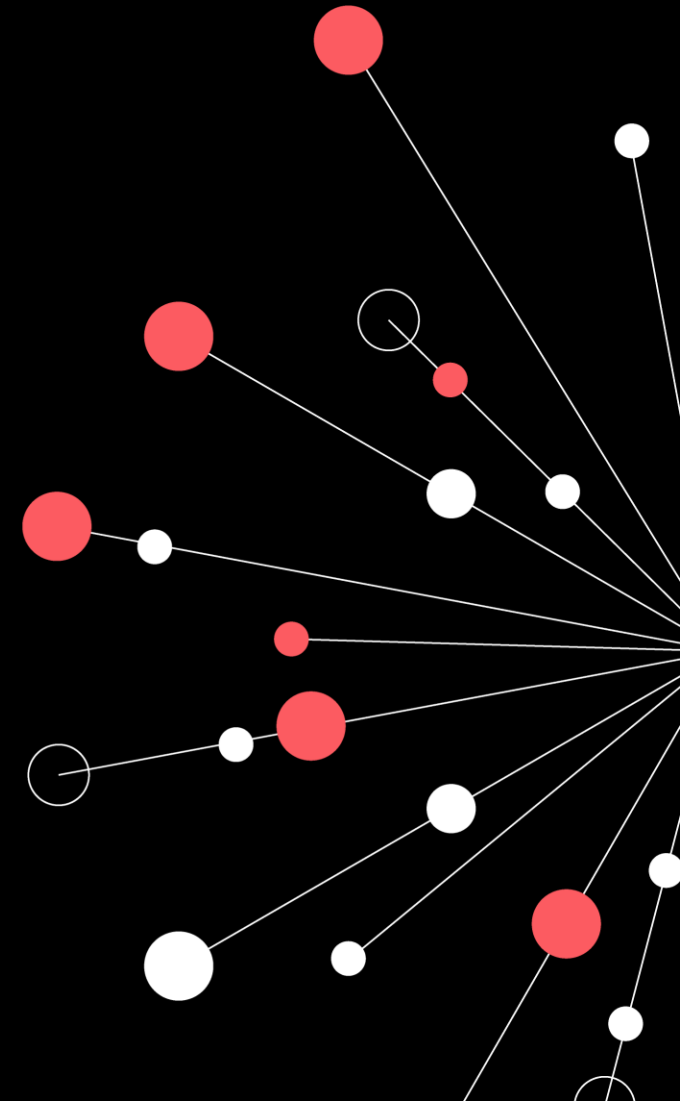
Summary and case studies: Reflection

- A company experiences a ransomware attack that locks critical business files. How should they respond?
- A major cloud service provider suffers an outage affecting thousands of businesses. What steps should the affected companies take?
- A small business faces data loss due to hardware failure. What disaster recovery strategies should they implement?
- A global corporation must recover from a natural disaster that destroys a data center. What should be their recovery priorities?
- A business finds that their backup system failed when trying to restore operations after a disaster. How could this have been prevented?



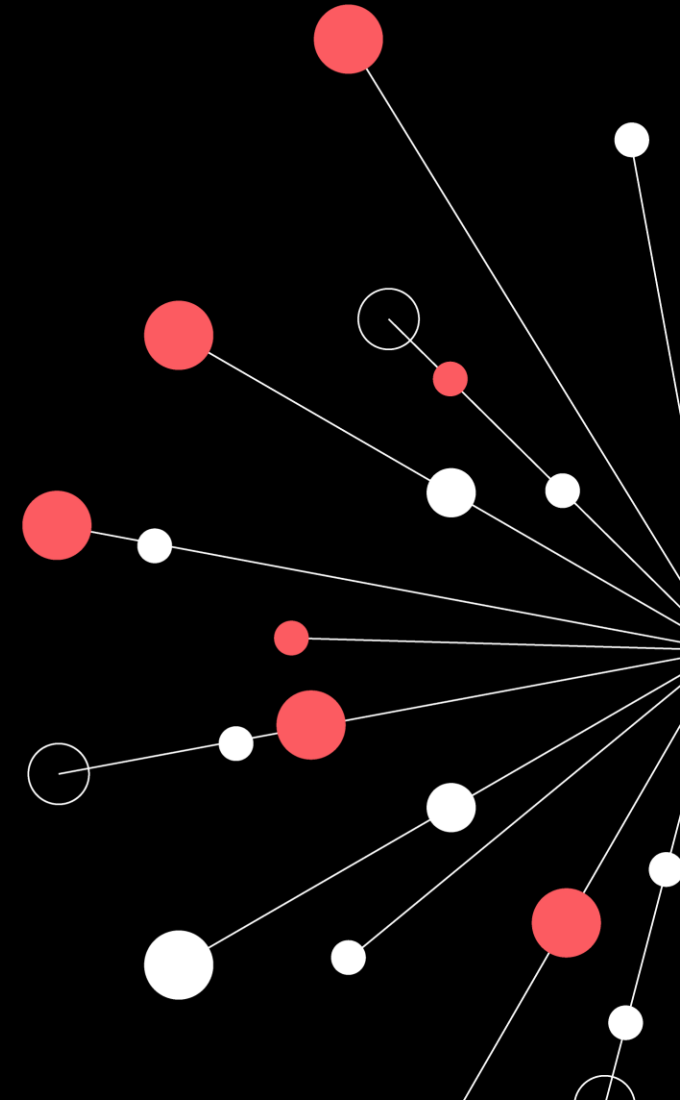
Activity 16: Server backup

Find the activity in your **Learner Activity Document** and read the instructions.



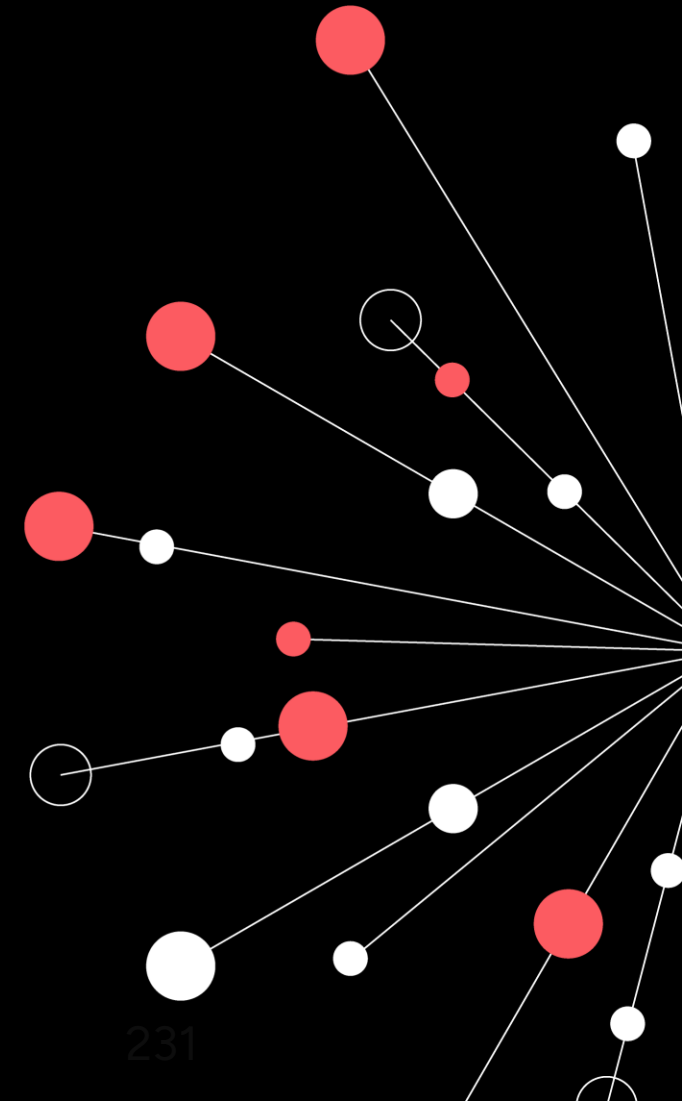
Activity 17: Client remote backup

Find the activity in your **Learner Activity Document** and read the instructions.

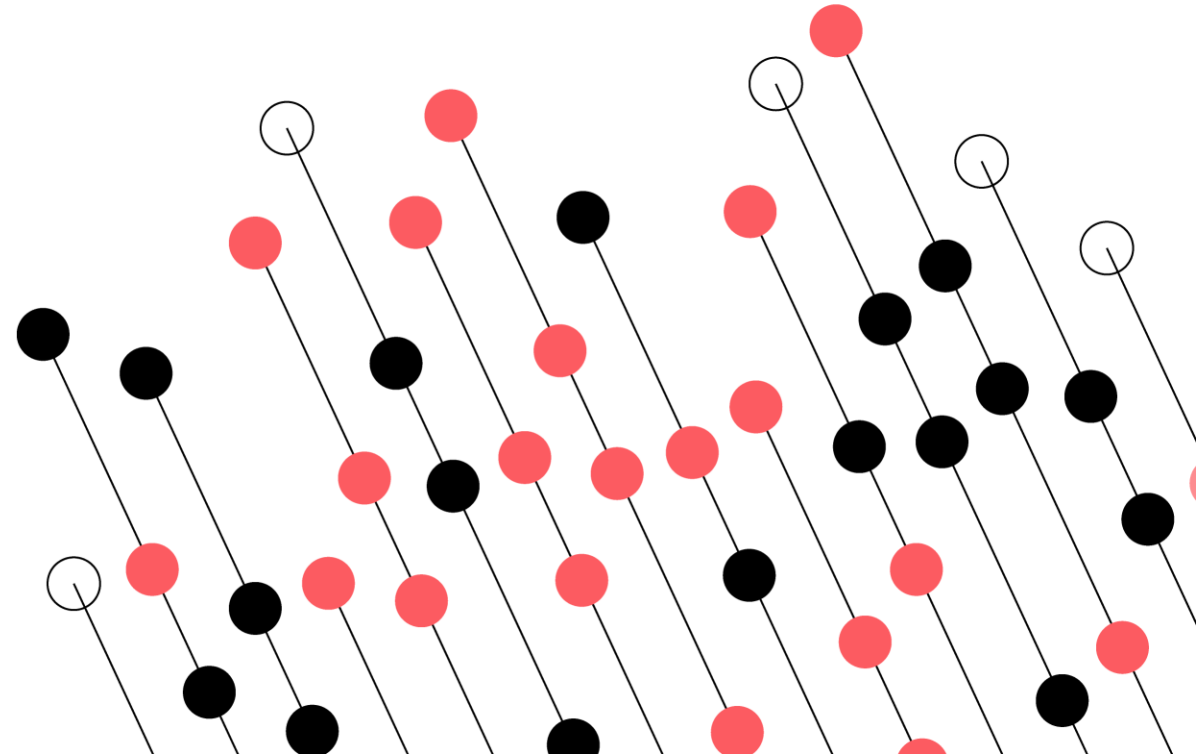


Activity 18: Disaster Recovery Planning (DRP)

Find the activity in your **Learner Activity Document** and read the instructions.



Approaches to change management in a network environment



Lesson objectives

- Explain at least one change management approach in network environments.
- Describe the importance of risk assessment, planning, and approval processes.
- Identify some challenges in relation to network change management.
- Apply change management best practices in a given case scenario.



Change management: Discussion and Q&A

- What is change management in relation to making changes in a network environments?
- Why should businesses use change management for maintaining a stable network?
- What are some of the potential risks of making changes without going through change management strategy?
- In what ways does change management reduce downtime and improve efficiency?



What is change management?

- Align changes to business strategy
- Systematic approach service transition from current state to desired state
- Better approach to formalized control
- Holistic method to implement changes to business internal and external processes
- Transformation of business goal with minimal risks and disruptions

Allow businesses to identify:

- Planning stages.
- When to schedule the change.
- How implement change.
- Potential impact.

Reasons for change management

- Reduce the impact of change to business operations.
- Provides seamless transition between systems.
- Integrate newer technologies.
- Provide enhance communication between stakeholders.
- Improve deployment efficiencies.
- Aligns changes to business objectives.
- Identify and mitigate risk associated with system / network changes.



Approaches to change management: Discussions and Q&A

- What steps should be implemented for effective change management?
- What is the best approach to change management (proactive or reactive approach)?
- If you were in-charge of a network, how would you balance rapid change requirements whilst ensuring network stability and security?
- Should agile approach be adapted into change management processes?
- How do you define approval process to reduce impact of downtime when implementing changes in a network?
- What best practices would you recommend when creating a change request document?
-



Risks and challenges in change management: Discussion and Q&A

- What do you think are some challenges businesses will face when implementing changes?
- What measures would you recommend to minimise risks associated with network changes?
- What role does backup / rollback play in change management, and why is it a critical part of change management?
- What should businesses do if an urgent change is required (except change management)?
- How can businesses quantify the success of change management process?



SASE in change management

Secure Access Service Edge (SASE): A network architecture that combines network security functions with wide area networking (WAN) capabilities to support the dynamic secure access needs of organizations.

SASE components:

- SD-WAN
- Firewall as a Service (FWaaS)
- Secure Web Gateway (SWG)
- Cloud Access Security Broker (CASB)
- Zero Trust Network Access (ZTNA)



How SASE aligns with change management

- Supports agile and scalable change
- Enhances security during transitions
- Centralized policy management
- Reduces complexity of change implementation
- Improves visibility and monitoring



Use cases of SASE

- Remote workforce security
- Secure cloud access
- Branch office connectivity
- IoT security
- Secure edge computing



Benefits of SASE

- Improved security posture
- Simplified management
- Enhanced user experience
- Cost savings
- Scalability

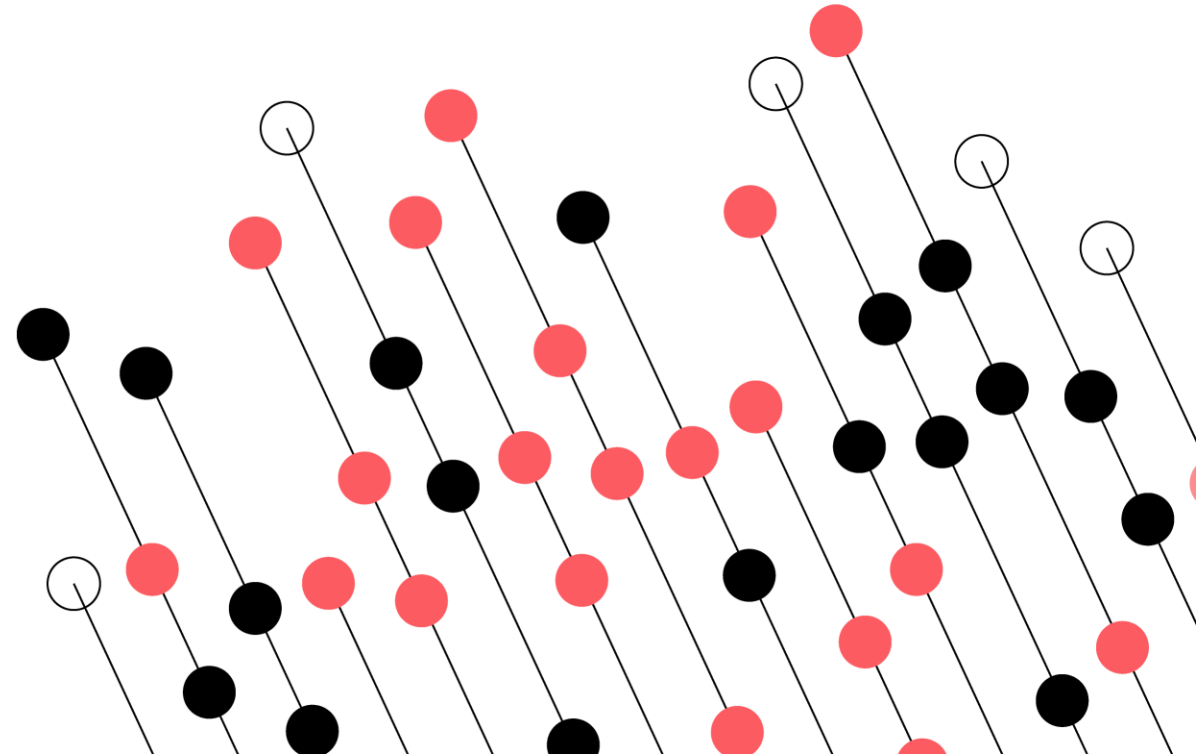


Challenges of SASE

- Integration complexity
- Performance concerns
- Vendor lock-in
- Skill gaps
- Regulatory compliance



Principles of change management



Objectives: Principles of change management

- Describe the core principles of change management
- Identify the impact of change management on business operations.
- Apply best practice in managing change effectively.
- Identify risks when carrying out changes to system / network.



Justification for change

- Why is an upgrade to the current system required?
 - Security
 - Performance
 - Compliance
 - Functionalities
- Give clear goals of what needs to be achieved by the change.
- Identify the impact of the change to the organization.
- How will it have impact on the employees (operations)?



Plan the change

- Identify and evaluate potential risks (downtime, costs, security, vulnerability).
- Identify stakeholders (IT, end-users, management).
- What will be achieved by the upgrade (KPI)?
- Get management support.
- Create contingency plan for mitigating potential downtime (backups).
- Identify pre and post change testing requirements.
- Create a communication channel.
- Define a deployment schedule.



Implement change

- Use a structured approach / framework during the change implementation (ITIL).
- Ensure employee engagement with the change.
- Implement any training requirements.
- Provide adequate support for employees.
- Regular update with stakeholders.
- Create detailed documentations of change (version control, audit logs).
- Use phased rollouts before business wide deployment.
- Deploy changes during off-peak to minimise disruptions.



Testing post change deployment

- Review the change outcomes to the change objectives.
- Continuously monitor to identify anomalies.
- Gather feedback from users to identify any issues.
- Identify a structured approach to rollback the system in the event of deployment failures.
- Maintain document review for compliance.
- Regular review to modify the change management process.

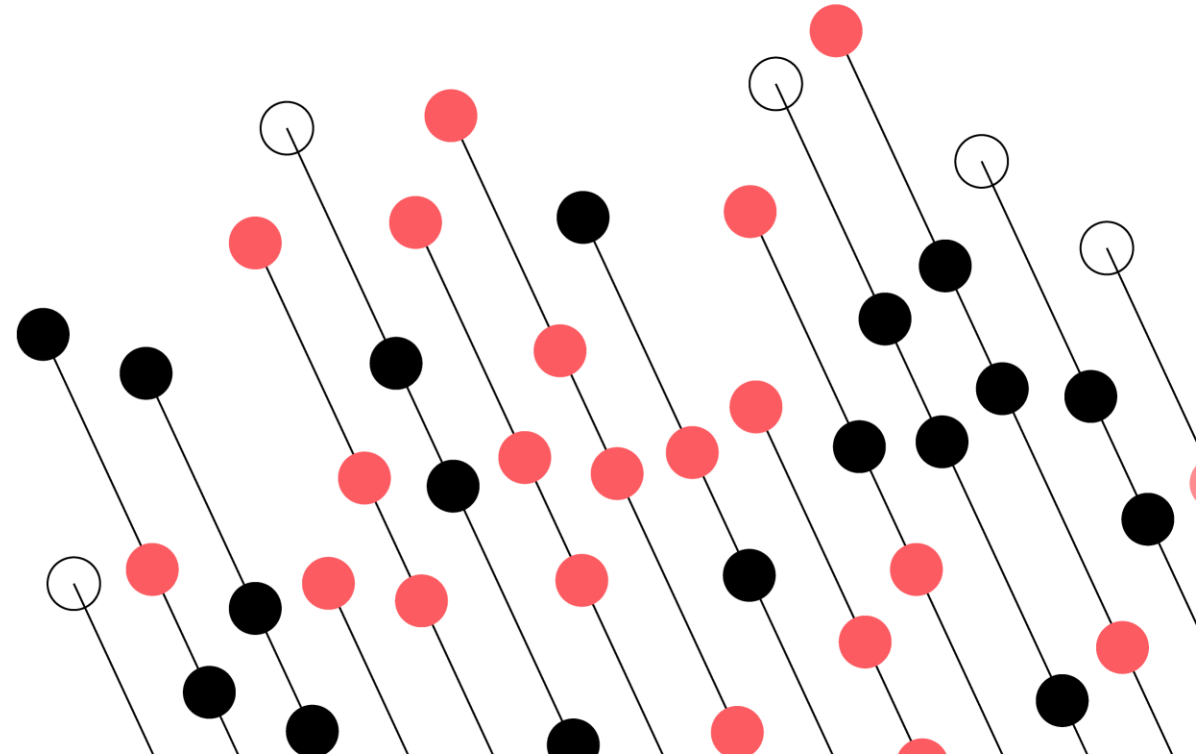


Summary and case studies: Reflection

- Has anyone experience a network / system changes that went wrong due to poor / lack change management process?
- What would you do if there was a network / system downtime due to a change? How could it have been prevented?
- What can businesses learn from network outages caused by poor change management?



**Ensure compliance of network
engineering outputs with change
management processes**



Objectives: Ensure compliance of network engineering outputs with change management processes

- Explain the importance of compliance in network change management.
- Apply best practices for maintaining compliance when making network changes.
- Describe the impact of failure of non-compliance on network change management.
- Develop procedures as network engineer to enforce compliance and minimise risk to business processes.



Compliance in network change management: Discussion and Q&A

- What is compliance in relation to network engineering and change management?
- Why is it critical for network engineering outputs to align with change management policies and procedures?
- What common standards do businesses use when creating change management (ITIL, ISO 27001, NIST)?
- What role does regulatory compliance play in network engineering changes (GDPR, DPA)?
- What measures should be put in place to ensure network engineers document and review changes and get approval before implementation?
- How would change logs help in providing audit trails for network modifications?



Compliance in network change management

- A process of ensuring modifications of system / network adhere to regulatory, security, and organisation policies.
- Make sure there is adequate approvals before changes are made.
- Ensure there is adequate security and data protection during system / network changes .
- Keeping documents of changes for audit trails.
- Consider regulatory requirements (GDPR, DPA, ISO).
- Minimise downtime to business operations.



Challenges in maintaining compliance: Discussions and Q&A

- What challenges do network engineers face when ensuring compliance in change management?
- How can businesses balance the need for immediate deployment with strict compliance requirements?
- What should be put in place to prevent unauthorised network changes? Can you give examples of issue that can occur because of unauthorised changes?
- What measures should be put in place to block unauthorised changes?
- How do businesses ensure compliance is maintained in a hybrid network environment?



Challenges in change management compliance

- Inconsistent change procedures
- Failure to document changes
- No audit trail to identify compliance requirements
- Lack of security and risk assessment prior to change
- Unauthorised changes to system / network
- Resistance to change by end-users
- No or poor clear communication channel
- Lack of sufficient testing prior to change
- Failure to align business policies to change



The role of network engineers in change management compliance: Discussions and Q&A

- What responsibilities do network engineers have in ensuring compliance with change management processes?
- How should network engineers respond when stakeholders request urgent changes that bypass standard procedures?
- What training and documentation should network engineers have to stay updated on change management policies?
- What is Zero Trust Access (ZTA) and how does it relate to change management compliance?



The role of a network engineer in change management

- Apply best practice for effective change management.
- Ensure security, performance, and compliance in line with change management.
- Evaluate need for change / upgrades to system / network.
- Identify and analyse potential risks.
- Configure network updates and apply security patches.
- Carry out pre and post testing in an isolated environment.
- Measure outcomes and track performance.
- Identify and resolve any issues.
- Maintain documentations of change.
- Ensure changes align with expected outcomes.
- Ensure secure deployment of changes.



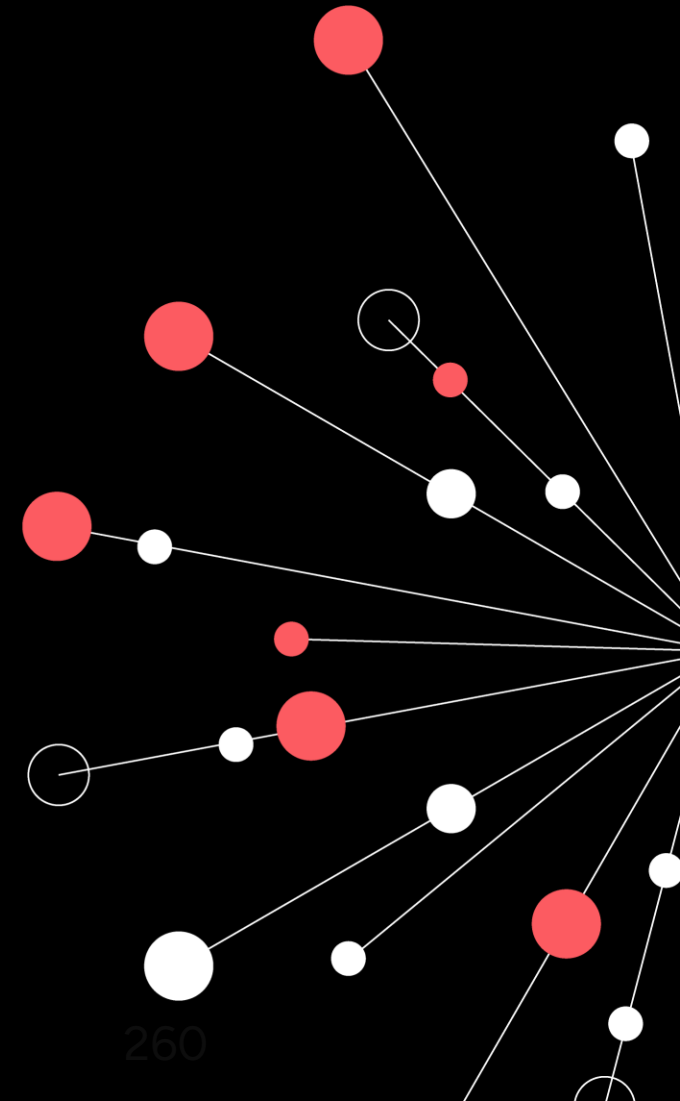
Summary and case studies: Reflection

- Can you provide some examples of where poor or lack of compliance with change management has led to network failures or security breach?
- Can you list some policies that you would include in change management policy to ensure compliance?
- How would you enforce compliance in your organisation for change management processes?
- How would you audit your business change management processes to ensure they remain relevant?



Activity 19: Change management compliance

Find the activity in your **Learner Activity Document** and read the instructions.



End of module review

- Explain server hardware and operating system (OS) roles, features, and functions.
- Compare server types, storage options, Redundant Array of Independent Disks (RAID) levels, and disaster recovery (DR) sites.
- Evaluate system configurations, storage solutions, and network maintenance strategies.
- Identify tools and steps for server installation, configuration, and post-installation.
- Apply virtualisation, secure installation, and RAID setup in case studies and labs.
- Discuss automation, Simple Network Management Protocol (SNMP), and network automation risks / benefits.
- Describe OS deployment, monitoring, and performance optimisation for Linux and Windows.
- Construct and optimise file systems, partitions, memory, and troubleshoot server OS, Active Directory, print servers, and performance tools.
- Recommend best practices for Service Level Agreements (SLAs), change management (CM), and documentation.

End of module: Review

Upload your completed Delegate Guide and *Certification of Completion to Bud under the following module:

- **Activity (Live event)**

*Certification of Completion can be obtained by going to the Live event module in Bud, pressing 'start now'. Then, tick off the modules accordingly.

Learning instructions & materials

To complete this activity:

1. Attend a two-day live event (*), led by your tutor | [Start now](#) | - please don't start this learning path before the live event.
2. Submit your work in the 'Messages and Submissions' area above

